

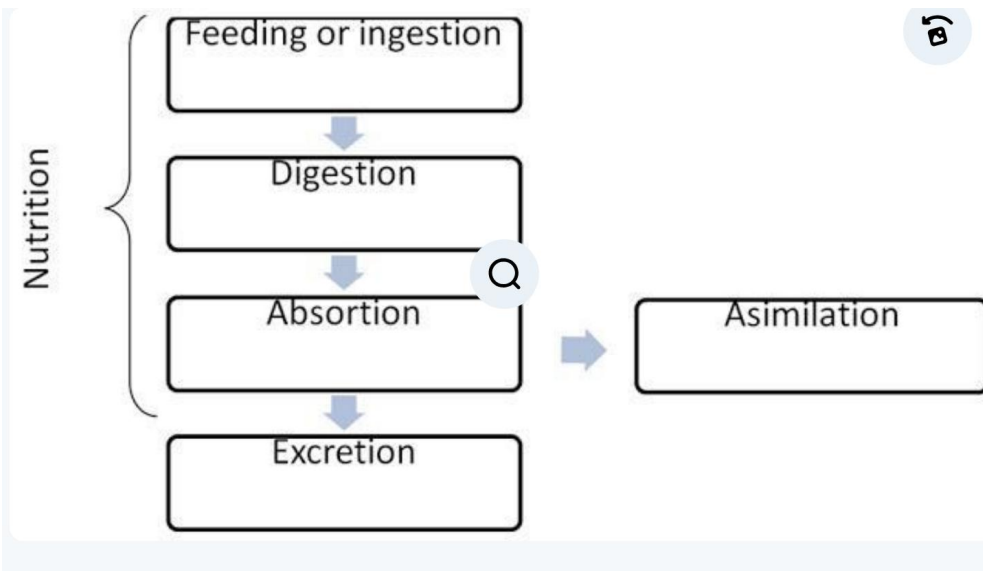
SS2 BIOLOGY LESSON NOTES

FIRST TERM

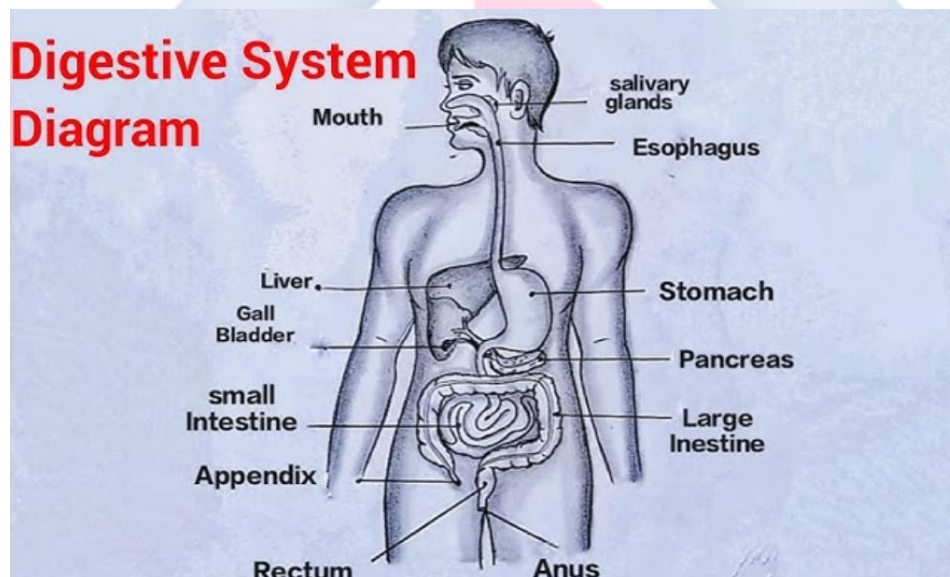
WEEK 1: THE DIGESTIVE SYSTEM I - STRUCTURE AND MACRONUTRIENT JOURNEY

1. Introduction to Nutrition: Beyond "Eating" Nutrition is the sophisticated biochemical and physiological process by which an organism utilises food to support its life. It's not merely about consumption, but a series of coordinated steps:

- **Ingestion:** The act of taking food into the body. This is the starting point of the digestive journey.
- **Digestion:** The transformative process where large, complex, and insoluble food molecules (macromolecules) are broken down into their small, simple, and soluble building blocks (monomers). This is essential because only these small molecules can be absorbed into the bloodstream.
- **Absorption:** The passage of these digested end-products from the lumen of the alimentary canal into the blood and lymph capillaries.
- **Assimilation:** The journey of absorbed nutrients from the bloodstream into the body's cells, where they are utilised for energy, growth, and repair.
- **Egestion (Defecation):** The elimination of undigested material and metabolic waste (like cellulose fiber and dead bacteria) from the body. **Crucially, this is not excretion**, as the material has never entered the body's tissues.



2. The Alimentary Canal: A Specialised Tubular System The alimentary canal is a continuous, muscular tube extending from the mouth to the anus, approximately 9 meters long in adults. Its structure is highly specialised in different regions for specific functions.



- **Mouth (Buccal Cavity):**

- **Teeth:** Perform **mechanical digestion** (mastication), breaking down food into a soft, flexible bolus and dramatically increasing its surface area for enzyme action.
- **Tongue:** A muscular organ that manipulates food, mixes it with saliva, and initiates the swallowing reflex.

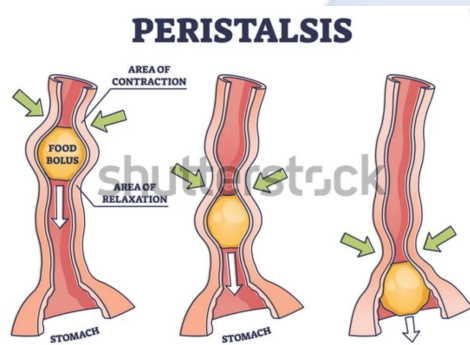
- **Salivary Glands:** Three pairs (parotid, sublingual, submandibular) secrete **saliva**.

Saliva contains:

- **Water:** Moistens food.
- **Mucin:** A glycoprotein that lubricates the bolus for easy swallowing.
- **Salivary Amylase (Ptyalin):** The first digestive enzyme, which begins the chemical digestion of starch into maltose.

- **Oesophagus (Gullet):**

- A muscular tube connecting the pharynx to the stomach.
- **Peristalsis:**



This is the involuntary, wave-like contraction of the smooth muscles in the alimentary canal wall that propels the bolus forward. It is so effective it works even against gravity.

- **Stomach:**

- A J-shaped, distensible muscular sac acting as a temporary storage and mixing chamber.
- **Gastric Glands** in the stomach lining secrete **gastric juice**, which contains:
 - **Hydrochloric Acid (HCl):**
 - Creates a highly acidic environment (pH ~1.5-2.0).
 - Denatures proteins (unfolding their complex structure).
 - Kills most ingested pathogens.
 - Activates the enzyme precursor **pepsinogen** into its active form, **pepsin**.
 - **Pepsin:** A protease that breaks down proteins into shorter polypeptide chains.
 - **Rennin:** Present in infants, coagulates milk protein (caseinogen) to slow its passage and allow better digestion.

- The churning action of the stomach's three muscular layers mixes the food with gastric juice, turning it into a semi-liquid paste called **chyme**. The **pyloric sphincter** regulates the slow release of chyme into the small intestine.
- **Small Intestine:** The primary site for digestion and absorption, divided into three parts:
 - **Duodenum (approx. 25 cm):** The C-shaped first part. This is the "mixing bowl" where chyme meets digestive juices from the liver and pancreas. Most chemical digestion is completed here.
 - **Jejunum (approx. 2.5 m):** The middle section, where some absorption begins.
 - **Ileum (approx. 3.5 m):** The final and longest section, specialised for the absorption of digested food molecules. Its internal structure is highly adapted for this role.
- **Large Intestine (Colon):**
 - Wider but shorter than the small intestine.
 - Primary functions are:
 - **Water and Mineral Absorption:** Reclaims water from the indigestible residue, solidifying it into faeces.
 - **Vitamin Synthesis:** Hosts symbiotic bacteria (gut flora) that produce essential vitamins like Vitamin K and some B vitamins.
 - **Egestion:** The rectum stores faeces until expulsion via the anus.

3. The Associated Glands: The Chemical Factories These accessory organs produce secretions vital for chemical digestion.

- **Liver: The Body's Metabolic Powerhouse**
 - Produces **bile**, a greenish-yellow, alkaline fluid.
 - **Bile is not an enzyme; it has a physical/mechanical role:**
 - **Emulsification:** Bile salts act as a biological detergent, breaking down large fat globules into a fine emulsion of tiny droplets. This increases the surface area for the enzyme **lipase** to work on by over 1000 times.
 - **Neutralization:** The alkaline bile neutralizes the acidic chyme from the stomach, providing the optimal pH for pancreatic and intestinal enzymes to function.

- The **gallbladder** stores and concentrates bile, releasing it into the duodenum via the bile duct when fat is present.
- **Pancreas: The Enzyme Powerhouse**
 - An exocrine and endocrine gland.
 - Its exocrine part produces **pancreatic juice**, a cocktail of enzymes and alkaline salts:
 - **Sodium Bicarbonate:** Neutralises stomach acid.
 - **Pancreatic Amylase:** Continues starch digestion.
 - **Trypsin:** A protease that breaks down proteins and polypeptides.
 - **Pancreatic Lipase:** The main fat-digesting enzyme.
 - These enzymes are released into the duodenum via the pancreatic duct.

Evaluation:

1. Create a table comparing the five stages of nutrition, defining each and stating its primary location in the human body.
2. Explain the statement: "The stomach is a site of both mechanical and chemical digestion." Provide specific evidence for each.
3. Describe the journey of a piece of bread from the mouth to the stomach, detailing the physical and chemical changes it undergoes.
4. Why is bile crucial for fat digestion even though it contains no enzymes? Use the term "surface area" in your answer.

Assignment:

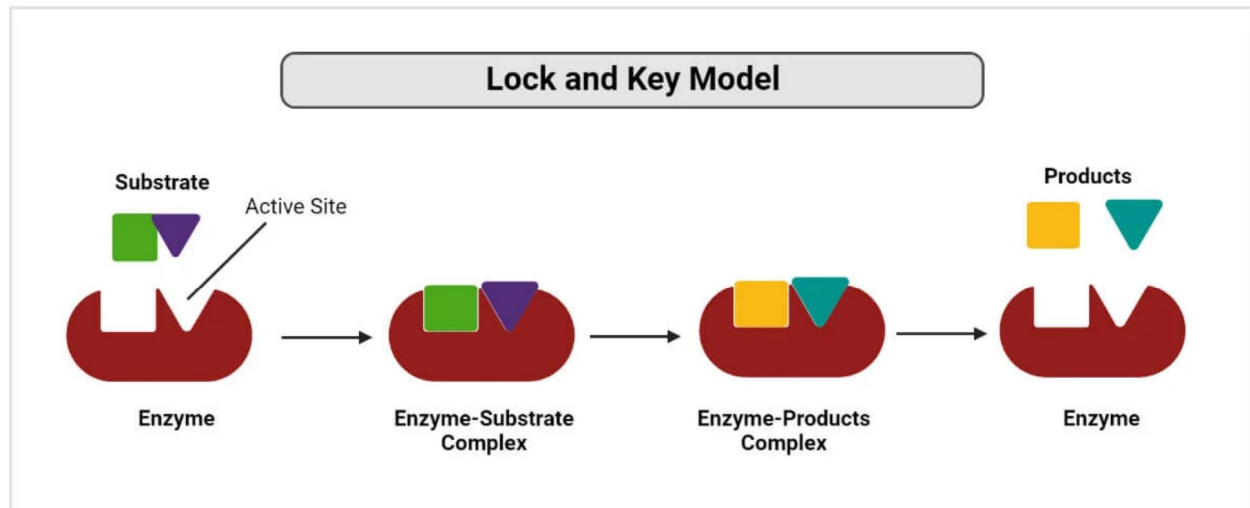
1. **Model Building:** Create a 3D model or a highly detailed poster of the human digestive system, labeling all major organs and associated glands. Include a short description of each part's function.
2. **Research and Report:** Investigate one common disorder of the digestive system (e.g., Gastric Ulcers, Gallstones, Lactose Intolerance). Write a one-page report on its causes, symptoms, and effects on digestion.

3. **Practical Connection:** Keep a 24-hour food log. For one meal, identify the primary macronutrients (carbohydrates, proteins, fats) and trace their hypothetical path through the digestive system, noting where each would be digested and absorbed.



WEEK 2: THE DIGESTIVE SYSTEM II - ENZYMES, ABSORPTION, AND UTILIZATION

1. Enzymes of Digestion: The Molecular Scissors. Enzymes are highly specific biological catalysts, almost always proteins, that speed up biochemical reactions without being consumed. Digestive enzymes are primarily **hydrolases**; they break down molecules by adding water (hydrolysis).



Lock and Key Model

Detailed Enzyme Action Table:

Location	Gland/Source	Enzyme	Substrate	Products	Optimal pH & Notes
Mouth	Salivary Glands	Salivary Amylase	Starch (Polysaccharide)	Maltose (Disaccharide)	Neutral (6.7-7.0). Denatured by stomach acid.
Stomach	Gastric Glands	Pepsin	Proteins	Peptides (shorter chains)	Acidic (1.5-2.5). Secreted as inactive pepsinogen to avoid self-digestion.

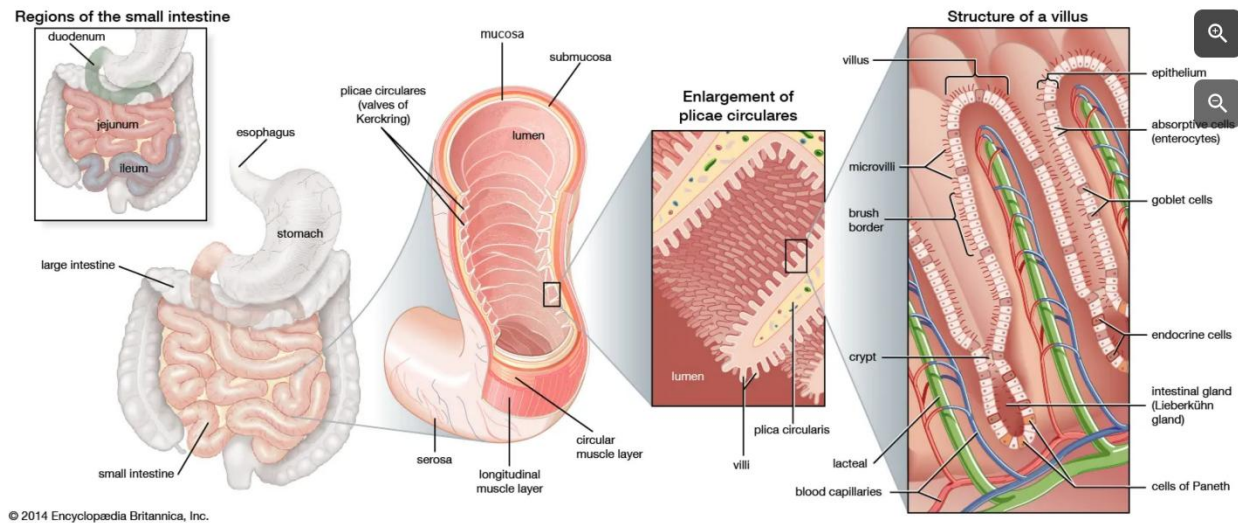
Location	Gland/Source	Enzyme	Substrate	Products	Optimal pH & Notes
Stomach (Infants)	Gastric Glands	Rennin	Soluble Caseinogen	Insoluble Casein (curd)	Acidic. Slows the passage of milk for better protein digestion.
Duodenum	Pancreas	Pancreatic Amylase	Starch	Maltose	Alkaline (8.0). Continues carbohydrate digestion.
Duodenum	Pancreas	Trypsin	Proteins/Peptides	Shorter Polypeptides	Alkaline. Secreted as inactive trypsinogen , activated by an intestinal enzyme.
Duodenum	Pancreas	Lipase	Emulsified Fats	Fatty Acids + Glycerol	Alkaline. Requires bile for emulsification to be effective.
Ileum	Intestinal Glands	Maltase	Maltose	Glucose + Glucose	Alkaline. These "brush border" enzymes complete digestion.

Location	Gland/Source	Enzyme	Substrate	Products	Optimal pH & Notes
Ileum	Intestinal Glands	Sucrase	Sucrose	Glucose + Fructose	Alkaline.
Ileum	Intestinal Glands	Lactase	Lactose	Glucose + Galactose	Alkaline. Lactose intolerance results from lactase deficiency.
Ileum	Intestinal Glands	Peptidases (Erepsin)	Polypeptides/Peptides	Amino Acids	Alkaline. The final step in protein digestion.

2. Absorption: Crossing the Intestinal Barrier The **ileum** is the primary site of absorption, and its structure is a masterpiece of biological engineering for maximising surface area.

Adaptations of the Ileum for Absorption:

1. **Great Length:** Over 6 meters long.
2. **Folds of Kerckring:** The inner wall is folded into large, circular folds.
3. **Villi:** Millions of finger-like projections cover the folds. Each villus is about 1mm long.



4. **Microvilli:** The epithelial cells lining each villus have a "brush border" of microvilli, tiny projections on their cell membrane. This is the final, massive increase in surface area.
5. **Excellent Blood Supply:** Each villus contains a dense network of blood capillaries and a central **lacteal** (a lymphatic vessel).

Mechanisms of Absorption:

- **Simple Diffusion:** Movement down a concentration gradient (e.g., fatty acids, glycerol, some vitamins).
- **Facilitated Diffusion:** Uses carrier proteins but no energy (e.g., fructose).
- **Active Transport:** Requires energy (ATP) to move substances against a gradient (e.g., glucose, galactose, amino acids). This ensures complete absorption even when their concentration in the gut is low.

Pathways of Absorbed Products:

- **Monosaccharides (Glucose, Fructose, Galactose) and Amino Acids:** Absorbed into the **blood capillaries** of the villi. These capillaries converge into the **hepatic portal vein**, which carries these nutrients directly to the **liver** for processing and regulation.
- **Fatty Acids and Glycerol:** They are small and lipid-soluble, so they diffuse easily into the epithelial cells of the villus. Inside these cells, they are recombined into **triglycerides**. These fats are then coated with proteins to form **chylomicrons**, which are too large to enter capillaries. Instead, they enter the **lacteal**. The lacteal, part of the lymphatic system, eventually drains its contents into the bloodstream near the heart.

3. Assimilation: From Blood to Body Assimilation is the utilization of absorbed nutrients by the body's cells.

- **The Liver's Role (The Body's Processing Plant):**

- **Glucose Regulation:** The liver maintains blood glucose levels. Excess glucose is converted to the storage polysaccharide **glycogen** (glycogenesis). When blood sugar is low, glycogen is broken down to glucose (glycogenolysis).
- **Amino Acid Management:** Amino acids are used to build body proteins. Excess amino acids cannot be stored. The liver removes the amino group ($-NH_2$) in a process called **deamination**, converting it to **urea**, which is excreted by the kidneys. The remaining carbon skeleton is used for energy or converted to fat.
- **Fat Processing:** The liver modifies and repackages fats, produces cholesterol, and breaks down old red blood cells.

Evaluation:

1. Explain why a different enzyme is needed to digest starch, proteins, and fats. Use the term "enzyme specificity."
2. Create a flowchart tracing the path of a protein molecule from an ingested piece of meat to being used to build muscle tissue in your arm. Include digestion, absorption, and assimilation.
3. A patient has a blocked lacteal. Predict the likely effect on their absorption of dietary fats and explain your reasoning.
4. Why is the liver considered a vital organ in the process of assimilation? Describe two of its key roles.

Assignment:

1. **Enzyme Experiment Design:** Design a simple experiment to demonstrate the action of amylase on starch. Outline the hypothesis, materials, method, and how you would test for the presence of starch and sugar.

2. **Villus Model:** Build a physical or digital 3D model of a villus. Your model must clearly show and label the epithelium, blood capillaries, and lacteal, and explain the function of each.
3. **Case Study Analysis:** Research and write a report on Celiac Disease. Explain how the autoimmune response damages the villi and the subsequent wide-ranging effects on health due to malabsorption.



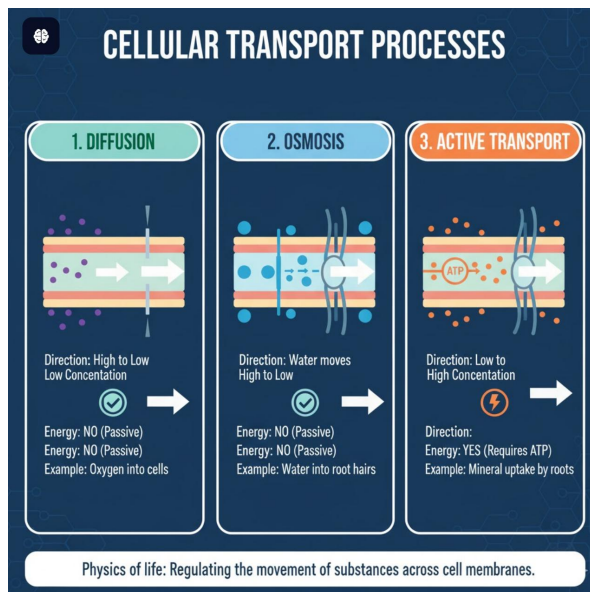
WEEK 3: TRANSPORTATION SYSTEM I - THE CELLULAR CROSSROADS

1. The Fundamental Need for Transport: The SA:V Ratio Dilemma The necessity for a transport system boils down to a fundamental principle of geometry and physics: the **Surface Area to Volume Ratio (SA:V)**.

(Illustration: A diagram comparing a single, large cube to multiple small cubes with the same total volume.)

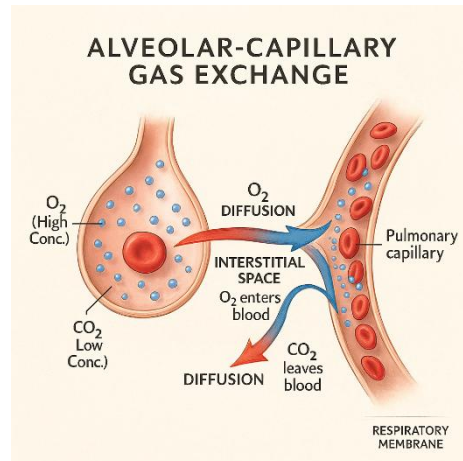
- **Unicellular Organisms (e.g., Amoeba):**
 - Are small, so they have a **large Surface Area relative to their Volume (High SA:V)**.
 - Calculation Example: A cube with 1mm sides has a SA:V ratio of 6:1. The same volume split into 1000 cubes of 0.1mm sides has a SA:V of 60:1.
 - This large surface area allows for efficient exchange of gases, nutrients, and wastes directly across the cell membrane via simple diffusion.
 - **Cytoplasmic Streaming (Cyclosis):** The movement of cytoplasm within the cell helps distribute materials.
- **Multicellular Organisms (e.g., Humans):**
 - Are large, so they have a **small Surface Area relative to their Volume (Low SA:V)**.
 - Calculation Example: A cube with 10cm sides has a SA:V ratio of 0.6:1.
 - The cells deep within the body are too far from the external environment for diffusion to be fast enough to meet their metabolic needs.
 - **Solution:** Evolution of specialized, internal transport systems (e.g., circulatory system, xylem/phloem) to act as an "internal environment" and a delivery network.

2. Transport Processes: The Physics of Life Substances move across membranes and within organisms through three primary physical processes.

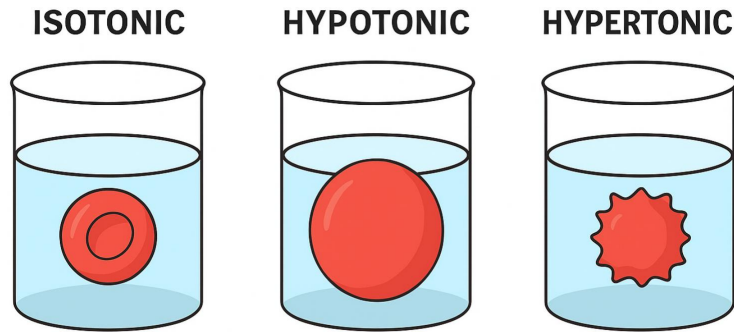


- **A. Diffusion: The Downhill Journey**

- **Definition:** The net movement of molecules or ions from a region of their **higher concentration** to a region of their **lower concentration** (down a concentration gradient).
- **Energy:** Passive process (does not require ATP).
- **Mechanism:** Driven by the intrinsic kinetic energy of particles.
- **Factors Affecting Rate:**
 - Concentration Gradient:** The steeper the gradient, the faster the diffusion.
 - Temperature:** Higher temperature increases particle kinetic energy, speeding up diffusion.
 - Surface Area:** A larger surface area allows for more particles to cross per unit time.
 - Distance:** Shorter distances allow for faster diffusion.
 - Size/Mass of Molecule:** Smaller, lighter molecules diffuse faster.
- **Biological Examples:**
 - Gaseous exchange in the alveoli of the lungs



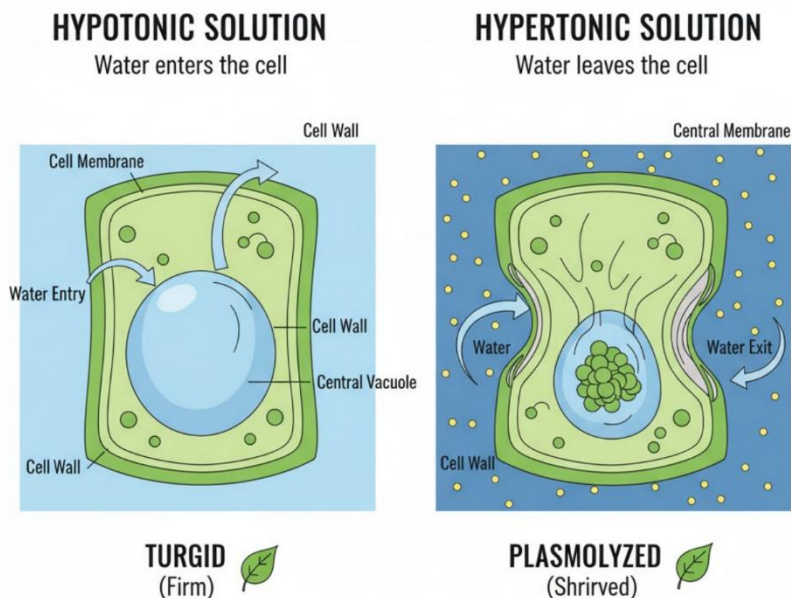
- Movement of neurotransmitters across a synapse.
- Absorption of some digested food in the ileum.
- **B. Osmosis: The Special Case of Water**
 - **Definition:** The net movement of **water molecules** from a region of **higher water potential** (dilute solution) to a region of **lower water potential** (concentrated solution), through a **selectively permeable membrane**.
 - **Key Concepts:**
 - **Water Potential (Ψ):** A measure of the potential energy of water. Pure water has the highest water potential ($\Psi=0$). Adding solutes lowers the water potential (makes it more negative).
 - **Selectively Permeable Membrane:** Allows water and some small molecules to pass, but blocks larger solute molecules.
 - **Energy:** Passive process (does not require ATP).
 - **Biological Examples & Effects:**
 - **In Animal Cells (e.g., Red Blood Cells):**



- **Isotonic:** Same water potential inside and outside. Cell remains normal.
 - **Hypotonic:** Outside has higher water potential (more dilute). Water moves *into* the cell, causing it to swell and potentially **burst (lysis)**.
 - **Hypertonic:** Outside has lower water potential (more concentrated). Water moves *out of* the cell, causing it to **shrink (crenation)**.
- **In Plant Cells:**

EFFECTS OF OSMOSIS ON A PLANT CELL

Illustrative Diagram: Hypotonic vs. Hypertonic Solutions



Water Potential Gradient

- **Hypotonic:** Water enters the cell. The cytoplasm pushes against the rigid cell wall, making the cell **turgid**. This is essential for support in non-woody plants.
- **Hypertonic:** Water leaves the vacuole and cytoplasm. The cytoplasm shrinks away from the cell wall, a process called **plasmolysis**. The plant wilts.
- **C. Active Transport: The Uphill Battle**
 - **Definition:** The movement of molecules or ions from a region of their **lower concentration** to a region of their **higher concentration** (against a concentration gradient).
 - **Energy:** Active process (requires ATP) from cellular respiration.
 - **Mechanism:** Involves specific **carrier proteins** embedded in the cell membrane that act as "pumps." These proteins change shape to move the substance across the membrane, a process that consumes energy.
 - **Biological Examples:**
 - **Sodium-Potassium Pump:** Crucial for nerve impulse transmission.
 - **Uptake of Mineral Ions by Root Hair Cells:** Soil water is very dilute, so ions must be pumped into the root against a steep gradient.
 - **Reabsorption of Glucose and Amino Acids** from the kidney tubules.

3. Comparative Analysis of Transport Processes

Feature	Diffusion	Osmosis	Active Transport
Substance Moved	Any particle (solutes, gases)	Water molecules only	Specific ions/molecules (e.g., Na ⁺ , K ⁺ , glucose)
Concentration Gradient	Down the gradient (High → Low)	Down the water potential gradient	Against the gradient (Low → High)

Feature	Diffusion	Osmosis	Active Transport
Membrane Requirement	No (but can occur across one)	Yes (Selectively Permeable)	Yes (with Carrier Proteins)
Energy (ATP) Requirement	No (Passive)	No (Passive)	Yes (Active)
Saturation Kinetics	No (rate proportional to gradient)	No	Yes (limited by number of carrier proteins)

Evaluation:

1. Using the concept of SA:V ratio, explain in detail why an amoeba does not need a heart, but an elephant does.
2. A student places a dried apricot in pure water and it swells up. They then place a fresh cucumber in concentrated salt water and it becomes limp. Explain these observations using the principles of osmosis.
3. Describe the role of ATP and carrier proteins in the process of active transport. Why is this process essential for the absorption of minerals by plant roots?
4. Create a table to highlight three key differences between diffusion and active transport.

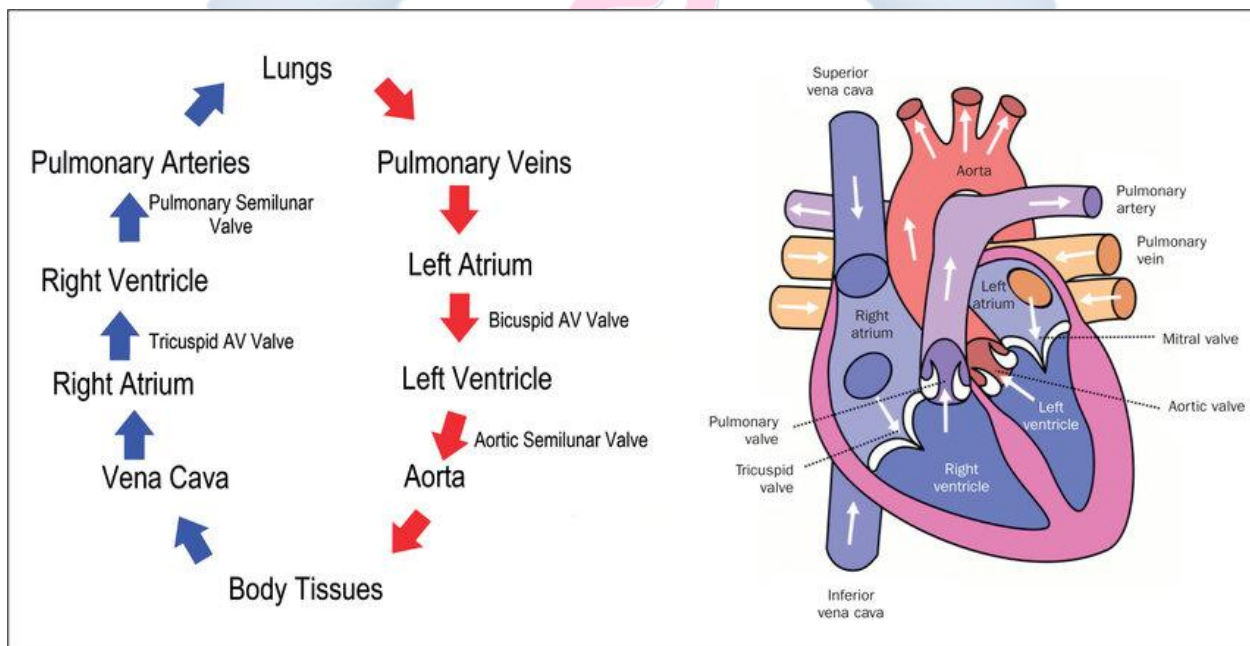
Assignment:

1. **Practical Investigation:** Design and conduct a simple experiment to demonstrate osmosis using a raw potato and sugar/salt solutions. Write a full lab report including aim, hypothesis, method, results, and conclusion.
2. **Research and Report:** Investigate cystic fibrosis, a genetic disorder that affects active transport mechanisms in the body. Explain how a faulty chloride ion channel leads to the symptoms of the disease.
3. **Concept Map:** Create a detailed concept map linking the three transport processes to their roles in the human digestive, respiratory, and excretory systems.

WEEK 4: TRANSPORTATION SYSTEM II - THE CIRCULATORY SUPERHIGHWAY

1. The Mammalian Circulatory System: A Closed, Double Circuit Mammals have a highly efficient system characterized by two key features:

- **Closed System:** Blood is confined to vessels (arteries, veins, capillaries) and does not directly bathe the tissues. This allows for higher blood pressure and more precise control of blood flow.
- **Double Circulation:** The blood passes through the heart twice for one complete circuit of the body. This separates oxygenated and deoxygenated blood and creates two distinct circuits:
 - a. **Pulmonary Circulation: Right Heart → Lungs → Left Heart.** This circuit's purpose is solely for gas exchange. Deoxygenated blood is pumped to the lungs to become oxygenated.
 - b. **Systemic Circulation: Left Heart → Body Tissues → Right Heart.** This circuit delivers oxygen and nutrients to every cell in the body and removes waste products.

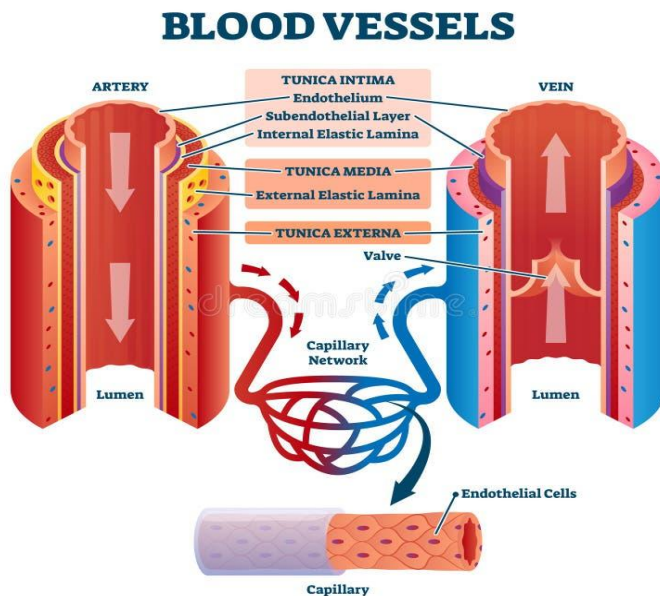


2. The Heart: The Unceasing Pump The heart is a four-chambered, muscular organ located in the thoracic cavity, slightly to the left, and protected by the rib cage.

- **Chambers and Their Functions:**

- **Atria (singular: Atrium):** The thin-walled, upper chambers that **receive** blood.
 - **Right Atrium:** Receives deoxygenated blood from the body via the **Superior and Inferior Vena Cava**.
 - **Left Atrium:** Receives oxygenated blood from the lungs via the **Pulmonary Veins**.
- **Ventricles:** The thick-walled, lower chambers that **pump** blood out of the heart.
 - **Right Ventricle:** Pumps deoxygenated blood to the lungs via the **Pulmonary Artery**. Its wall is moderately thick.
 - **Left Ventricle:** Pumps oxygenated blood to the entire body via the **Aorta**. It has the **thickest, most muscular wall** because it must generate enough force to overcome the high resistance of the systemic circulation.
- **Valves: The One-Way Gates**
 - **Atrioventricular (AV) Valves:** Between atria and ventricles. Prevent backflow into atria when ventricles contract.
 - **Tricuspid Valve:** On the right side (3 flaps).
 - **Bicuspid (Mitral) Valve:** On the left side (2 flaps).
 - **Semi-lunar Valves:** At the base of the pulmonary artery and aorta. Prevent backflow into the ventricles when they relax.
- **The Cardiac Cycle: "Lub-Dub"** The sequence of events in one complete heartbeat (approx. 0.8 seconds).
 - a. **Atrial Systole:** Atria contract, pushing blood through the AV valves into the ventricles.
 - b. **Ventricular Systole:** Ventricles contract. AV valves close ("Lub" sound). High pressure forces open the semi-lunar valves, ejecting blood into the arteries.
 - c. **Diastole:** All chambers relax. Semi-lunar valves close ("Dub" sound) to prevent backflow. Heart fills passively with blood from the veins.
- **Coronary Arteries:** The first branches off the aorta, these vital vessels supply the heart muscle itself with oxygenated blood. A blockage here causes a myocardial infarction (heart attack).

3. Blood Vessels: The Arteries, Veins, and Capillaries The structure of each vessel is perfectly adapted to its function.

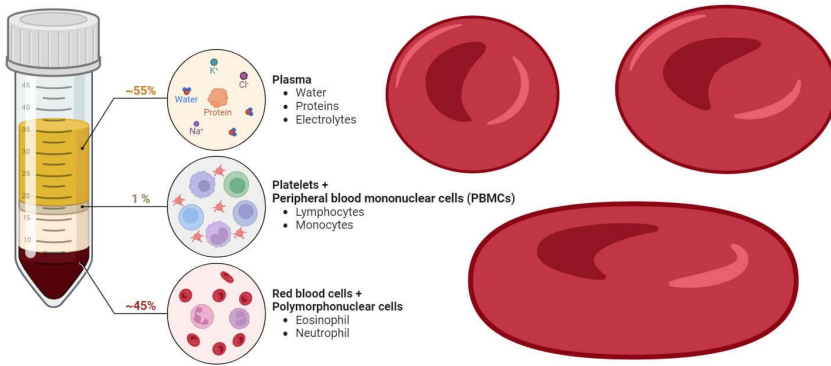


Feature	Artery	Capillary	Vein
Function	Carry blood away from heart	Site of exchange with tissues	Carry blood towards heart
Wall Structure	Thick, muscular, elastic	Very thin (one cell thick)	Thin , less muscular
Lumen Diameter	Narrow	Microscopic (one RBC wide)	Wide
Blood Pressure	High	Drops significantly	Low
Valves	Absent (except aorta/pulmonary)	Absent	Present (prevent backflow)
Blood Carried	Oxygenated (except Pulmonary)	Mixed (O ₂ delivered, CO ₂ collected)	Deoxygenated (except Pulmonary)

4. Blood: The River of Life Blood is a complex connective tissue composed of cells suspended in a liquid matrix called plasma.

- **Plasma (55%):** A straw-colored liquid, about 92% water.
 - **Functions:** Transport medium for blood cells, platelets, nutrients (glucose, amino acids), hormones, waste (CO₂, urea), and distributes heat.
- **Red Blood Cells (Erythrocytes):**
 - **Structure:**

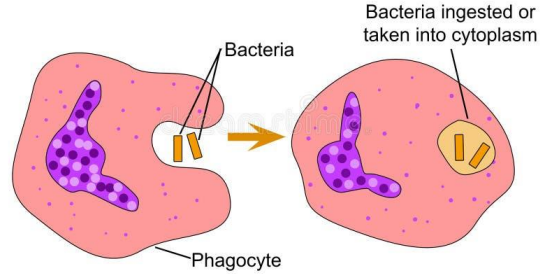
Red Blood Cells (RBCs)



No nucleus (in mammals), packed with **hemoglobin**. The biconcave shape maximizes surface area for gas exchange and allows flexibility.

- **Function:** Transport oxygen. Hemoglobin binds reversibly with O₂ in the lungs to form **oxyhemoglobin** and releases it in the tissues.
- **White Blood Cells (Leucocytes):** The Army of the Immune System.
 - **Structure:** Larger than RBCs, have a nucleus.
 - **Types and Functions:**
 - **Phagocytes (e.g., Neutrophils, Macrophages):** Engulf and digest pathogens (phagocytosis). Part of the **non-specific** immune response.

Phagocytosis



- **Lymphocytes (B-cells and T-cells):** [Diagram: B-cell producing antibodies.]
Responsible for **specific** immunity. **B-cells** produce **antibodies** that neutralize specific pathogens. **T-cells** destroy infected body cells.
- **Platelets (Thrombocytes):**
 - **Structure:** Small, irregularly shaped cell fragments.
 - **Function: Blood Clotting (Hemostasis).**
 - i. **Vasoconstriction:** Vessel narrows.
 - ii. **Platelet Plug:** Platelets adhere to the damaged site and form a temporary plug.
 - iii. **Coagulation:** Platelets and damaged tissue release chemicals that trigger a cascade, resulting in the conversion of soluble **fibrinogen** into insoluble **fibrin** threads. These threads form a mesh that traps blood cells, forming a stable clot (scab).

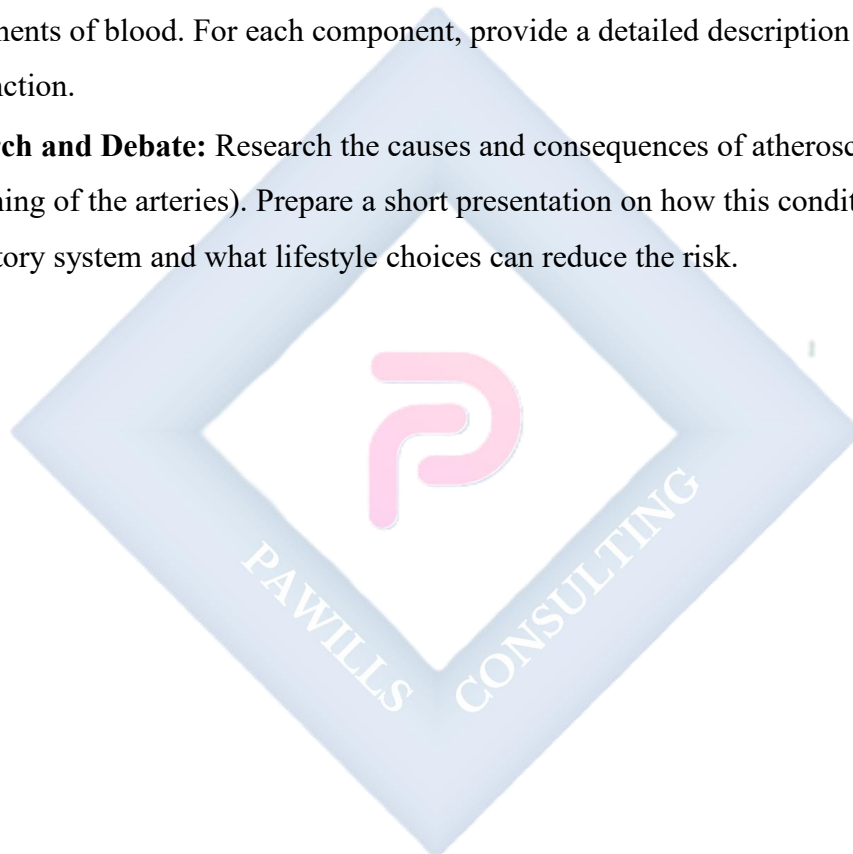
Evaluation:

1. Explain the physiological advantage of a double circulatory system over a single circulatory system (like in fish).
2. Trace the path of a single red blood cell from the right atrium, through the pulmonary and systemic circuits, and back to the right atrium. Name all chambers and major vessels it passes through.
3. Justify the following statement: "The structure of an artery is perfectly suited to withstand high pressure, while the structure of a vein is suited for low-pressure return of blood."

4. Describe the role of lymphocytes in the immune response and contrast it with the role of phagocytes.

Assignment:

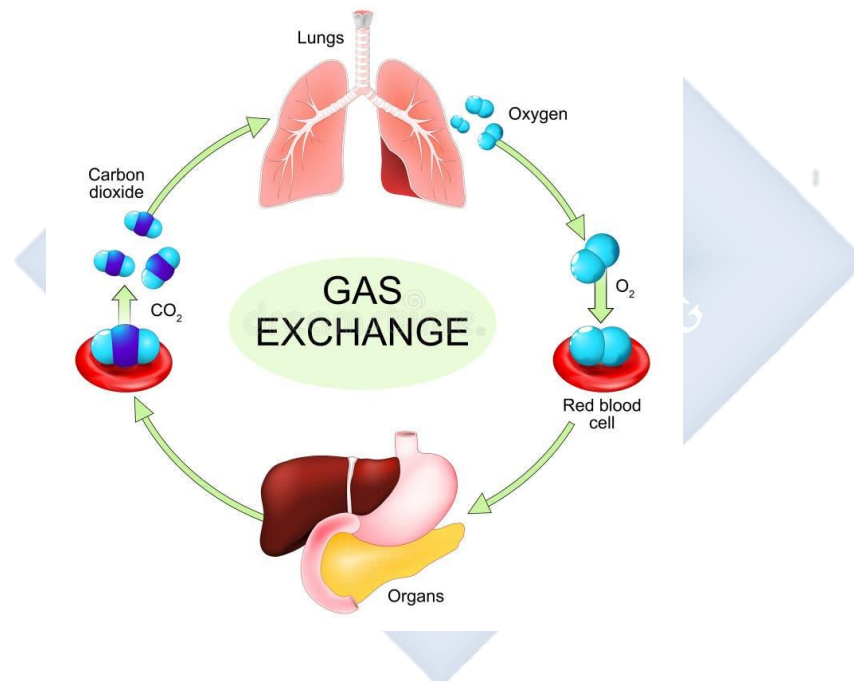
1. **Heart Dissection Report:** (If a heart is available for dissection) Conduct a dissection of a mammalian heart (e.g., sheep's). Identify all external and internal structures and relate them to their function. Submit a labeled drawing and a written report.
2. **Blood Cell Model:** Create a 3D model or a detailed poster showing the different components of blood. For each component, provide a detailed description of its structure and function.
3. **Research and Debate:** Research the causes and consequences of atherosclerosis (hardening of the arteries). Prepare a short presentation on how this condition affects the circulatory system and what lifestyle choices can reduce the risk.



WEEK 6: RESPIRATORY SYSTEM I - THE MECHANICS OF BREATHING AND GASEOUS EXCHANGE

1. The Dual Meaning of "Respiration": A Critical Distinction It is crucial to understand that "respiration" in biology has two distinct but interconnected meanings:

- **Gaseous Exchange (Breathing):** The **physical process** of inhaling and exhaling, and the subsequent exchange of oxygen (O_2) and carbon dioxide (CO_2) between an organism and its environment. This is the focus of this week.
- **Cellular Respiration:** The **biochemical process** occurring within cells where glucose is broken down to release energy (ATP), consuming O_2 and producing CO_2 .



2. The Human Respiratory System: An Airway Highway The respiratory system is designed to conduct air from the external environment to the delicate sites of gas exchange, while simultaneously conditioning the air.

(Illustration: A large, detailed, labeled diagram of the human respiratory system, from the nose to the alveoli.)

- **1. Nostrils & Nasal Cavity:**

- **Structure:** Lined with hairs (vibrissae) and a ciliated mucous membrane.
- **Function:**
 - **Filtering:** Hairs trap large dust particles; mucus traps finer particles and microbes.
 - **Warming:** A rich network of capillaries warms the incoming air to body temperature.
 - **Moistening:** Mucus humidifies the air, preventing the delicate lung tissues from drying out.
 - **Olfaction:** The roof of the nasal cavity contains olfactory receptors for smell.

- **2. Pharynx (Throat):**

- A common passage for both air and food.
- **The Epiglottis:** A flap of elastic cartilage that acts like a trapdoor. During swallowing, it closes over the **glottis** (the entrance to the larynx), directing food into the esophagus and preventing choking.

- **3. Larynx (Voice Box):**

- **Structure:** Composed of cartilage plates (the "Adam's apple" is the prominent thyroid cartilage). Contains the **vocal cords** – two folds of tissue that can be stretched and vibrated.
- **Function:** Production of sound. The pitch of sound is controlled by the tension of the vocal cords.

- **4. Trachea (Windpipe):**

- **Structure:** A tube about 10-12 cm long, reinforced with C-shaped rings of **hyaline cartilage**.
- **Function of Cartilage Rings:** Prevents the trachea from collapsing during inhalation when the pressure inside drops.
- **The Mucociliary Escalator:** The trachea is lined with ciliated columnar epithelium and goblet cells. Goblet cells secrete mucus to trap debris, and the cilia beat in a

coordinated, upward motion to sweep this mucus towards the pharynx, where it is swallowed or expectorated. This is a vital defense mechanism.

- **5. Bronchi and Bronchioles:**

- **Bronchi:** The trachea divides into the right and left primary bronchi, each leading to a lung. These also contain cartilage for support.
- **Bronchioles:** The bronchi undergo multiple divisions, becoming smaller bronchioles. The amount of cartilage decreases and smooth muscle increases, allowing for regulation of airflow.

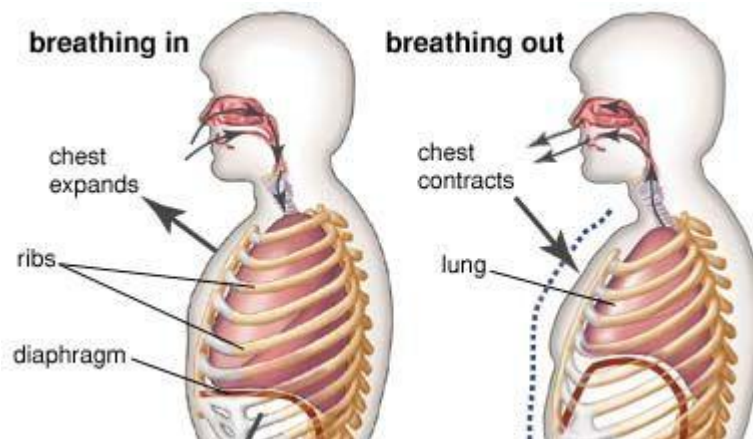
- **6. Alveoli: The Site of Gaseous Exchange**

- **Structure:** The bronchioles terminate in clusters of tiny, grape-like air sacs called **alveoli**. There are approximately 300-500 million alveoli in the human lungs, providing a massive surface area (~70-100 m²).
- **The Respiratory Membrane:** The wall of an alveolus is extremely thin (one cell thick) and is in close contact with the wall of a blood capillary. This combination of the alveolar wall, the capillary wall, and their fused basement membranes is the respiratory membrane, where gas exchange occurs.

3. Adaptations of the Alveoli for Maximum Efficiency The alveoli are perfectly engineered for their role:

1. **Large Surface Area:** Millions of alveoli create a surface area roughly the size of a tennis court.
2. **Thin Diffusion Pathway:** The respiratory membrane is only 0.5-1.0 micrometers thick, allowing for rapid diffusion.
3. **Moist Surface:** The inner lining of the alveoli is coated with a thin film of water, essential for gases to dissolve before diffusing.
4. **Rich Blood Supply:** A dense network of capillaries surrounds each alveolus, maintaining a steep concentration gradient for O₂ and CO₂ by constantly bringing deoxygenated blood and removing oxygenated blood.
5. **Surfactant:** A special phospholipid secretion that reduces the surface tension of the alveolar fluid, preventing the alveoli from collapsing during exhalation.

4. The Breathing Mechanism: A Pressure-Driven Pump Breathing is the result of pressure changes in the thoracic cavity, driven by the diaphragm and intercostal muscles.



- **A. Inhalation (Inspiration) - An Active Process**
 - a. The **external intercostal muscles** contract, pulling the rib cage upwards and outwards.
 - b. The **diaphragm** contracts and flattens, moving downwards.
 - c. These actions **increase the volume** of the thoracic cavity.
 - d. According to Boyle's Law (pressure decreases as volume increases), the **intrapulmonary pressure drops below atmospheric pressure**.
 - e. Air rushes into the lungs down this pressure gradient until the pressures equalize.
- **B. Exhalation (Expiration) - Usually a Passive Process**
 - a. The **external intercostal muscles** relax, and the rib cage moves downwards and inwards due to gravity.
 - b. The **diaphragm** relaxes and returns to its dome shape.
 - c. These actions **decrease the volume** of the thoracic cavity.
 - d. The **intrapulmonary pressure rises above atmospheric pressure**.
 - e. Air is forced out of the lungs.
 - **Forced Exhalation:** During exercise or coughing, this becomes active. The **internal intercostal muscles** and abdominal muscles contract to force air out more vigorously.

Evaluation:

1. Explain the difference between breathing and cellular respiration. How are these two processes related?
2. Trace the path of an oxygen molecule from the nostrils to an alveolus, describing how the air is modified along the way.
3. Describe three adaptations of the alveoli that make them efficient organs for gaseous exchange.
4. Explain the mechanism of inhalation, detailing the roles of the diaphragm and intercostal muscles and the associated pressure changes.

Assignment:

1. **Model Building:** Create a functional model of the lung and diaphragm using a plastic bottle, balloons, and rubber sheeting. Use your model to demonstrate and explain the process of inhalation and exhalation.
2. **Research and Report:** Investigate the condition of cystic fibrosis. Explain how a defect in chloride ion transport affects the mucociliary escalator and leads to severe respiratory problems.
3. **Practical Investigation:** Design an experiment to measure the effect of exercise on breathing rate. Formulate a hypothesis, outline your method for data collection, and predict your results.

WEEK 8: RESPIRATORY SYSTEM II - CELLULAR RESPIRATION: THE ENERGY CURRENCY OF LIFE

1. Cellular Respiration: The Universal Energy-Harvesting Process Cellular respiration is the controlled, enzymatic release of energy from organic compounds (primarily glucose) within cells to produce Adenosine Triphosphate (ATP). ATP is the universal "energy currency" of the cell; energy is stored in the high-energy bonds between its phosphate groups.

(Illustration: A diagram showing the structure of an ATP molecule, highlighting the three phosphate groups and the high-energy bonds.)

- **ATP/ADP Cycle:** When a cell needs energy, it hydrolyzes ATP: $\text{ATP} \rightarrow \text{ADP} + \text{P}_i + \text{Energy}$. This energy drives cellular work. Respiration recharges ADP back into ATP.

2. The Overall Equation and Its Significance

- **Word Equation:** Glucose + Oxygen $\rightarrow$ Carbon Dioxide + Water + Energy (as ATP)
- **Balanced Chemical Equation:** $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \sim 38 \text{ ATP}$
- **Significance:** This is an oxidation reaction. The energy that is released was originally captured from the sun and stored in the chemical bonds of glucose during photosynthesis.

3. The Two Main Types of Respiration Respiration can occur with or without oxygen, leading to vastly different energy yields.

(Illustration: A flowchart showing the decision point: "Oxygen Present?" leading to Aerobic or Anaerobic pathways.)

• A. Aerobic Respiration

- **Definition:** The complete breakdown of glucose into CO_2 and H_2O in the **presence of oxygen**.
- **Location:** Begins in the cytoplasm but is completed in the **mitochondria**.
- **Energy Yield:** Highly efficient, producing a maximum of **38 molecules of ATP per glucose molecule**.
- **Stages:** (A deeper look)

- i. **Glycolysis:** (In cytoplasm) One glucose molecule (6C) is partially broken down into two molecules of pyruvate (3C), producing a net gain of 2 ATP and 2 NADH.
- ii. **Krebs Cycle (Citric Acid Cycle):** (In mitochondrial matrix) Pyruvate is completely oxidized, releasing CO₂ and generating high-energy electron carriers (NADH, FADH₂) and 2 ATP.
- iii. **Electron Transport Chain (ETC):** (On inner mitochondrial membrane)
The high-energy electrons from NADH and FADH₂ are passed along a chain of proteins. This process pumps protons (H⁺) across the membrane, creating a gradient. As protons flow back through the enzyme **ATP synthase**, it drives the production of ~34 ATP (oxidative phosphorylation).
Oxygen acts as the final electron acceptor, forming water.

- **B. Anaerobic Respiration**

- **Definition:** The partial breakdown of glucose in the **absence of oxygen**.
- **Location:** **Cytoplasm only**.
- **Energy Yield:** Very inefficient, producing only **2 molecules of ATP per glucose molecule** (from glycolysis alone).
- **Types and End-Products:**
 - **In Animal Cells (e.g., Muscle Cells):**
 - **Equation:** Glucose → Lactic Acid + 2 ATP
 - **Mechanism:** When O₂ is scarce during intense exercise, pyruvate from glycolysis is converted to **lactic acid** by accepting electrons from NADH. This regenerates NAD⁺, which is essential for glycolysis to continue.
 - **Oxygen Debt:** The build-up of lactic acid causes muscle fatigue and cramps. After exercise, the heavy breathing continues to repay the "oxygen debt" – the extra oxygen is used to convert the lactic acid in the liver back into glucose or to oxidize it completely in the heart.
 - **In Plant and Yeast Cells (Alcoholic Fermentation):**
 - **Equation:** Glucose → Ethanol + Carbon Dioxide + 2 ATP

- **Mechanism:** Pyruvate is first converted to acetaldehyde, releasing CO₂. Acetaldehyde is then reduced by NADH to form **ethanol**.
- **Economic Importance:**
 - **Baking:** The CO₂ produced by yeast causes dough to rise.
 - **Brewing/Wine Making:** The ethanol produced is the desired alcoholic beverage.
 - **Biofuel Production:** Yeast is used to produce ethanol from plant material.

4. Comprehensive Comparison: Aerobic vs. Anaerobic Respiration

Feature	Aerobic Respiration	Anaerobic Respiration
Oxygen Requirement	Essential	Absent
Location	Cytoplasm & Mitochondria	Cytoplasm only
Completeness of Breakdown	Complete (to CO ₂ & H ₂ O)	Incomplete (to lactic acid or ethanol)
Energy Yield (ATP/Glucose)	High (36-38 ATP)	Low (2 ATP)
End Products	Carbon Dioxide, Water	Lactic Acid (animals) or Ethanol & CO ₂ (plants/yeast)
Efficiency	~40%	~2%
Long-Term Sustainability	Sustainable for hours	Only sustainable for short bursts (minutes)

Evaluation:

1. Explain why aerobic respiration produces significantly more ATP than anaerobic respiration.
2. Describe what happens in a muscle cell during vigorous exercise when oxygen delivery cannot keep up with demand. What is the consequence of this, and how is it resolved after the exercise?

3. Write the balanced chemical equations for both aerobic respiration and alcoholic fermentation.
4. A student claims that "anaerobic respiration is a waste of glucose." Evaluate this statement, discussing both its accuracy and the biological importance of anaerobic pathways.

Assignment:

1. **Practical Investigation: Yeast Fermentation:** Design and conduct an experiment to investigate the effect of temperature on the rate of anaerobic respiration in yeast. Measure CO₂ production as an indicator. Present your data in a graph and write a conclusion.
2. **Metabolic Pathways Poster:** Create a detailed, color-coded poster that compares the pathways of aerobic and anaerobic respiration. Include locations, inputs, outputs, and ATP yields for each stage.
3. **Case Study Analysis:** Research the phenomenon of "hitting the wall" or "bonking" in endurance athletics. Explain the physiological and biochemical reasons for this, linking it to the depletion of glucose/glycogen stores and the body's reliance on different energy systems.

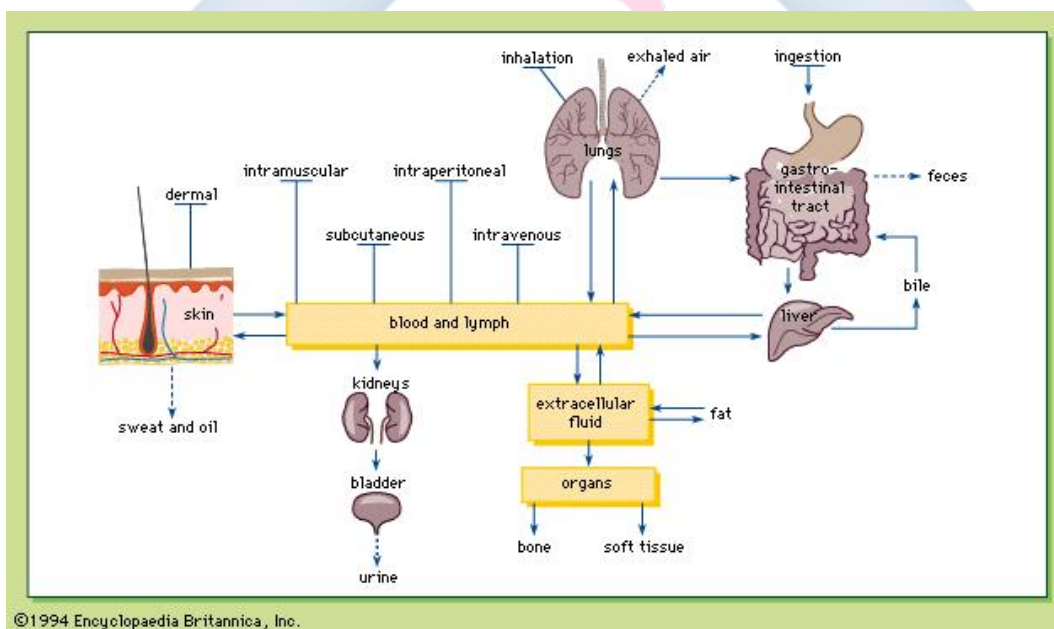
WEEK 9: EXCRETORY SYSTEM - THE BODY'S FILTRATION AND PURIFICATION SYSTEM

1. The Concept of Excretion: More Than Just Waste Removal Excretion is the biological process by which an organism separates and expels **metabolic wastes**—toxic by-products of cellular metabolism—from its body fluids. It is a fundamental process for maintaining homeostasis.

- **Crucial Distinction: Excretion vs. Egestion**

- **Excretion:** Removal of wastes formed from *chemical reactions inside cells* (e.g., CO₂ from respiration, urea from protein deamination).
- **Egestion (Defecation):** Removal of *undigested food material* that has never been part of the internal tissue environment (e.g., cellulose fiber in faeces).

2. The Excretory System: A Multi-Organ Effort The human body employs several organs to manage different types of waste.

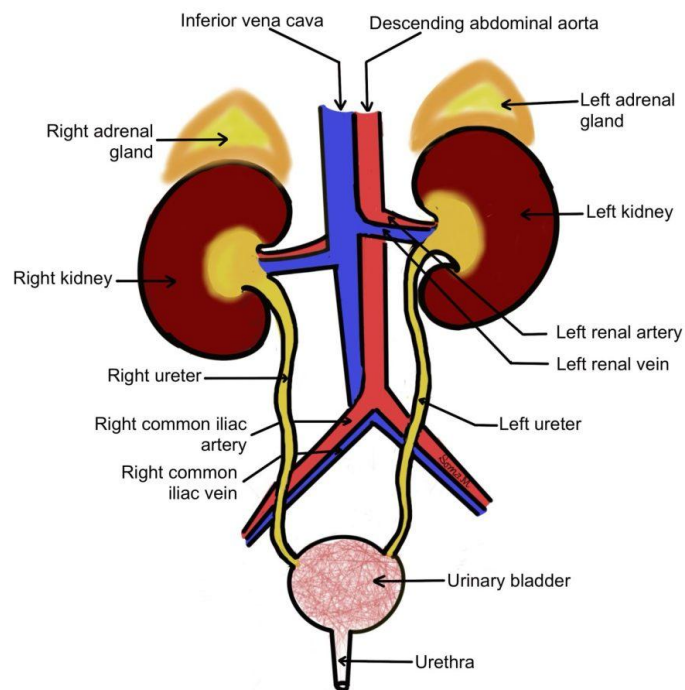


A. The Lungs

- **Waste Product: Carbon Dioxide (CO₂)** and water vapor from cellular respiration.

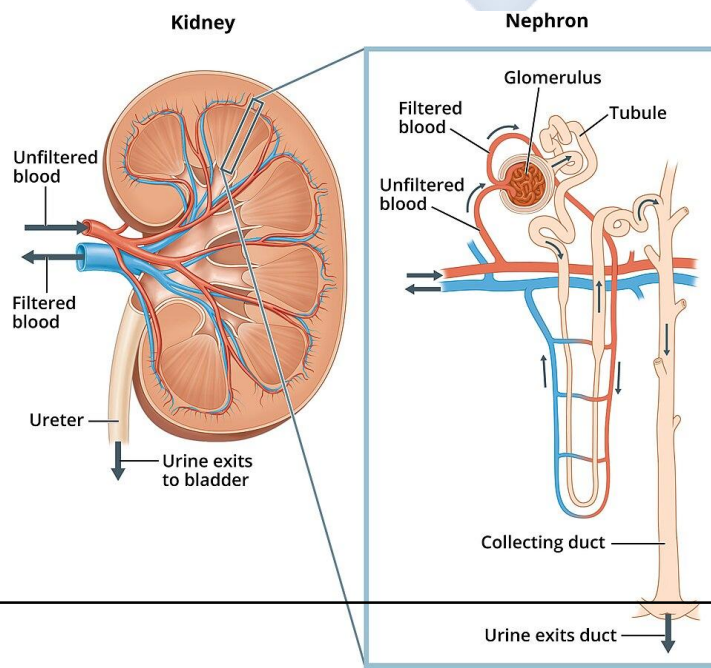
- **Mechanism:** CO₂ diffuses from the blood in the pulmonary capillaries into the alveoli and is exhaled.
- **B. The Skin (Sudoriferous Glands)**
 - **Waste Product: Sweat**, composed of water, dissolved salts (NaCl), urea, and lactic acid.
 - **Mechanism:** Sweat is secreted onto the skin surface for evaporative cooling. While thermoregulation is its primary role, it also serves a minor excretory function.
- **C. The Liver: The Body's Detox Center**
 - The liver is not primarily an organ of expulsion, but a vital processor of wastes.
 - **Key Functions:**
 - **Deamination of Amino Acids:** [Diagram: An amino acid molecule losing its amino group (-NH₂).] Excess amino acids cannot be stored. The liver removes the toxic amino group (-NH₂) in a process called **deamination**. The -NH₂ group is converted to **ammonia (NH₃)** and immediately combined with CO₂ to form **urea**, a less toxic compound. The remaining carbon skeleton is used for energy or stored as fat.
 - **Breakdown of Hemoglobin:** Old red blood cells are broken down in the liver and spleen. The iron is recycled, and the heme group is converted into **bile pigments (bilirubin)**, which are excreted as part of the bile into the faeces, giving it its characteristic brown color.
 - **Detoxification:** It converts other metabolic toxins and drugs (e.g., alcohol) into less harmful forms that can be excreted by the kidneys.
- **D. The Urinary System: The Primary Excretory System** This system is specialized for the removal of nitrogenous wastes (especially urea) and the precise regulation of water and salt balance (osmoregulation).

3. The Human Urinary System: Structure and Function



- **Kidneys (2):** Bean-shaped organs located against the back muscles in the abdominal cavity. They filter blood and form urine.
- **Ureters (2):** Muscular tubes that propel urine from the kidneys to the bladder via peristalsis.
- **Urinary Bladder:** A hollow, muscular, and distensible sac that acts as a temporary storage reservoir for urine.
- **Urethra:** The tube that carries urine from the bladder to the exterior of the body. It is longer in males than in females.

4. The
Unit of
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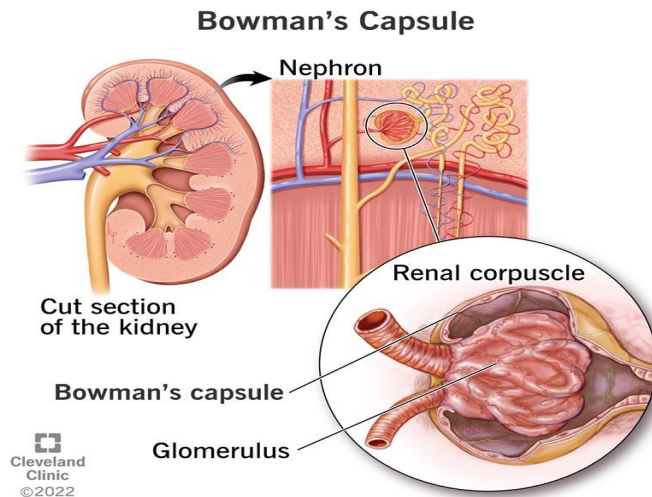
**Nephron: The Functional
the Kidney** Each kidney
about one million
microscopic nephrons, where
magic of filtration and urine
formation happens.

A nephron consists of two main parts:

1. **The Malpighian Body (Renal Corpuscle):** The filter.
 - **Glomerulus:** A dense knot of capillaries where blood is filtered under high pressure.
 - **Bowman's Capsule:** A cup-shaped structure that surrounds the glomerulus and collects the filtrate.
2. **The Renal Tubule:** The modifier.
 - **Proximal Convoluted Tubule (PCT):** Twisted section right after Bowman's capsule.
 - **Loop of Henle:** A long, U-shaped loop that dips into the medulla.
 - **Distal Convoluted Tubule (DCT):** Another twisted section after the loop.
 - **Collecting Duct:** Receives urine from multiple nephrons and passes it into the renal pelvis.

5. The Mechanism of Urine Formation: A Two-Step Process

- **Step 1: Ultrafiltration in the Glomerulus**



- The afferent arteriole bringing blood into the glomerulus is wider than the efferent arteriole taking it out. This creates high hydrostatic pressure.
- This pressure forces water and small molecules (urea, glucose, amino acids, salts, ions) out of the blood and through the filtration membrane (capillary walls + basement membrane + podocytes of Bowman's capsule) into the Bowman's capsule.
- This fluid is called the **glomerular filtrate**. It is similar to blood plasma but lacks cells and large proteins.
- **Step 2: Selective Reabsorption in the Renal Tubule** The body cannot afford to lose all the water and useful substances in the filtrate. Over 99% of the filtrate is reabsorbed back into the blood in the peritubular capillaries.
 - **Proximal Convoluted Tubule (PCT):**
 - **Active Transport:** All glucose, amino acids, vitamins, and most ions (Na^+ , K^+ , Cl^-) are actively reabsorbed.
 - **Osmosis:** The reabsorption of solutes creates an osmotic gradient, so about 80% of the water follows passively.
 - **Loop of Henle:** Its primary role is to create a high salt concentration (hypertonicity) in the medulla, which is crucial for water conservation.
 - **Distal Convoluted Tubule (DCT):** Fine-tuning occurs here. More water and salts (like Na^+) are reabsorbed under hormonal control.

- **Collecting Duct (CD):** The Final Checkpoint for Water.
 - **Hormonal Control (ADH):**
 - **Dehydration:** The pituitary gland releases **Antidiuretic Hormone (ADH)**. ADH makes the walls of the collecting duct more permeable to water. Water is drawn out by the salty medulla, leading to the production of **small volumes of concentrated urine**.
 - **Hydration:** Less ADH is released. The collecting duct is impermeable to water, so it is not reabsorbed, leading to the production of **large volumes of dilute urine**.

6. Common Excretory System Disorders

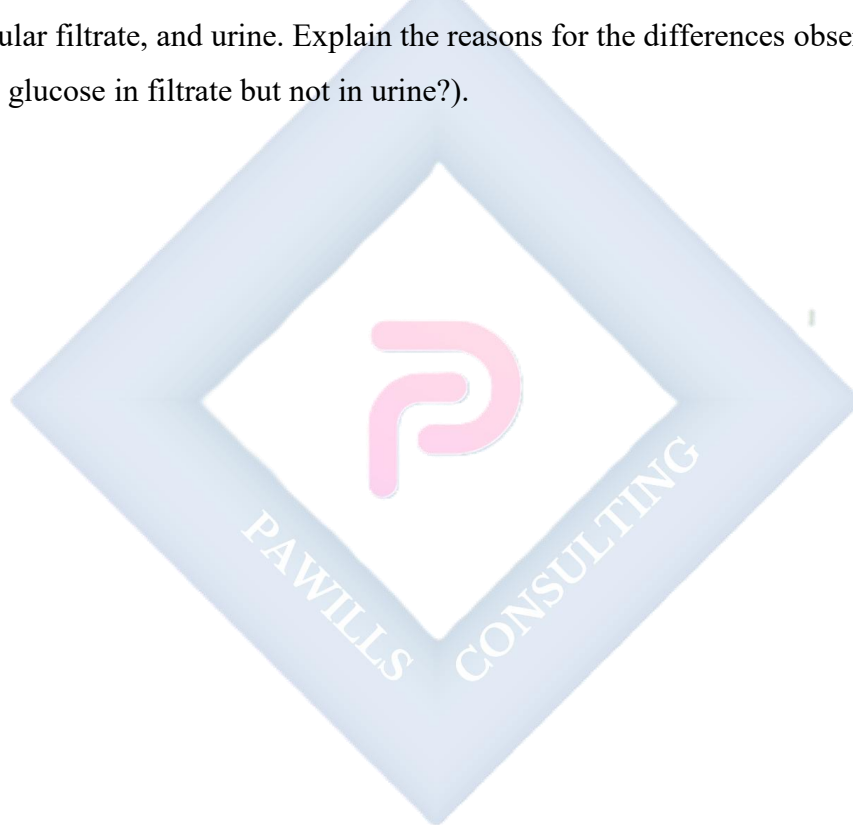
- **Kidney Stones (Renal Calculi):** Crystals of salts (e.g., calcium oxalate) that form in the kidney. They can block the ureter, causing intense pain.
- **Nephritis:** Inflammation of the kidneys, often due to infection or an autoimmune response, which can damage the nephrons.
- **Kidney (Renal) Failure:** A life-threatening condition where the kidneys lose their filtering ability, leading to the accumulation of urea and other wastes in the blood (uremia).
 - **Treatment:**
 - **Dialysis:** A machine acts as an artificial kidney, filtering the patient's blood.
 - **Kidney Transplant:** Surgical replacement of a failed kidney with a healthy one from a donor.

Evaluation:

1. Differentiate clearly between excretion and egestion, providing a specific example of a waste product for each process.
2. Describe the role of the liver in the excretion of nitrogenous waste. Use the term 'deamination' in your answer.
3. Trace the path of a urea molecule from the liver to the exterior of the body, naming all organs and tubes it passes through.
4. Explain how the hormone ADH regulates the water content of the blood. What would be the effect on urine output if a person drank a large volume of water?

Assignment:

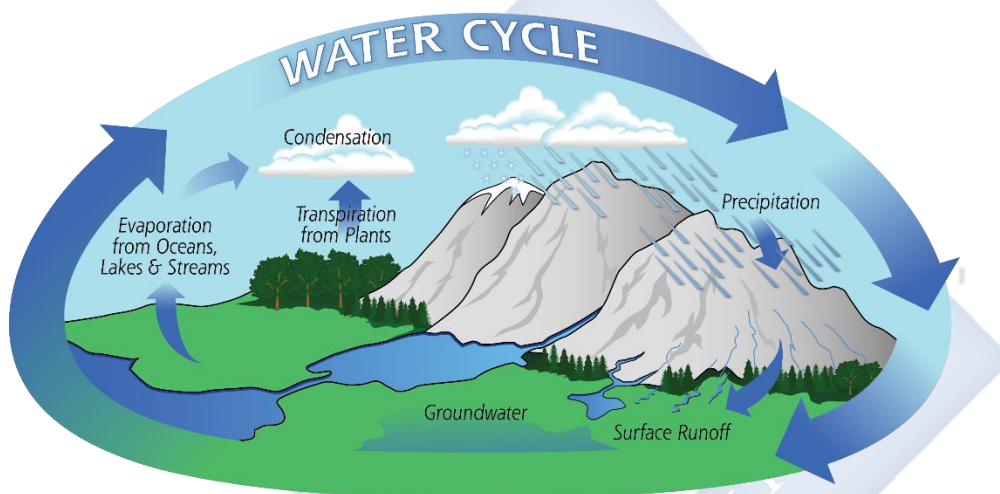
1. **Nephron Model:** Create a detailed, labeled model of a nephron using different colored pipe cleaners, strings, or clay. Use your model to explain the processes of ultrafiltration and selective reabsorption.
2. **Research and Debate:** Investigate the ethical and medical challenges surrounding organ transplantation, specifically for kidneys. Prepare a one-page report discussing the shortage of donors and the issue of organ trafficking.
3. **Data Analysis:** Analyze data on the concentration of various substances in blood plasma, glomerular filtrate, and urine. Explain the reasons for the differences observed (e.g., why is there glucose in filtrate but not in urine?).



WEEK 10: NUTRIENT CYCLING IN NATURE - THE EARTH'S RECYCLING SYSTEM

1. The Principle of Nutrient Cycling The total amount of matter on Earth is fixed. Life depends on the constant recycling of key elements like carbon, nitrogen, and water. These **biogeochemical cycles** describe how elements move from the environment into living organisms and back again, ensuring a continuous supply.

2. The Water Cycle (Hydrological Cycle) This cycle describes the continuous movement of water on, above, and below the surface of the Earth.

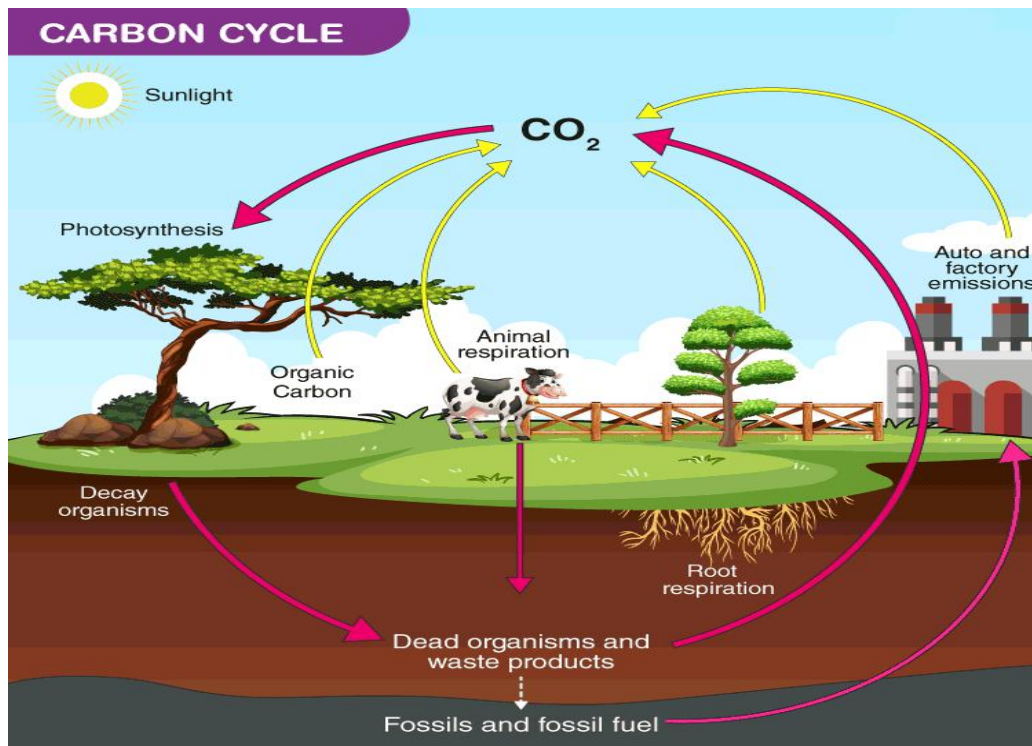


Key Processes:

- **Evaporation:** The change of water from liquid to vapor (gas), primarily from oceans, lakes, and rivers, driven by solar energy.
- **Transpiration:** The loss of water vapor from the aerial parts of plants, mainly through stomata in the leaves. This combined process is often called **evapotranspiration**.
- **Condensation:** As water vapor rises and cools in the atmosphere, it condenses around dust particles to form clouds (liquid water droplets).
- **Precipitation:** Water returns to the Earth's surface as rain, snow, sleet, or hail when the water droplets in clouds become too heavy.
- **Collection & Flow:**
 - **Run-off:** Water flows over the land surface into streams and rivers.

- **Infiltration/Percolation:** Water soaks into the soil, recharging groundwater aquifers.
- **Biological Importance:** Water is the universal solvent for biochemical reactions, a transport medium, and essential for maintaining turgor in plants.

3. The Carbon Cycle: The Cycle of Life and Energy Carbon is the backbone of all organic molecules. The carbon cycle is fundamentally about the flow of energy, as carbon is exchanged between reservoirs.



Key Processes:

- Photosynthesis (The Entry Point into the Biosphere):**
 - **Equation:** $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Light Energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
 - **Role:** Green plants and algae absorb atmospheric **carbon dioxide (CO_2)** and convert it into **organic carbon (glucose)**. This is the only process that reduces atmospheric CO_2 on a large scale.
- Cellular Respiration (The Return Path):**
 - **Equation:** $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$

- **Role:** All living organisms (plants, animals, decomposers) break down organic compounds during respiration, releasing CO_2 back into the atmosphere.

c. **Decomposition:**

- **Role:** Bacteria and fungi (decomposers) break down dead organisms and waste products, releasing CO_2 through respiration and returning nutrients to the soil.

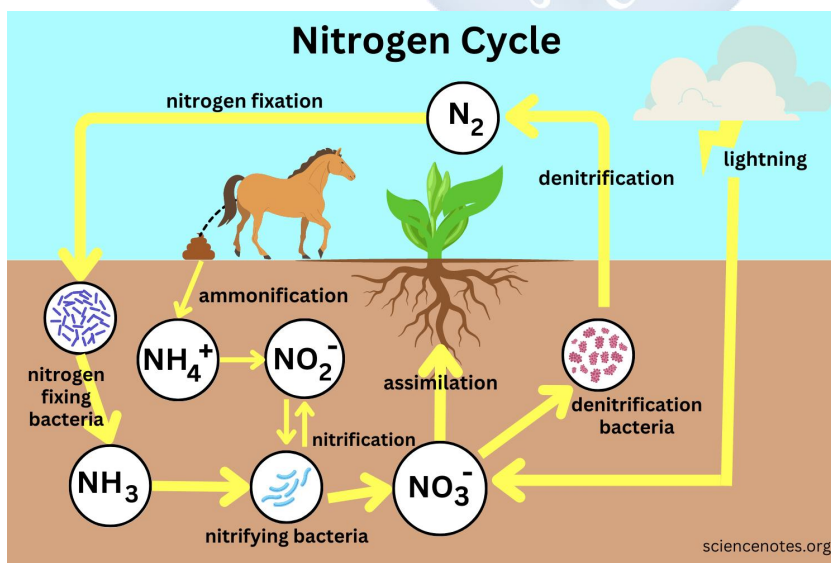
d. **Combustion:**

- **Natural:** Forest fires.
- **Anthropogenic (Human-Induced):** The burning of **fossil fuels** (coal, oil, natural gas), which are ancient, stored carbon from prehistoric biomass. This releases vast amounts of "old" CO_2 into the atmosphere very rapidly.

- **Human Impact and Global Warming:**

- The massive increase in combustion, coupled with deforestation (which reduces photosynthesis), has significantly increased atmospheric CO_2 levels.
- CO_2 is a **greenhouse gas**; it traps heat in the atmosphere, leading to the **Enhanced Greenhouse Effect** and global climate change.

4. The Nitrogen Cycle: From Inert Gas to Life's Building Blocks Although nitrogen gas (N_2) makes up 78% of the atmosphere, it is inert and unusable by most organisms. The nitrogen cycle converts N_2 into biologically useful forms.



Key Processes (Driven by Microbes):

a. Nitrogen Fixation:

- **Definition:** The conversion of atmospheric nitrogen (N_2) into ammonia (NH_3) or ammonium (NH_4^+).
- **Agents:**
 - **Biological:**
 - **Symbiotic Bacteria (e.g., Rhizobium):** Live in root nodules of leguminous plants (e.g., beans, peas). The plant provides sugars; the bacteria provide fixed nitrogen.
 - **Free-Living Bacteria (e.g., Azotobacter, Clostridium):** Live in the soil.
 - **Non-Biological:**
 - **Lightning:** The high energy of lightning fixes nitrogen directly.
 - **Industrial (Haber Process):** Human-made production of nitrogen fertilizers.

b. Ammonification (or Mineralization):

- **Definition:** Decomposers (bacteria and fungi) break down dead organic matter and nitrogenous wastes (urea, uric acid) into ammonia (NH_3) or ammonium (NH_4^+).

c. Nitrification:

- **Definition:** The conversion of ammonia/ammonium into nitrates (NO_3^-), the form most easily absorbed by plant roots.
- **A Two-Step Process:**
 - **Step 1:** *Nitrosomonas* bacteria oxidize $NH_3/NH_4^+ \rightarrow$ **Nitrites (NO_2^-)**.
 - **Step 2:** *Nitrobacter* bacteria oxidize $NO_2^- \rightarrow$ **Nitrates (NO_3^-)**.
- *Note: This process requires oxygen, so it only occurs in well-aerated soils.*

d. Assimilation:

- **Definition:** Plants absorb nitrates (and ammonium) from the soil through their roots and use them to build proteins, nucleic acids, and chlorophyll. Animals obtain nitrogen by eating plants or other animals.

e. **Denitrification:**

- **Definition:** The conversion of nitrates (NO_3^-) back into atmospheric nitrogen gas (N_2).
- **Agents: Denitrifying bacteria (e.g., *Pseudomonas*).** These bacteria are anaerobic and are active in waterlogged, poorly aerated soils.
- **Role:** Returns nitrogen to the atmosphere, completing the cycle.

Evaluation:

1. Explain why nutrient cycling is essential for the sustainability of an ecosystem.
2. Describe the role of transpiration in the water cycle.
3. Identify two major human activities that are disrupting the carbon cycle and explain the environmental consequence of this disruption.
4. Create a table to summarize the five main processes of the nitrogen cycle, naming the key bacteria involved in each where applicable.

Assignment:

1. **Cycle Poster:** Create a large, integrated poster that shows the water, carbon, and nitrogen cycles interacting in a single ecosystem. Use color coding and clear labels for all processes and reservoirs.
2. **Local Ecosystem Analysis:** Investigate a local environmental issue (e.g., use of fertilizers in farming, deforestation, a polluted water body). Write a report explaining how this issue disrupts one or more of the nutrient cycles you have studied.
3. **Debate Preparation:** "The invention of the Haber-Bosch process for fixing nitrogen is the most important discovery for human survival." Research the pros (enabling global food production) and cons (environmental pollution, eutrophication) and prepare arguments for a class debate.

WEEK 11: REVISION OF FIRST TERM'S WORK

This week is dedicated to a strategic and thorough review of all topics covered this term. The goal is to move knowledge from short-term to long-term memory by actively connecting concepts, identifying areas for improvement, and practicing application.

A. Thematic Integration: Connecting the Dots Biology is not a series of isolated facts but an interconnected web of processes. Use this framework to see the "big picture":

(Illustration: A central concept map with "The Organism" in the middle, linked to all major systems studied: Digestion -> Circulation -> Respiration -> Excretion, with Nutrient Cycles as the external environment.)

- **The Story of a Glucose Molecule:** Trace its path from a piece of bread.
 - a. **Digestion:** Broken down by amylases (mouth/duodenum) into glucose.
 - b. **Absorption:** Glucose is actively transported into the villi of the ileum.
 - c. **Transport:** Travels via hepatic portal vein to liver, then is distributed by systemic circulation to a muscle cell.
 - d. **Respiration:** In the muscle cell's mitochondria, it is used in aerobic respiration to produce ATP.
 - e. **Waste Product:** CO₂ from respiration is transported back to the lungs and excreted.
- **The Story of a Protein Molecule:** Trace its path from a piece of fish.
 - a. **Digestion:** Broken down by pepsin (stomach) and trypsin/peptidases (duodenum/ileum) into amino acids.
 - b. **Absorption & Transport:** Amino acids are absorbed and travel to the liver.
 - c. **Assimilation:** Used to build muscle tissue.
 - d. **Waste Product:** Excess amino acids are **deaminated** in the liver, producing **urea**.
 - e. **Excretion:** Urea is filtered by the kidneys and excreted in urine.

B. Guided Topic-by-Topic Revision

1. The Digestive System (Weeks 1 & 2)

- **Core Concepts:** Nutrition (Ingestion, Digestion, Absorption, Assimilation, Egestion), Enzyme specificity, Adaptations for absorption.
- **Key Diagrams You MUST Be Able to Draw & Label:**
 - The human alimentary canal and associated glands.
 - A single villus, showing the lacteal and blood capillaries.
- **Active Recall Tasks:**
 - **Create an Enzyme Table:** For Amylase, Pepsin, Trypsin, Lipase, and Maltase, state the source, substrate, products, and optimal pH.
 - **Explain Adaptations:** Write a paragraph explaining how the structure of the ileum is adapted for its function.
 - **Compare and Contrast:** Differentiate between the roles of bile and lipase in fat digestion.

2. Transport Systems (Weeks 3 & 4)

- **Core Concepts:** SA:V ratio, Diffusion, Osmosis, Active Transport, Double Circulation, Heart structure, Blood components, Blood clotting.
- **Key Diagrams You MUST Be Able to Draw & Label:**
 - The human heart (cross-section, showing blood flow).
 - Comparative structure of an artery, vein, and capillary.
 - Red blood cell in hypertonic, hypotonic, and isotonic solutions.
- **Active Recall Tasks:**
 - **SA:V Ratio:** Explain why an elephant needs a circulatory system but a flatworm does not.
 - **Process Comparison:** Create a table comparing Diffusion, Osmosis, and Active Transport.
 - **Trace the Pathway:** Trace a drop of blood from the right ventricle to the aorta, naming all chambers and vessels.
 - **Function Match:** Link each component of blood (Plasma, RBC, Phagocyte, Lymphocyte, Platelet) to its primary function.

3. Respiratory System (Weeks 6 & 8)

- **Core Concepts:** Gaseous Exchange vs. Cellular Respiration, Mechanism of Breathing, Adaptations of Alveoli, Aerobic vs. Anaerobic Respiration, Oxygen Debt.
- **Key Diagrams You MUST Be Able to Draw & Label:**
 - The human respiratory system (path of air).
 - An alveolus surrounded by a capillary.
 - The mechanism of inhalation and exhalation (ribs/diaphragm).
- **Active Recall Tasks:**
 - **Distinguish:** Write clear definitions for *Breathing*, *Gaseous Exchange*, and *Cellular Respiration*.
 - **Alveolar Adaptations:** List four adaptations of the alveoli and explain how each one facilitates efficient gas exchange.
 - **Equation Proficiency:** Write the balanced chemical equations for both aerobic respiration and lactic acid fermentation.
 - **Explain Oxygen Debt:** Describe what causes it and how it is "repaid."

4. Excretory System (Week 9)

- **Core Concepts:** Excretion vs. Egestion, Structure of the Urinary System, The Nephron, Ultrafiltration, Selective Reabsorption, Osmoregulation (ADH).
- **Key Diagrams You MUST Be Able to Draw & Label:**
 - The human urinary system.
 - A nephron and its associated blood supply.
- **Active Recall Tasks:**
 - **Waste Management:** List three excretory organs and the specific waste product each removes.
 - **Nephron Function:** Describe, in detail, what happens during ultrafiltration and what substances are present in the glomerular filtrate.
 - **Explain ADH:** Explain how the presence and absence of ADH affects the volume and concentration of urine.

5. Nutrient Cycling (Week 10)

- **Core Concepts:** Importance of recycling, Processes in the Water, Carbon, and Nitrogen cycles, Role of Bacteria, Human Impact.
- **Key Diagrams You MUST Be Able to Draw & Label:**
 - A simple diagram of the Carbon Cycle.
 - A simple diagram of the Nitrogen Cycle, highlighting the role of specific bacteria.
- **Active Recall Tasks:**
 - **Process Identification:** Define: Nitrogen Fixation, Nitrification, Denitrification, and Combustion.
 - **Bacterial Roles:** Name the bacteria responsible for: (a) nitrogen fixation in roots, (b) converting ammonia to nitrites, (c) converting nitrates to nitrogen gas.
 - **Human Impact:** Explain how the burning of fossil fuels disrupts the carbon cycle.

C. Effective Revision Strategies

1. **Active Recall:** Don't just re-read notes. Close your book and try to write out a process or draw a diagram from memory. Then check for accuracy.
2. **Spaced Repetition:** Review topics multiple times over several days. This is more effective than one long "cramming" session.
3. **Teach Someone:** Explain a complex topic like the Nitrogen Cycle or the Cardiac Cycle to a friend or family member. Teaching forces you to clarify your understanding.
4. **Past Questions:** Actively solve past exam questions under timed conditions. This familiarizes you with the format and application of knowledge.

Evaluation (Self-Assessment Checkpoint)

Answer these questions under exam conditions to test your integrated understanding.

1. **Integrated Question:** Describe how the circulatory system and the respiratory system work together to supply oxygen to a working muscle cell and remove the waste products of respiration.

2. **Comparison Question:** Create a table to compare the composition and function of arteries and veins. Include three differences.
3. **Diagram-Based Question:** Draw a labeled diagram of a nephron. On your diagram, indicate with an 'F' where ultrafiltration occurs and with an 'R' where the majority of selective reabsorption takes place.
4. **Application Question:** A farmer applies excessive nitrate fertilizer to his field. Following a heavy rain, the fertilizer washes into a nearby lake. Using your knowledge of nutrient cycling, explain the sequence of events that would lead to the death of fish in the lake (eutrophication).
5. **Problem-Solving Question:** A student places a sample of pure distilled water into a beaker and adds a single celery stalk. She leaves it for 24 hours. Predict and explain what will happen to the firmness (turgidity) of the celery stalk.

Assignment:

1. **Create a "Master Summary Poster":** Design a single, large poster for ONE of the following systems: The Digestive System, The Circulatory System, or The Excretory System. Your poster must include all major structures, a clear description of the key processes, and how the system is linked to at least one other system in the body.
2. **Write a "Concept Explanation":** Choose one of the following concepts and write a one-page explanation as if you were teaching it to a JSS2 student: *The Importance of Enzymes*, *How the Kidney Regulates Water Content*, or *Why We Breathe Heavily After a Sprint*. Use simple language and analogies.
3. **Self-Assessment Checklist:** Create a table with two columns: "Topic" and "Confidence Level (1-5)". List all the major topics from the term. Rate your understanding of each from 1 (Very Poor) to 5 (Excellent). For every topic you rate 3 or below, write one specific action you will take to improve (e.g., "Redraw the heart diagram 3 times," "Make flashcards for digestive enzymes").
4. **Past Paper Drill:** Complete a full past examination paper for SS2 Biology (First Term). Time yourself strictly. After marking it, write a reflection analyzing your three most common errors and how you will avoid them in the future.

5. **Glossary Creation:** Create a personal glossary of key biological terms from this term (minimum 30 terms). For each term, write a precise definition and use it in a sentence that demonstrates its meaning.

Final Note: Consistent, active revision is the key to mastery. Use this week to transform your understanding from a collection of facts into a cohesive and powerful knowledge framework. Identify your weaknesses now and turn them into strengths. Good luck



SS2 BIOLOGY LESSON NOTES

SECOND TERM

WEEK 1: ECOLOGICAL MANAGEMENT

1. Meaning of Ecology Ecology is the scientific study of the interactions between organisms and their environment. The term comes from the Greek words *oikos* (meaning "house" or "dwelling") and *logos* (meaning "study"). Essentially, it is the study of the "house" where living things reside.

Explanation: Think of an ecosystem as a complex neighbourhood. Each organism has a "job" (its niche) and interacts with its neighbours (biotic factors) and the physical environment like the weather and landscape (abiotic factors). Ecology studies these relationships at different levels of organization, from a single individual to the entire planet.

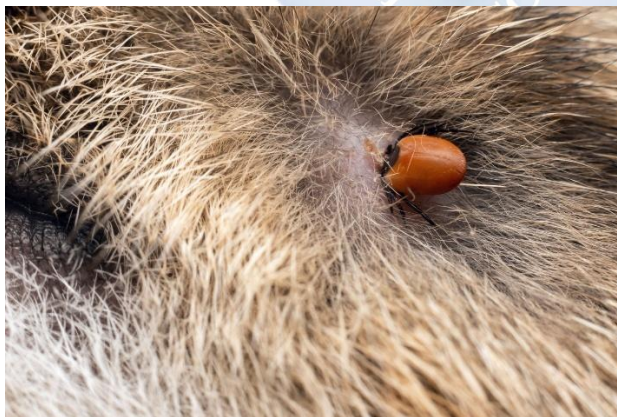
Levels of Ecological Organization: (Illustration: A set of concentric circles or a pyramid, with the smallest level at the bottom/top and the largest at the top/bottom)

- **Individual:** A single organism. *Example: One lion.*
- **Population:** A group of individuals of the *same species* living in the same area. *Example: A pride of lions in Serengeti National Park.*
- **Community:** All the *different populations* (different species) living and interacting in the same area. *Example: The lions, zebras, grasses, trees, and bacteria in the Serengeti.*
- **Ecosystem:** The community of living organisms (biotic factors) interacting with their non-living (abiotic) environment. *Example: The Serengeti ecosystem, including the animals, plants, soil, rainfall, and sunlight.*
- **Biome:** A large geographical area with a distinct climate, and specific types of plants and animals. *Example: The Tropical Savannah biome, which includes the Serengeti.*
- **Biosphere:** The part of the Earth where life exists, encompassing all biomes. It extends from the deepest ocean trenches to the highest mountains.

2. Concept of Association (Symbiosis) In a community, no organism exists in isolation. Species form close, long-term interactions known as **associations** or **symbiosis**.

Types of Associations:

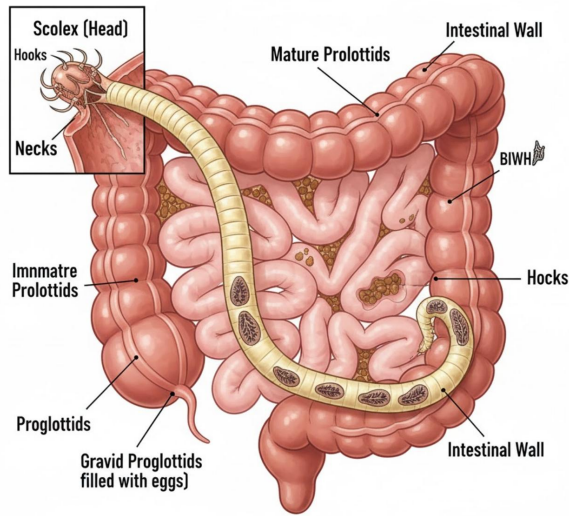
- **Mutualism (+/+):** Both species benefit.
 - **Example & Illustration:**
 - **Bee and Flower:** [Image: A bee on a flower] The bee gets nectar (food), and the flower gets pollinated (its pollen is transferred to other flowers).
 - **Rhizobium Bacteria and Legume Root Nodules:** [Diagram: A legume plant root with enlarged view showing nodules] The bacteria convert atmospheric nitrogen into a form the plant can use, and the plant provides the bacteria with sugars and a home.
- **Commensalism (+/0):** One species benefits, and the other is unaffected.
 - **Example & Illustration:**
 - **Orchid (Epiphyte) on a Tree:** [Image: An orchid growing on a tree branch] The orchid gets access to sunlight by growing on the tree, but the tree is not harmed or helped.
 - **Remora and Shark:** The remora fish attaches to the shark, getting free transport and leftover food scraps, while the shark is unaffected.
- **Parasitism (+/-):** One species (the parasite) benefits at the expense of the other (the host), which is harmed.
 - **Types:**
 - **Ectoparasites:** Live *on* the host. *Example: Ticks, lice, fleas.*



- **Endoparasites:** Live *inside* the host. *Example: Tapeworm in the human gut, Plasmodium (malaria parasite) in the blood.*

TAPEWORM IN HUMAN INTESTINE

Illustrative Diagram: Parasitic Infection



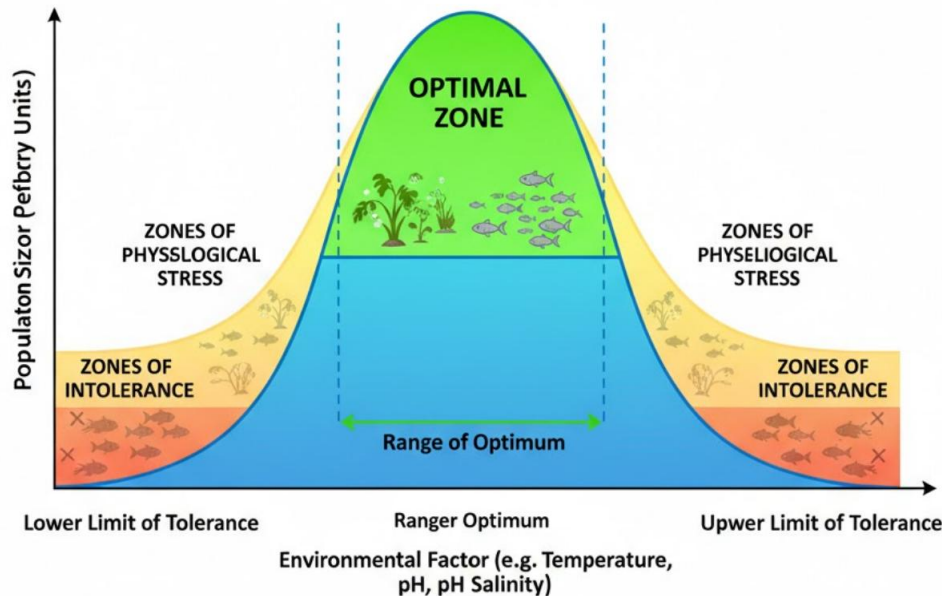
- **Predation (+/-):** One organism (predator) hunts, kills, and eats another (prey).
 - **Example:** A lion (predator) hunting a zebra (prey). This is a key interaction for controlling population sizes.
- **Competition (-/-):** Two or more species require the same limited resource (e.g., food, water, space), and both are negatively affected.
 - **Example:** Lions and hyenas competing for the same prey.

3. Concept of Tolerance (Shelford's Law of Tolerance) Tolerance is the ability of an organism to survive and reproduce within a specific range of environmental conditions.

Shelford's Law of Tolerance states that for each environmental factor (e.g., temperature, pH), an organism has a range within which it can survive. Outside this range, it cannot live.

SHELFORD'S LAW OF TOLERANCE

Illustrative Diagram



- **Zone of Intolerance:** At the far left and right. The organism cannot survive here.
- **Zone of Physiological Stress:** The areas where the factor is too high or too low. The organism is weakened, grows poorly, and cannot reproduce.
- **Optimum Range:** The central peak. The environmental factor is at its ideal level, and the population is at its maximum size and health.

Explanation: A fish species, for example, has a specific tolerance range for water temperature. If the water becomes too cold or too hot, it will move into stress zones and eventually die if it cannot escape.

4. Concept of Adaptation An adaptation is an inherited trait that improves an organism's chance of survival and reproduction in its environment.

- **Structural Adaptations (Physical Form):**
 - **Cactus (Xerophyte):** Spines (reduce water loss), thick fleshy stem (stores water), deep roots (access water).

- **Fish:** Streamlined body (reduces drag), gills (extract oxygen from water), fins (propulsion and steering).
- **Physiological Adaptations (Internal Processes):**
 - **CAM Photosynthesis in Cacti:** Opens stomata at night to take in CO₂, minimizing water loss during the hot day.
 - **Hibernation in Bears:** Slows metabolism to conserve energy during winter when food is scarce.
- **Behavioral Adaptations (Actions):**
 - **Migration in Wildebeest:** Moving to areas with better food and water.
 - **Nocturnal Activity in Desert Animals:** Avoiding the extreme heat of the day.

Evaluation:

1. Define ecology and distinguish between biotic and abiotic factors with two examples each.
2. With specific examples, explain the following associations: Mutualism, Commensalism, and Parasitism.
3. Using a well-labeled diagram, explain Shelford's Law of Tolerance.
4. Describe two structural adaptations of a camel for desert life and one physiological adaptation of a saltwater fish.

Assignment:

1. Create a detailed poster showing the six levels of ecological organization, using a specific local ecosystem as your example.
2. Observe and document three different symbiotic relationships in your school compound or home garden. Classify them and explain your reasoning.
3. Research and write a one-page report on a specific animal, focusing on its structural, physiological, and behavioral adaptations.

WEEK 2: POLLUTION

1. Meaning of Pollution

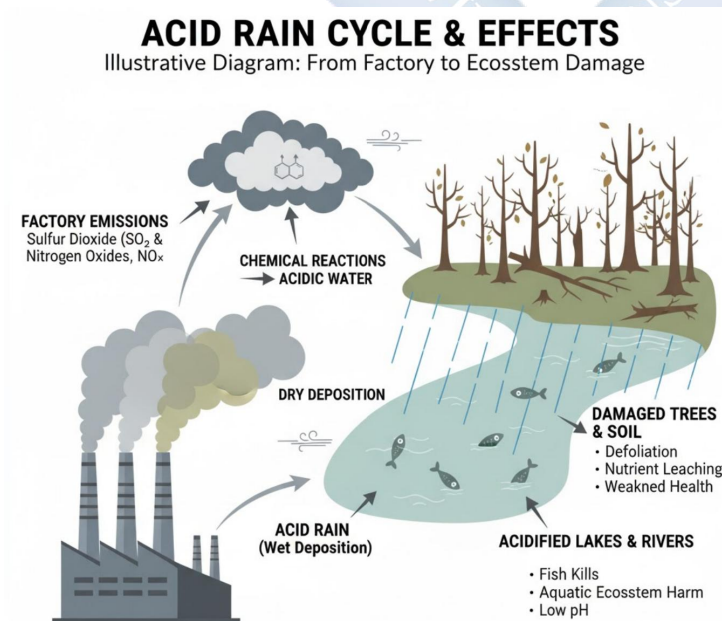
Pollution is the introduction of harmful substances or contaminants into the natural environment, causing instability, harm, or discomfort to the ecosystem. These harmful substances are called **pollutants**.

2. Types, Causes, Effects, and Control of Pollution

(Illustration: Use icons for each type of pollution for visual recall)

A. AIR POLLUTION

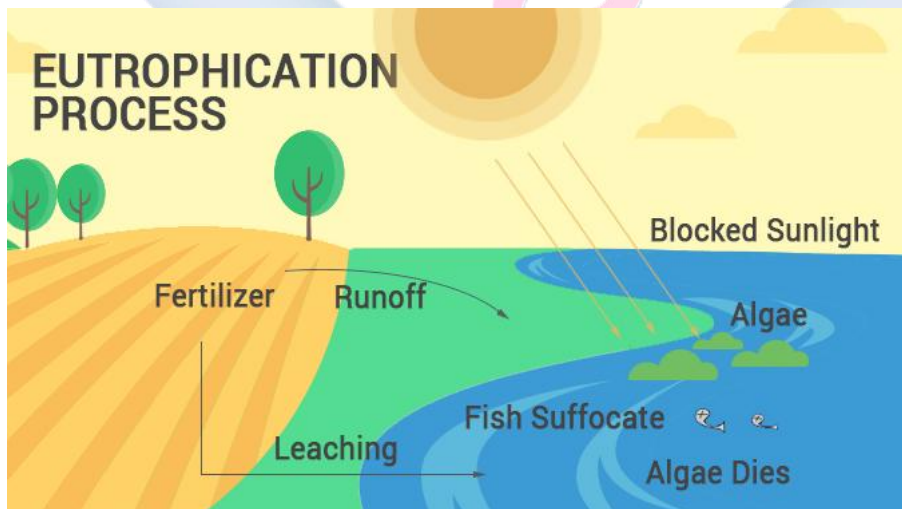
- **Causes (Sources):**
 - Industrial emissions (SO_2 , NO_x).
 - Burning fossil fuels in vehicles and power plants.
 - Bush burning and deforestation.
- **Effects:**
 - **Respiratory Diseases:** Asthma, bronchitis, lung cancer.
 - **Acid Rain:** SO_2 and NO_x mix with atmospheric water to form acids, damaging buildings, forests, and aquatic life.



- **Global Warming & Climate Change:** Increased CO₂ and other greenhouse gases trap heat in the atmosphere.
- **Control:**
 - Use of cleaner energy sources (solar, wind).
 - Installing catalytic converters in vehicles.
 - Government regulations on industrial emissions.
 - Afforestation (planting trees).

B. WATER POLLUTION

- **Causes (Sources):** [Icon: Pipe discharging waste] [Icon: Oil spill]
 - Discharge of untreated sewage and industrial waste.
 - Agricultural runoff (fertilizers, pesticides).
 - Oil spills from tankers and pipelines.
- **Effects:**
 - **Eutrophication:**



Excess nutrients cause algal blooms. Decomposing algae use up oxygen, suffocating fish.

- **Water-borne Diseases:** Cholera, typhoid.
- **Death of Aquatic Life.**
- **Control:**

- Building and maintaining sewage treatment plants.
- Proper treatment of industrial effluent before discharge.
- Laws against dumping waste in water bodies.
- Public awareness on proper waste disposal.

C. SOIL/LAND POLLUTION

- **Causes (Sources):**
 - Improper disposal of domestic and industrial solid waste.
 - Excessive use of agrochemicals (pesticides, fertilizers).
 - Mining activities.
- **Effects:**
 - Reduced soil fertility and crop yield.
 - Contamination of groundwater through leaching.
 - Harm to soil organisms essential for fertility.
- **Control:**
 - **The 3 R's: Reduce, Reuse, Recycle.**
 - Use of biodegradable products.
 - Bioremediation (using microorganisms to clean up pollutants).
 - Organic farming.

D. NOISE POLLUTION

- **Causes (Sources):**
 - Loud sounds from traffic, industries, generators, and loudspeakers.
- **Effects:**
 - Hearing impairment.
 - Stress, high blood pressure, and sleep disturbance.
 - Disruption of animal communication.
- **Control:**
 - Planting trees as sound barriers.

- Use of soundproofing materials in buildings.
- Regulations on noise levels in residential areas.

Evaluation:

1. Define pollution and list three major pollutants of air.
2. Describe the process of eutrophication, linking each step to its consequence.
3. Explain how the "3 R's" can help in controlling land pollution.
4. What are the physiological effects of noise pollution on humans?

Assignment:

1. Conduct a mini-survey of your immediate environment. Identify and list five sources of pollution and classify them.
2. Design an awareness flyer on one type of pollution, highlighting its causes, effects, and control measures.
3. Write a letter to your local government chairman, suggesting three practical ways to reduce pollution in your community.

WEEK 3: CONSERVATION OF NATURAL RESOURCES

1. Meaning of Conservation

Conservation is the wise, sustainable management and use of natural resources to prevent their exploitation, destruction, or extinction, ensuring their availability for future generations.

- **Conservation vs. Preservation:**

- **Conservation:** Sustainable *use* of resources. *Example: Selective logging, where only mature trees are cut.*
- **Preservation:** Protecting nature from *any* use. *Example: A strict nature reserve where no human activity is allowed.*

2. Importance and Methods of Conservation

A. SOIL CONSERVATION

- **Importance:** Soil is the foundation for agriculture; it filters water and supports biodiversity.
- **Methods:**
 - **Contour Ploughing:** Ploughing along the contours of a slope to reduce water runoff and soil erosion.
 - **Crop Rotation:** Planting different crops in sequence on the same land to maintain soil fertility.
 - **Cover Cropping:** Planting crops like legumes to cover the soil, protecting it from erosion.

B. WATER CONSERVATION

- **Importance:** Water is essential for all life processes, agriculture, and industry.
- **Methods:**
 - **Drip Irrigation:** Delivering water directly to plant roots, minimizing evaporation and waste.
 - **Rainwater Harvesting:** Collecting and storing rainwater for later use.
 - **Building Dams and Reservoirs:** Storing water during the rainy season for use during dry periods.

C. FOREST CONSERVATION

- **Importance:** Forests produce oxygen, absorb CO₂, regulate climate, protect watersheds, and house immense biodiversity.
- **Methods:**
 - **Afforestation/Reforestation:** Planting trees in new areas or replanting deforested areas.
 - **Selective Felling:** Cutting only mature trees instead of clear-cutting entire forests.
 - **Establishing Protected Areas:** Creating forest reserves and national parks.
 - **Using Alternative Energy:** Promoting gas and solar energy to reduce reliance on firewood.

D. WILDLIFE CONSERVATION

- **Importance:** Maintains biodiversity, ecological balance, and has aesthetic and economic (tourism) value.
- **Methods:**
 - **Game Reserves and National Parks:** Providing safe habitats for animals.
 - **Anti-Poaching Laws:** Enforcing laws against illegal hunting and trade of wildlife.
 - **Captive Breeding:** Breeding endangered species in protected environments for later reintroduction into the wild.
 - **Public Education:** Creating awareness on the importance of wildlife.

Evaluation:

1. Differentiate with examples between conservation and preservation.
2. Explain how contour ploughing helps in soil conservation.
3. Why is afforestation considered a critical method for both soil and forest conservation?
4. List three reasons for conserving wildlife.

Assignment:

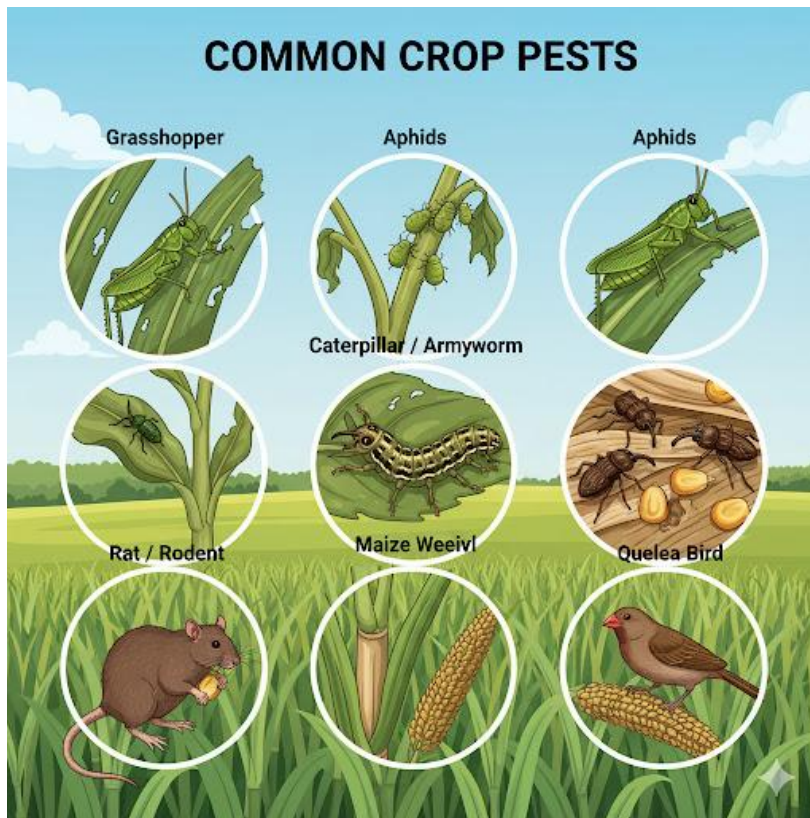
1. Propose a detailed conservation plan for a forest area near your community that is facing deforestation.

2. Create a comparative table showing four natural resources, the threats they face, and a conservation method for each.
3. Research and write a brief profile on one successful wildlife conservation project in Nigeria (e.g., the Cross River Gorilla project).



WEEK 4: PESTS AND DISEASES OF CROPS

1. Types of Crop Pests A pest is any organism that causes economic damage or nuisance to crops, livestock, or humans.



- **Insects (Most Common Pests):**

- **Biting and Chewing Insects:**

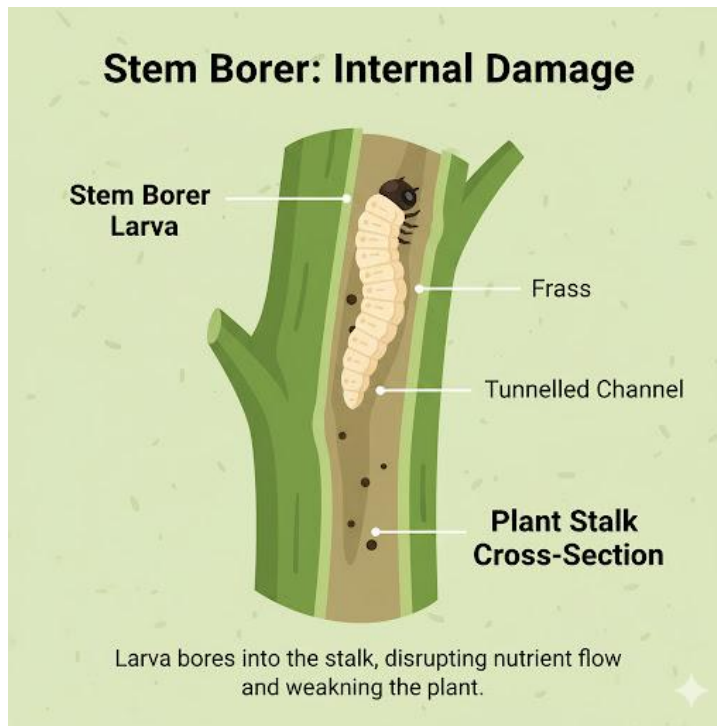
- **Mode of Action:** Have strong mandibles to bite off and chew plant tissues like leaves, stems, and fruits.
 - **Examples:** Grasshoppers, locusts, armyworms, beetles, caterpillars.
 - **Damage:** Notched leaves, defoliation, bored holes in fruits and stems.

- **Piercing and Sucking Insects:**

- **Mode of Action:** Have needle-like mouthparts (stylets) that pierce plant epidermis and suck sap from phloem or xylem.
 - **Examples:** Aphids, mealybugs, whiteflies, leafhoppers, scale insects.

- **Damage:** Yellowing of leaves (chlorosis), wilting, stunted growth, and honeydew secretion which promotes sooty mold. They are also primary **vectors** for viral diseases.

- **Boring Insects:**



- **Mode of Action:** Larvae or adults bore into stems, fruits, or seeds, feeding on the internal tissues.
- **Examples:** Stem borers (e.g., in maize), bean beetles, maize weevils.
- **Damage:** Hidden internal damage; plants may wilt and die; stored grains are destroyed from the inside.
- **Other Invertebrate Pests:**
 - **Nematodes (e.g., Root-Knot Nematodes):**
 - Microscopic worms that infect roots, causing galls (swellings) that disrupt water and nutrient uptake, leading to stunting and wilting.
- **Vertebrate Pests:**
 - **Birds (e.g., Quelea birds, weaver birds):** Feed on ripening grains and fruits.

- **Rodents (e.g., Rats, squirrels):** Gnaw on stems, eat stored grains, and can burrow, damaging root systems.

2. Types of Crop Diseases A plant disease is any deviation from the normal physiological and morphological development of a plant, caused by a pathogenic agent or environmental factor.



- **Fungal Diseases (Most Common):**

- **Spread via:** Spores (airborne, water-splashed, soil-borne).
- **Examples:**
 - **Black Pod of Cocoa:** Causes black lesions on pods, destroying the beans.
 - **Maize Smut:** Causes large, black, powdery galls on any part of the plant.
 - **Rusts:** Cause red, orange, or brown pustules on leaves (e.g., maize rust).
 - **Mildew (Powdery & Downy):** White, powdery, or fuzzy growth on leaf surfaces.

- **Bacterial Diseases:**

- **Spread via:** Water splash, infected tools, insects.
- **Examples:**

- **Bacterial Blight of Cassava:** Causes angular leaf spots, wilting, and dieback.
- **Citrus Canker:** Causes raised, corky lesions on fruits, leaves, and stems.
- **Viral Diseases:**
 - **Spread via:** Vectors (mainly piercing-sucking insects like aphids and whiteflies), infected plant material.
 - **Examples:**
 - **Cassava Mosaic Disease (CMD):** Causes a characteristic yellow mosaic pattern on leaves, leading to severe stunting and yield loss.
 - **Maize Streak Virus:** Causes yellow streaks running parallel to the leaf veins.

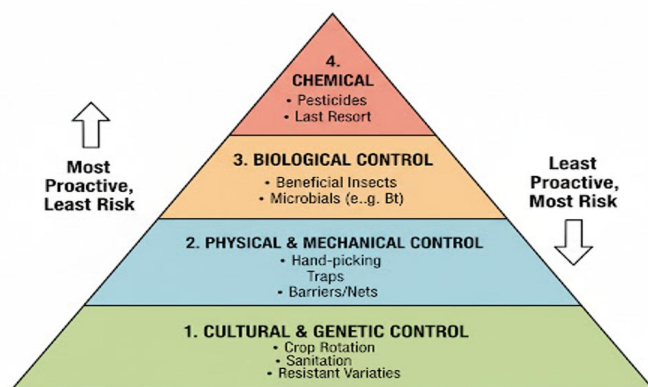
3. Effects of Pests and Diseases on Crop Production

- **Reduced Yield and Quality:** Direct damage reduces the amount and market value of the harvest.
- **Increased Production Costs:** Money spent on pesticides, fungicides, and control labor.
- **Post-Harvest Losses:** Pests like weevils can destroy stored grains, leading to food insecurity.
- **Total Crop Failure:** Severe infestations (e.g., locust swarms) or epidemics can wipe out entire farms.
- **Contamination:** Fungal diseases can produce mycotoxins, which are harmful to humans and animals.

4. Preventive and Control Measures:

INTEGRATED PEST MANAGEMENT (IPM)

Illustrative Diagram: Prioritizing Control Tactics



- **Cultural Control:** Modifying the environment and farming practices.
 - **Crop Rotation:** Prevents the buildup of pests/diseases specific to one crop.
 - **Field Sanitation:** Removing and burning crop residues to destroy overwintering pests and pathogens.
 - **Use of Clean Seeds/Planting Materials:** Prevents introducing diseases.
 - **Proper Spacing:** Ensures good air circulation, reducing humidity that favors fungal growth.
- **Biological Control:** Using natural enemies to control pests.
 - **Example 1:** Introducing **Ladybirds** to prey on **aphids**.
 - **Example 2:** Spraying **Bacillus thuringiensis (Bt)**, a bacterium that produces toxins lethal to specific caterpillars but harmless to other organisms.
 - **Example 3:** Encouraging birds that feed on insects by providing perches.
- **Chemical Control:** Using pesticides.
 - **Types:** Insecticides (insects), Fungicides (fungi), Nematicides (nematodes), Rodenticides (rodents).
 - **Caution:** Can lead to environmental pollution, harm non-target organisms (like bees), and cause pest resistance. **Always follow safety instructions.**
- **Physical/Mechanical Control:**
 - **Hand-picking:** Removing larger pests like caterpillars by hand.
 - **Traps:** Using light traps, pheromone traps, or sticky traps.
 - **Barriers:** Using nets to protect fruits from birds and insects.
- **Legislative/Quarantine Control:** Government laws to prevent the import and spread of foreign pests and diseases.
- **Plant Breeding (Genetic Control):** Developing and planting crop varieties that are genetically resistant to specific pests or diseases (e.g., CMD-resistant cassava varieties).

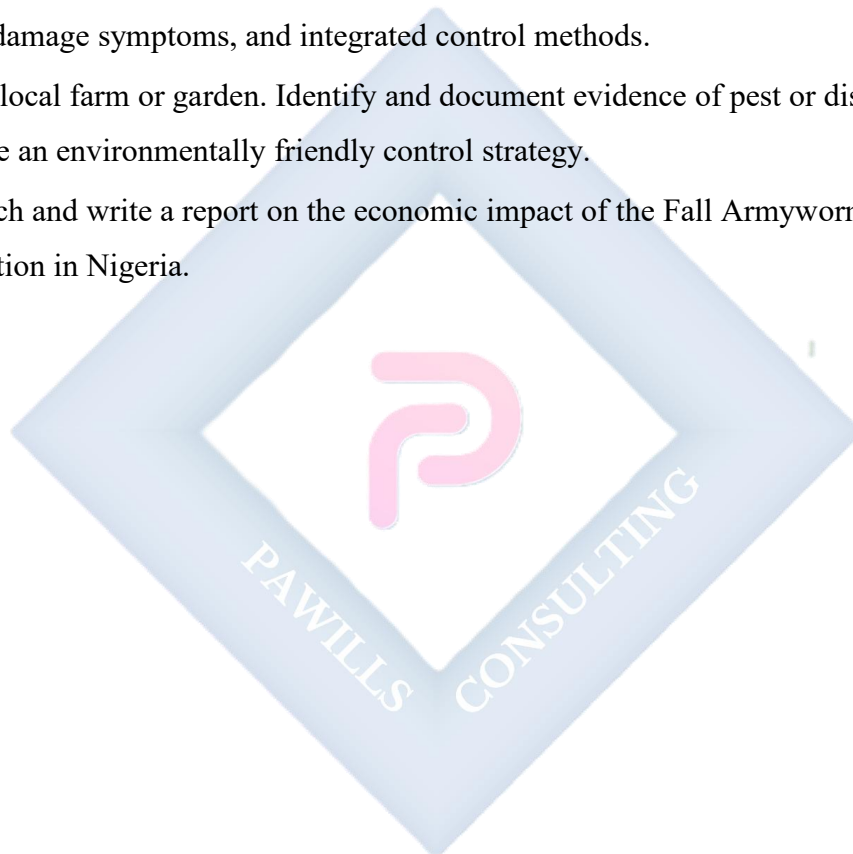
Evaluation:

1. Differentiate between a biting/chewing pest and a piercing/sucking pest, giving one example of each.

2. Explain why piercing/sucking insects are particularly dangerous as vectors.
3. Name the causal agent (fungus, bacterium, or virus) of: (a) Maize Smut, (b) Citrus Canker, (c) Cassava Mosaic Disease.
4. Compare and contrast biological control and chemical control, listing one advantage and one disadvantage of each.

Assignment:

1. Create a detailed poster on a major crop pest or disease in your locality, detailing its life cycle, damage symptoms, and integrated control methods.
2. Visit a local farm or garden. Identify and document evidence of pest or disease damage. Propose an environmentally friendly control strategy.
3. Research and write a report on the economic impact of the Fall Armyworm on maize production in Nigeria.



WEEK 6: REPRODUCTIVE SYSTEMS IN VERTEBRATES

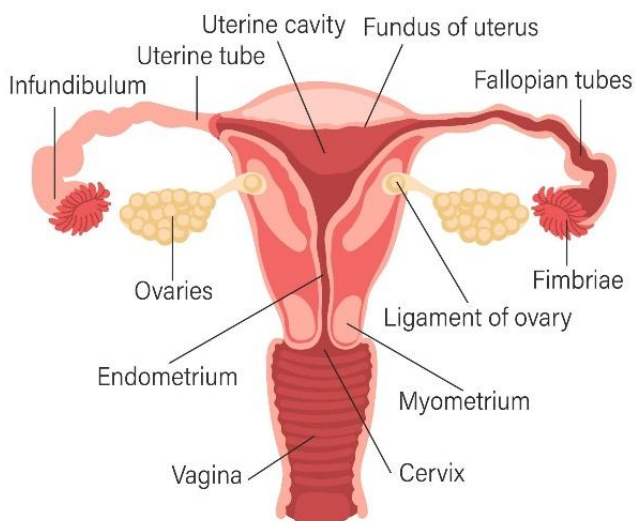
1. Introduction: Sexual Reproduction

Reproduction involves the creation of new individuals. In vertebrates, this is primarily sexual, requiring the fusion of male and female gametes.

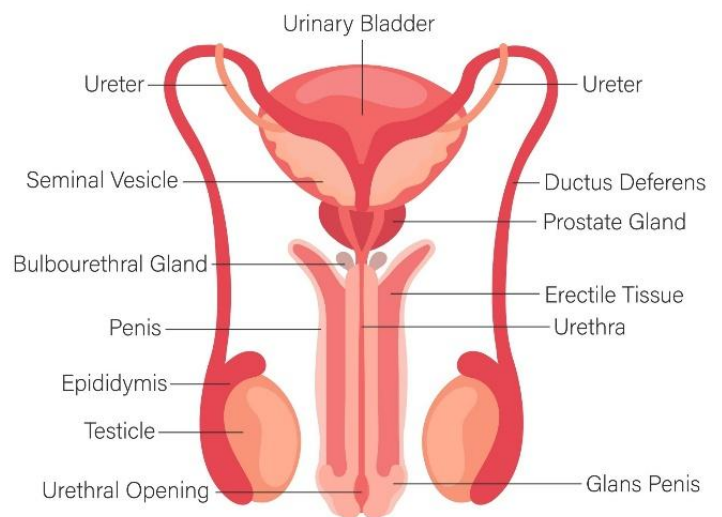
- **Gonads:** Primary sex organs (Testes in males, Ovaries in females) that produce gametes.
- **Fertilization:** The fusion of a sperm cell (male gamete) and an egg cell (female gamete) to form a zygote.
 - **External Fertilization:** Occurs outside the female's body, typically in water. (e.g., Fish, Amphibians).
 - **Internal Fertilization:** Occurs inside the female's reproductive tract. (e.g., Reptiles, Birds, Mammals).

2. Basic Anatomy (Using Mammals as a Model)

HUMAN REPRODUCTIVE SYSTEM



FEMALE REPRODUCTIVE SYSTEM



MALE REPRODUCTIVE SYSTEM

Male System:

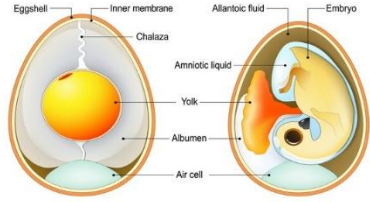
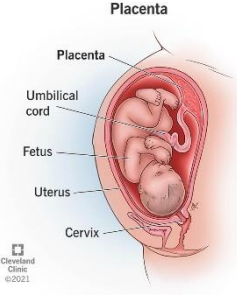
- **Testes:** Produce sperm and testosterone. Located in the scrotum for temperature regulation.
- **Epididymis:** Coiled tube where sperm mature and are stored.
- **Vas Deferens:** Tube that carries sperm from the epididymis to the urethra.
- **Urethra:** Dual-function tube that carries both sperm (during ejaculation) and urine.
- **Penis:** Organ for depositing sperm into the female vagina.

• Female System:

- **Ovaries:** Produce eggs (ova) and hormones (estrogen, progesterone).
- **Oviduct (Fallopian Tube):** Site of fertilization; transports the egg to the uterus.
- **Uterus (Womb):** Muscular organ where the embryo/fetus develops.
- **Cervix:** The neck of the uterus; it dilates during childbirth.
- **Vagina:** Birth canal and the receptacle for sperm during intercourse.

3. Comparative Reproduction in Vertebrates (Illustration: A comparative table with icons for each class)

Class	Fertilization	Development	Key Adaptations & Parental Care
Fish	External	External (Oviparous - egg-laying)	• Lays vast numbers of small, jelly-coated eggs. • No parental care in most species. High mortality.
Amphibians	External	External (Oviparous)	• Eggs laid in water or moist environments. • Metamorphosis: Larval stage (tadpole) is aquatic, adult is terrestrial. • Little to no parental care.

Class	Fertilization	Development	Key Adaptations & Parental Care
Reptiles	Internal	External (Oviparous)	Bird embryo  • Amniotic Egg: A revolutionary adaptation! The egg has a leathery shell and extra-embryonic membranes (amnion, chorion, allantois) that protect the embryo and allow gas exchange, freeing reproduction from water. • No larval stage.
Birds	Internal	External (Oviparous)	• Hard-Shelled Amniotic Egg. • Extensive Parental Care: Eggs must be incubated (kept warm). Parents feed and protect the altricial (helpless) or precocial (mobile) young.
Mammals	Internal	Internal (Viviparous - live-bearing)	 • Placenta: An organ that allows the exchange of nutrients, gases, and wastes between the mother and the developing fetus. • Lactation: Mammary glands produce milk to nourish the young after birth. • Extended parental care.

Evolutionary Trends:

1. **Fertilization:** Shift from **External** → **Internal** (an adaptation for terrestrial life).
2. **Egg Type:** Shift from **Soft Jelly Egg** → **Amniotic Egg** (providing a "private pond" for the embryo on land).
3. **Development:** Shift from **External** → **Internal** (Viviparity in most mammals, offering maximum protection).
4. **Parental Care:** A general increase in complexity and duration, ensuring higher offspring survival rates.

Evaluation:

1. Using specific examples, distinguish between external and internal fertilization.
2. Why is the evolution of the amniotic egg considered a major breakthrough for reptiles?
3. Compare and contrast reproduction in birds and mammals, focusing on development and parental care.
4. Draw a labeled diagram of the female mammalian reproductive system and state the function of the oviduct and uterus.

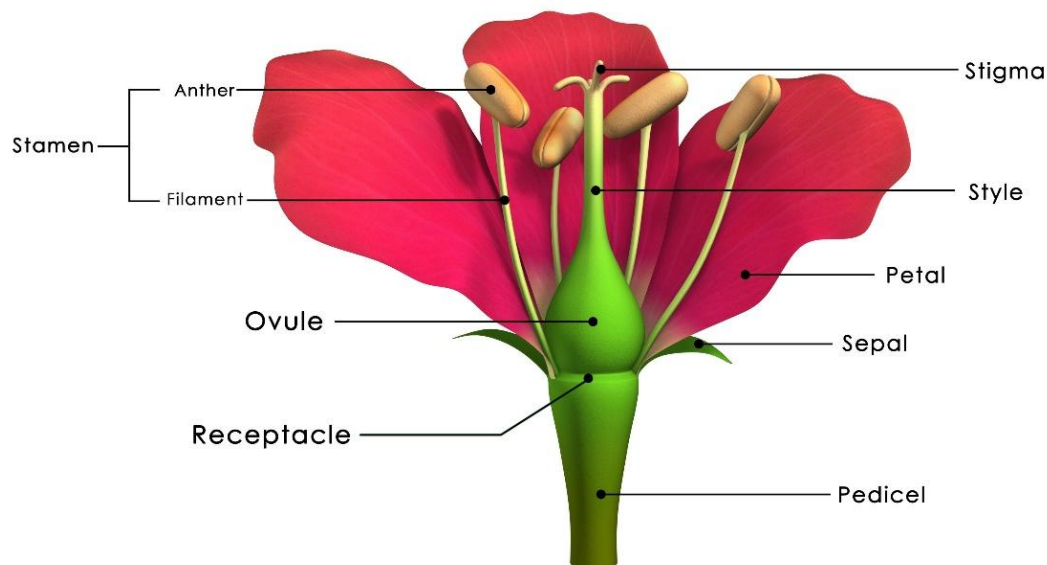
Assignment:

1. Create a detailed evolutionary timeline poster showing the progression of reproductive adaptations from fish to mammals.
2. Research and write a report on an unusual reproductive strategy in a vertebrate (e.g., ovoviviparity in some snakes, male pregnancy in seahorses).
3. Dissect a fertilized chicken egg (under supervision) to identify the blastodisc (embryo), yolk, albumen, and chalazae. Relate these parts to the diagram of the amniotic egg.

WEEK 8: REPRODUCTIVE SYSTEMS IN PLANTS (FLOWERING PLANTS)

1. The Flower: A Modified Shoot The flower is the reproductive structure of angiosperms (flowering plants). It is essentially a highly modified shoot, with leaves specialized for reproduction.

2. Floral Parts and Their Functions



- **Non-Reproductive Parts:**

- **Pedicel:** The stalk of the flower.
- **Receptacle:** The enlarged end of the pedicel where floral parts are attached.
- **Calyx (Sepals):** The outermost whorl. Usually green and leaf-like.
Function: To protect the flower in the bud stage.
- **Corolla (Petals):** The whorl inside the sepals. Often brightly colored and scented.
Function: To attract pollinators (insects, birds, etc.).

- **Reproductive Parts (Male):**

- **Androecium (Stamens):** The male part of the flower.
 - **Filament:** A slender stalk that holds up the anther.

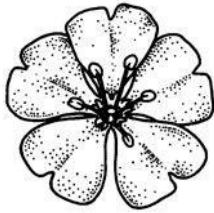
- **Anther:** [Microscopic Image: Pollen grains] A bilobed structure that produces **pollen grains**, which contain the male gametes.
- **Reproductive Parts (Female):**
 - **Gynoecium (Carpel/Pistil):** The female part of the flower.
 - **Stigma:** The sticky, receptive tip for receiving pollen grains.
 - **Style:** A long, slender tube that connects the stigma to the ovary.
 - **Ovary:** The swollen base containing **ovules**. Each ovule contains an embryo sac with the female gamete (egg cell). **The ovary develops into the FRUIT.**
 - **Ovule:** The ovule develops into the **SEED** after fertilization.

3. Types of Flowers

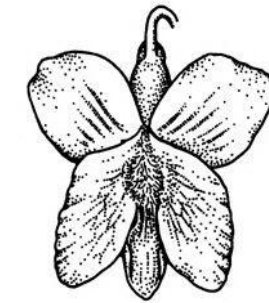
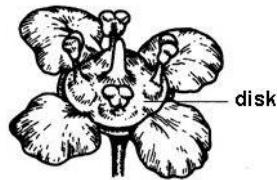


PLATE 9. FLORAL MORPHOLOGY

SYMMETRY



ACTINOMORPHIC / RADIAL SYMMETRY

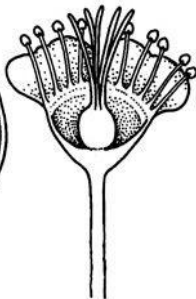


ZYGOMORPHIC / BILATERAL SYMMETRY

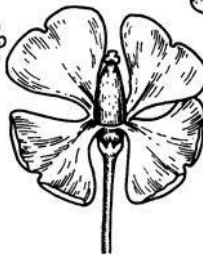
OVARY POSITION



SUPERIOR
HYPOGYNOUS



SUPERIOR
PERIGYNOUS



1/2 INFERIOR

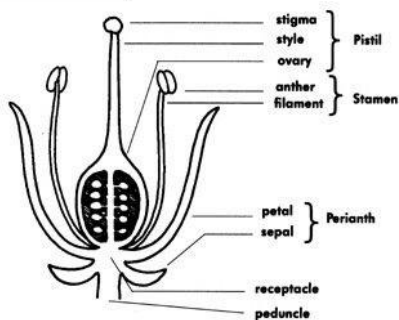


INFERIOR

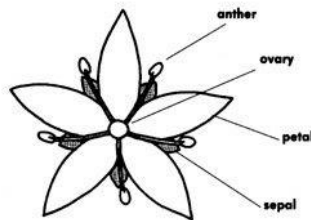


INFERIOR
EPIGYNOUS

COMPONENTS



FLOWER LONGITUDINAL SECTION



FLOWER TOP VIEW

modified from Swink, F. and G. Wilhelm. 1994. *Plants of the Chicago region*. 4th ed. Indianapolis: Indiana Academy of Science.

- **Based on Reproductive Parts:**

- **Bisexual (Perfect) Flower:** Has both stamens and carpels. *Example: Hibiscus, tomato.*
- **Unisexual (Imperfect) Flower:** Has either stamens or carpels, but not both.
 - **Staminate (Male) Flower:** Has stamens but no functional carpel.
 - **Pistillate (Female) Flower:** Has a carpel but no functional stamens.

- **Based on Plant Type (for Unisexual Flowers):**

- **Monoecious Plants:** ("One house") - Both male and female flowers are found on the same individual plant. *Example: Maize, coconut.*
- **Dioecious Plants:** ("Two houses") - An individual plant has either only male flowers or only female flowers. *Example: Pawpaw, date palm.*

4. Comparison of Monocot and Dicot Flowers:

Feature	Monocotyledonous Flower (e.g., Lily, Grass)	Dicotyledonous Flower (e.g., Hibiscus, Rose)
Number of Floral Parts	In multiples of 3 (Trimerous). E.g., 3 or 6 petals/stamens.	In multiples of 4 or 5 (Tetramerous/Pentamerous). E.g., 4 or 5 petals/stamens.
Sepals and Petals	Often similar in appearance and color, collectively called Tepals .	Usually distinct; sepals are green and petal-like, petals are colored and conspicuous.
Venation in Floral Parts	Parallel venation.	Reticulate (net-like) venation.

Evaluation:

1. Draw and label a longitudinal section of a bisexual flower, stating the function of the anther, stigma, and ovary.
2. What is the difference between a monoecious plant and a dioecious plant? Give one example of each.
3. A flower has 6 petals and 6 stamens. Is it a monocot or a dicot? Give a reason for your answer.
4. Explain the following statement: "The ovary becomes the fruit, and the ovule becomes the seed."

Assignment:

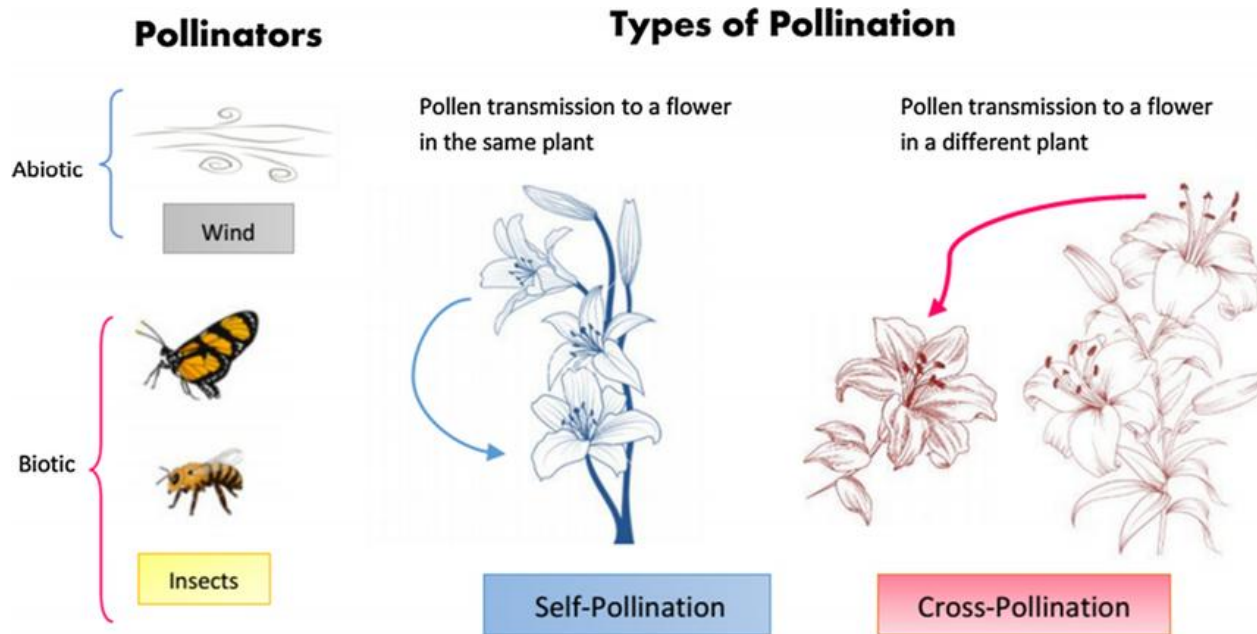
1. **Floral Dissection Practical:** Dissect a locally available flower (Hibiscus, Glorybower, etc.). Mount and label the parts on a sheet of paper. Classify the flower as complete/incomplete, bisexual/unisexual, and monocot/dicot.
2. Compare the inflorescence and flower structure of maize (a monocot) with that of a bean plant (a dicot).
3. Research and create a presentation on the economic importance of flowers (e.g., in agriculture, horticulture, perfumery).



WEEK 9: POLLINATION IN PLANTS

1. What is Pollination? Pollination is the transfer of pollen grains from the anther of a flower to the stigma of the same or another flower. It is a prerequisite for fertilization but is **not** fertilization itself.

2. Types of Pollination:



- **Self-Pollination (Autogamy):** Transfer of pollen from anther to stigma of the **same flower or another flower on the same plant**.
 - **Genetic Consequence:** No new gene combinations; offspring are genetically identical to the parent (clones).
 - **Advantages:** Guarantees seed production, preserves well-adapted genotypes.
 - **Disadvantages:** No genetic variation, leading to **inbreeding depression** (accumulation of harmful recessive traits).
- **Cross-Pollination (Allogamy):** Transfer of pollen from anther to stigma of a flower on a **different plant of the same species**.
 - **Genetic Consequence:** Leads to genetic variation and **hybrid vigor** (offspring are often stronger and more adaptable).
 - **Disadvantages:** Relies on pollinators, less certain, wasteful (much pollen is lost).

3. Mechanisms to Ensure/Promote Cross-Pollination Plants have evolved sophisticated mechanisms to prevent self-pollination and encourage cross-pollination:

- **Dichogamy:** The anthers and stigmas mature at different times.
 - **Protandry:** Anthers mature first (e.g., Sunflower).
 - **Protogyny:** Stigma matures first (e.g., Avocado pear).
- **Self-Incompatibility:** A genetic mechanism where pollen from the same plant fails to germinate on the stigma or the pollen tube is blocked in the style.
- **Structural Barriers (Herkogamy):** The anthers and stigma are positioned in such a way that self-pollination is physically unlikely (e.g., Stigma is much higher than the anthers).

4. Agents of Pollination:

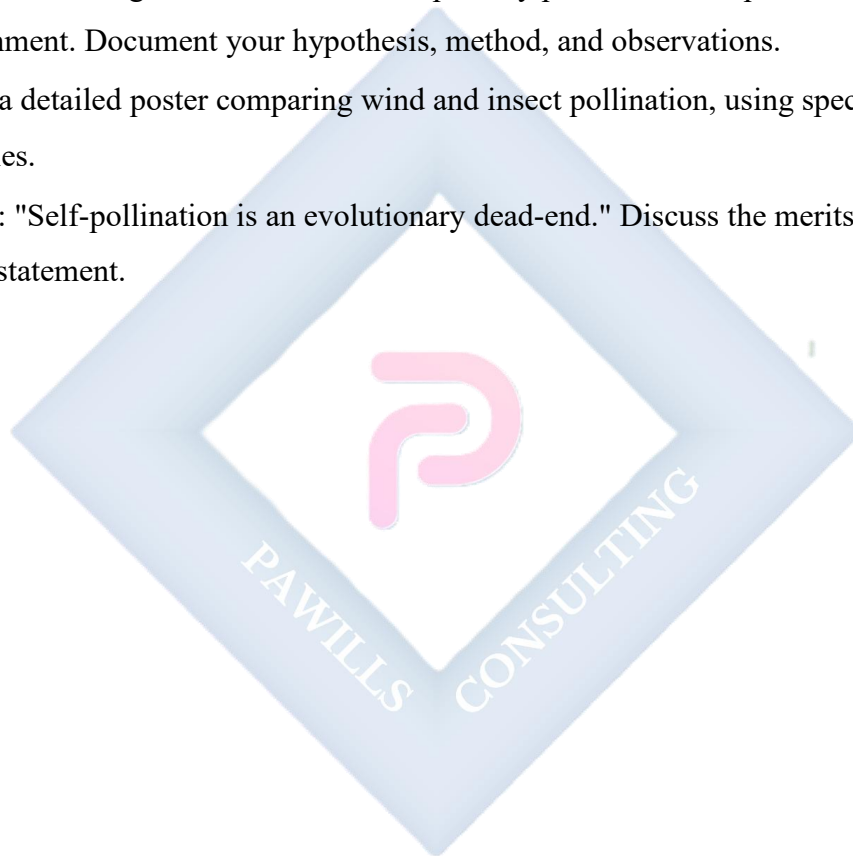
Agent	Flower Adaptations	Pollen Adaptations	Examples
Wind (Anemophily)	• Small, inconspicuous, no scent/nectar. • Feathery, large stigmas to catch pollen. • Stamens hang outside flower.	• Very light, smooth, and dry. • Produced in enormous quantities.	Grasses, Maize, Rice
Insects (Entomophily)	• Large, brightly colored petals. • Scented and produce nectar. • Have guide marks to direct insects.	• Sticky or spiny to cling to insect bodies. • Produced in smaller quantities.	Hibiscus, Sunflower, Orchids
Birds (Ornithophily)	• Bright red or orange color. • Tubular shape to fit beak. • Lots of nectar, but no scent (birds have poor smell).	• Stickier than wind-pollinated pollen.	<i>Hibiscus</i> , <i>Bombax</i> (Silk Cotton tree)
Bats (Chiropterophily)	• Open at night, large and robust. • Dull color (white, green). • Strong, fruity odor.		<i>Baobab</i> , <i>Agave</i>

Evaluation:

1. Define pollination and distinguish it from fertilization.
2. Explain two mechanisms that prevent self-pollination in flowering plants.
3. Describe three characteristics of an insect-pollinated flower and explain how each characteristic aids in pollination.
4. State two advantages and two disadvantages of cross-pollination.

Assignment:

1. Design an investigation to determine the primary pollinator of a specific flower in your environment. Document your hypothesis, method, and observations.
2. Create a detailed poster comparing wind and insect pollination, using specific local examples.
3. Debate: "Self-pollination is an evolutionary dead-end." Discuss the merits and limitations of this statement.



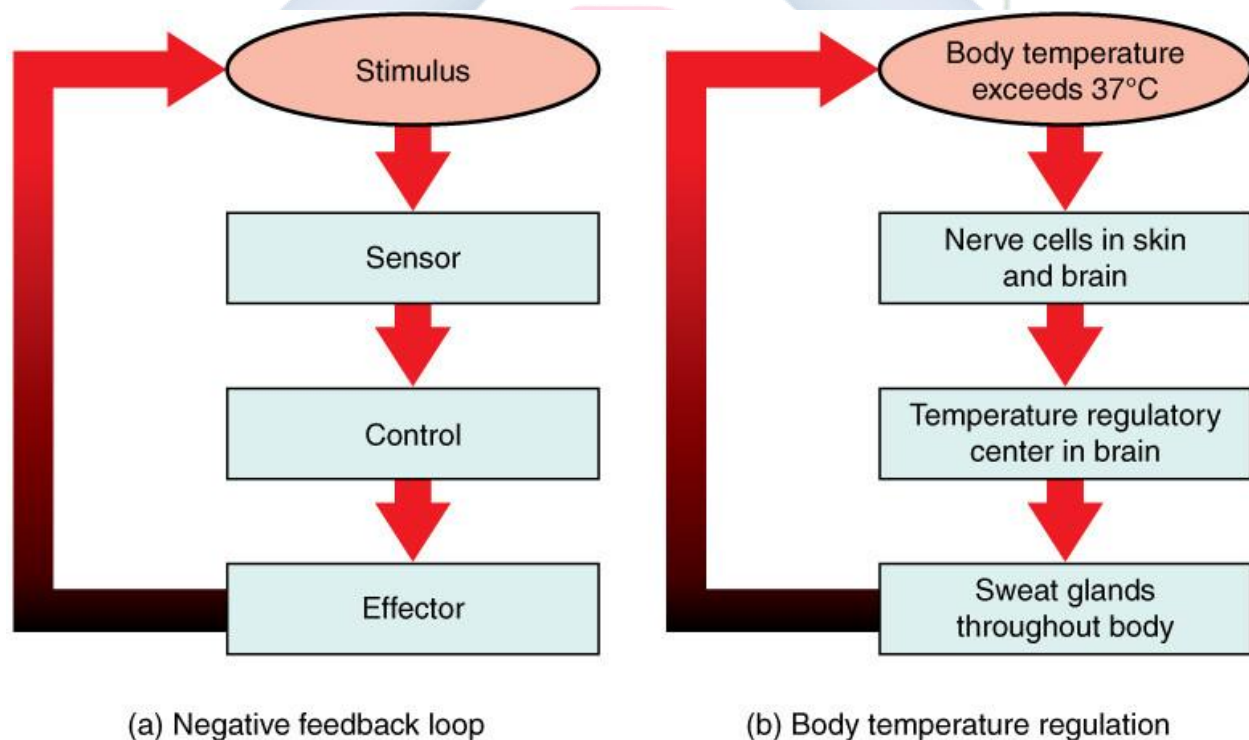
WEEK 10: REGULATION OF INTERNAL ENVIRONMENT (HOMEOSTASIS)

1. What is Homeostasis? Homeostasis is the maintenance of a relatively constant internal environment within the body, despite changes in the external environment. It is a dynamic process of equilibrium crucial for the optimal functioning of enzymes and cells.

- **Examples of Conditions Regulated:**

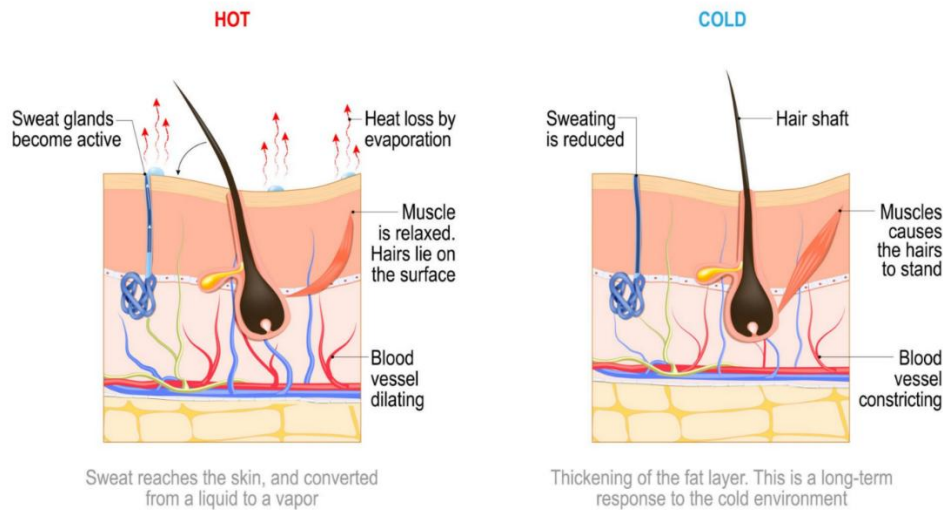
- **Body Temperature**
- **Blood Glucose Level**
- **Water and Salt (Ion) Concentration**
- **Blood pH**
- **Blood Pressure**

2. The Mechanism: Negative Feedback Loops Homeostasis is achieved primarily through **negative feedback loops**, where the body reverses a change to restore the normal set point.



3. Regulation of Body Temperature (Thermoregulation) Control Center: Hypothalamus in the brain.

Thermoregulation



- **Responses to HIGH Temperature (Cooling Mechanisms):**

- **Vasodilation:** Blood vessels in the skin dilate, allowing more blood to flow near the surface. Heat is lost by **radiation**.
- **Sweating:** Sweat glands secrete sweat. As sweat evaporates, it uses heat from the skin, causing **evaporative cooling**.
- **Hairs Lie Flat:** Erector pili muscles relax, reducing the insulating layer of air.

- **Responses to LOW Temperature (Warming Mechanisms):**

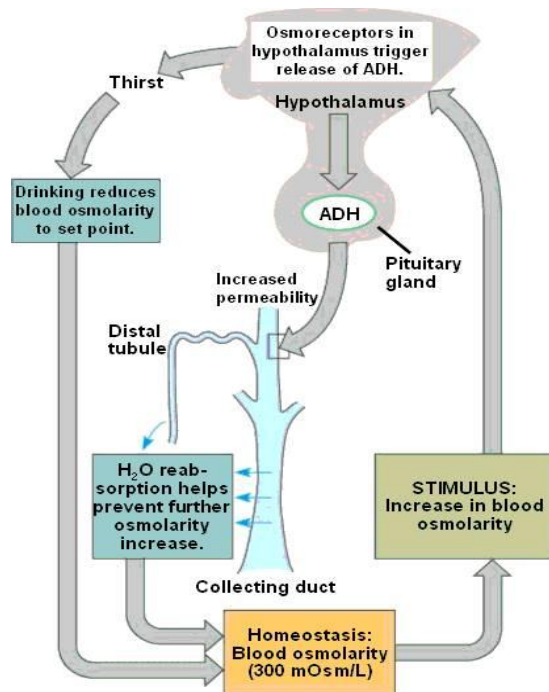
- **Vasoconstriction:** Blood vessels in the skin constrict, reducing blood flow to the surface and conserving heat in the body core.
- **Shivering:** Rapid, involuntary muscle contractions generate metabolic heat.
- **Hairs Stand Erect (Piloerection):** Erector pili muscles contract, trapping a layer of insulating air (causes "goosebumps").
- **Increased Metabolic Rate:** Hormones like thyroxine are released to increase the body's basal metabolic rate.

4. Regulation of Water and Salt Balance (Osmoregulation) Primary Organ: The Kidneys.

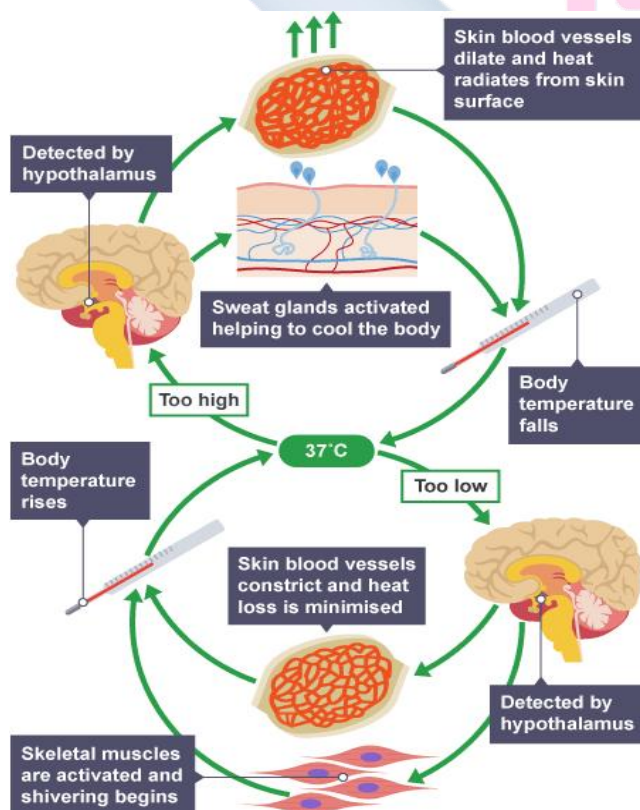
Key Hormone: ADH (Antidiuretic Hormone or Vasopressin), released by the pituitary gland.

ADH makes the walls of the kidney's collecting ducts more permeable to water.

- When **DEHYDRATED** (High Blood Solute Concentration):



- When **OVER-HYDRATED** (Low Blood Solute Concentration):



Explanation: This is a perfect example of a negative feedback loop. The hormone ADH regulates the *water content* of the blood by controlling how much water is reabsorbed in the kidneys.

Evaluation:

1. Define homeostasis and explain why it is important for living organisms.
2. Using the regulation of body temperature as an example, describe the sequence of events in a negative feedback loop.
3. What is the role of the skin in thermoregulation when the body is too hot?
4. Explain how the hormone ADH helps the body to conserve water.

Assignment:

1. Draw a large, labeled diagram of a nephron. Trace the path of filtrate and explain where water reabsorption occurs and how it is controlled by ADH.
2. Design an experiment to demonstrate the effect of a specific variable (e.g., exercise, drinking water) on urine production. Outline your method and predicted results.
3. Research and write a report on a homeostatic disorder (e.g., diabetes mellitus, relating to blood glucose control).

WEEK 11: REVISION OF SECOND TERM'S WORK

This week is dedicated to a comprehensive review and consolidation of all topics covered during the term. The goal is to strengthen your understanding, identify areas requiring further study, and integrate concepts across different topics.

A. Guided Revision of Key Topics

1. Ecological Management

- **Core Concepts:** Ecology, Biotic/Abiotic Factors, Levels of Organization (Individual -> Biosphere).
- **Associations (Symbiosis):**
 - **Master Table:** Create a table to compare Mutualism, Commensalism, Parasitism, Predation, and Competition. For each, define the interaction, the effect on each species (+, -, 0), and give a specific, detailed example.
 - **Illustration:** [Draw a simple web of interactions in a savanna, linking predators, prey, grazers, and plants]
- **Tolerance & Adaptation:**
 - Redraw and explain **Shelford's Law of Tolerance** graph from memory. Be able to label the zones of intolerance, stress, and optimum range.
 - **Adaptation Challenge:** Pick one organism from a harsh environment (e.g., camel, cactus, fish). List two structural, one physiological, and one behavioral adaptation, explaining how each aids survival.

2. Pollution

- **The Four Main Types:** Air, Water, Soil, Noise.
- **Revision Strategy:** For EACH type, create a summary card with:
 - **3 Major Causes**
 - **3 Specific Effects** (e.g., for water pollution, don't just say "kills fish," explain **eutrophication** in steps).
 - **3 Practical Control Measures**

- **Key Diagram:** [Redraw the Eutrophication flowchart]

3. Conservation of Natural Resources

- **Concept:** Remember the difference between **Conservation** (wise use) and **Preservation** (no use).
- **Resource Review:** For Soil, Water, Forests, and Wildlife:
 - State **two reasons** why it is important to conserve this resource.
 - Describe **two effective methods** of conservation.
- **Application:** Link conservation back to ecology and pollution. For example, deforestation (a conservation failure) leads to soil erosion (a pollution/management issue) and loss of biodiversity (an ecology concept).

4. Pests and Diseases of Crops

- **Classification is Key:**
 - **Pests:** Categorize them as Biting/Chewing, Piercing/Sucking, or Boring. Know an example and the specific damage for each.
 - **Diseases:** Classify by causative agent: **Fungal** (e.g., Smut), **Bacterial** (e.g., Blight), **Viral** (e.g., Mosaic). Know the symptoms.
- **Control Measures Matrix:** Create a table comparing Cultural, Biological, Chemical, and Physical control methods. For each, list one advantage and one disadvantage.

5. Reproductive Systems in Vertebrates

- **The Great Transition:** The key is to understand the **evolutionary trends**.
 - **Fertilization:** External (Fish/Amphibians) -> Internal (Reptiles/Birds/Mammals).
 - **Egg:** Soft Jelly -> **Amniotic Egg** (The major breakthrough for life on land).
 - **Development:** External -> Internal (Viviparity in mammals).
 - **Parental Care:** None -> Extensive care (birds and mammals).
- **Revision Task:** Draw a large, comparative table for Fish, Frog, Lizard, Bird, and Mammal. Columns should include: Fertilization Type, Development Type, Key Adaptations, and Parental Care.

6. Reproductive Systems in Plants (Flowering Plants)

- **Flower Power:**

- **Mandatory Diagram:** [From memory, draw and label a longitudinal section of a bisexual flower. Check your work against the textbook.] You must know the function of the **Anther, Filament, Stigma, Style, Ovary, and Ovule**.
- **Terminology:** Ensure you can define and give examples of: Bisexual vs. Unisexual flowers, Monoecious vs. Dioecious plants.
- **Monocot vs. Dicot:** Be able to distinguish them based on floral parts (multiple of 3 vs. 4 or 5).

7. Pollination in Plants

- **The Core Concept:** Pollination is the transfer of pollen from anther to stigma. It is NOT fertilization.
- **Self vs. Cross:** Make a list of advantages and disadvantages for each.
- **Agents of Pollination:** This is a high-yield topic. Create a detailed table for **Wind** and **Insects**. For each, describe the adaptations of the flower and the pollen.
- **Mechanisms Promoting Cross-Pollination:** Recall **Dichogamy** (protandry/protogyny) and **Self-Incompatibility**.

8. Regulation of Internal Environment (Homeostasis)

- **The Big Idea:** Homeostasis maintains a stable internal environment via **Negative Feedback Loops**.
- **Master the Loop:** Be able to draw and explain the generic negative feedback loop (Stimulus -> Receptor -> Control Center -> Effector -> Response).
- **Two Key Systems:**
 - a. **Thermoregulation (Skin):**
 - **Hot:** Vasodilation, Sweating.
 - **Cold:** Vasoconstriction, Shivering.
 - [Redraw the skin diagram showing these responses]
 - b. **Osmoregulation (Kidneys/ADH):**

- **Dehydrated:** More ADH → More water reabsorbed → Small, concentrated urine.
- **Hydrated:** Less ADH → Less water reabsorbed → Large, dilute urine.
- [Redraw the diagram showing ADH action on the kidney tubule]

B. Integrated Revision Activities

1. **Concept Mapping:** Pick a broad term like "**Energy Transfer**" or "**Genetic Variation**." Create a concept map that links it to as many topics from this term as possible. For example, "Genetic Variation" links to Cross-Pollination, Sexual Reproduction in vertebrates, and the disadvantage of Self-Pollination.
2. **Past Question Practice:** Actively solve past SSCE or other exam questions. Do not just read them. Time yourself and write out full answers.
3. **Teach-Back Method:** Explain a complex process (e.g., the role of ADH, or the process of eutrophication) to a friend or family member as if you are the teacher. This reveals gaps in your own understanding.

Evaluation (Revision Checkpoint):

Answer the following questions comprehensively without consulting your notes first.

1. **Integrated Question:** Describe how the structure of a wind-pollinated flower is adapted to its method of pollination, and explain one disadvantage of this method compared to insect pollination.
2. **Comparison Question:** Create a table to compare reproduction in a named reptile and a named mammal. Highlight four key differences.
3. **Application Question:** A farmer observes that his maize plants are stunted, and their leaves have a yellow-streaked pattern. Suggest the likely type of disease (causal agent) and the probable pest that acts as a vector. Propose two control measures for the pest.
4. **Diagram-Based Question:** Draw a labeled diagram to show how the human body responds to a decrease in external temperature. Your diagram should include the role of the brain, skin, and muscles.
5. **Conceptual Question:** Using the regulation of water content in the blood as an example, describe the principle of negative feedback.

Assignment:

1. **Create a "Summary Poster":** Design a single, large poster for ONE of the following topics: Pollution, Vertebrate Reproduction, or Homeostasis. The poster must use concise points, flowcharts, and diagrams to summarize the entire topic.
2. **Write Five "Challenge" Questions:** Based on the term's work, write five challenging exam-style questions that require application and analysis, not just recall. Provide a marking scheme for each question.
3. **Self-Assessment Checklist:** Create a checklist of all the sub-topics covered this term. Rate your understanding of each on a scale of 1 (Poor) to 5 (Excellent). Focus your final pre-exam study on the topics you rated 3 or below.
4. **Define and Link:** Choose 20 key terms from the term's work. For each term, write a clear definition and then write a sentence linking it to another term or concept from the syllabus.
5. **Past Paper Drill:** Complete one full past paper for SS2 Biology (Second Term). Mark your work strictly and write a reflection on your performance, noting three areas for final improvement.

SS2 BIOLOGY LESSON NOTES

THIRD TERM

WEEK 1: NERVOUS COORDINATION

SPECIFIC LEARNING OBJECTIVES: By the end of this week, students should be able to:

1. Analyze the ultrastructure of a neuron and correlate the structure of the axon/myelin sheath with the speed of impulse transmission.
2. Detail the ionic basis of the Resting and Action Potentials, explaining the role of the Sodium-Potassium pump.
3. Differentiate between the grey and white matter distribution in the Brain versus the Spinal Cord.
4. Compare and contrast the sympathetic and parasympathetic nervous systems with at least four physiological examples.
5. Trace the path of a reflex arc on a diagram and explain why the brain is bypassed for the immediate response.

I. INTRODUCTION TO COORDINATION

Living organisms must detect changes in their internal and external environments (stimuli) and respond appropriately to maintain **Homeostasis** (a stable internal environment). In animals, this is achieved by two systems:

1. **The Nervous System:** Rapid, electrical, short-lived responses.
2. **The Endocrine System:** Slower, chemical (hormonal), long-lasting responses.

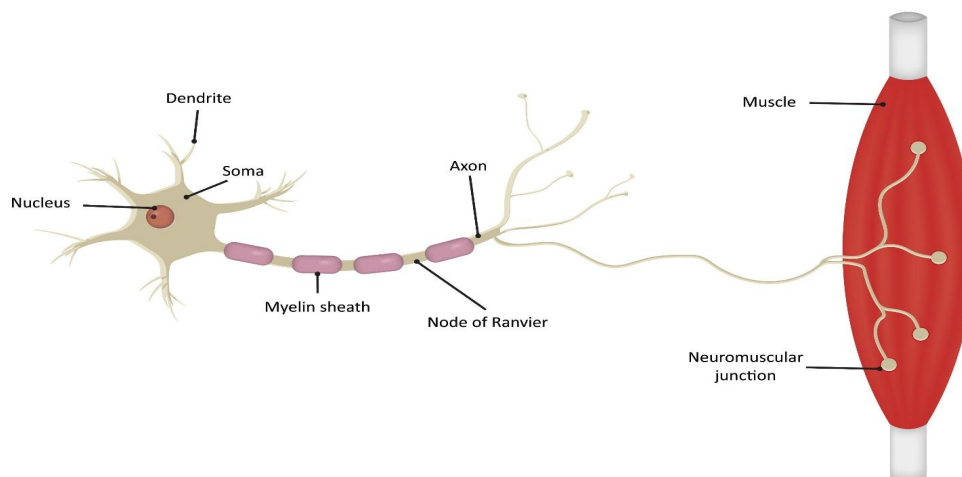
II. THE NEURON: THE STRUCTURAL UNIT

The nervous system consists of billions of specialized cells called neurons, supported by "glial cells" (neuroglia) which nourish and protect them.

1. Ultrastructure of a Neuron

- **Cell Body (Soma/Perikaryon):** The metabolic center. It contains the nucleus, mitochondria (for energy), and **Nissl Granules** (groupings of ribosomes for protein synthesis).
- **Dendrites:** Short, highly branched extensions that receive signals from other neurons and conduct them *centripetally* (towards the cell body).
- **Axon:** A single, long cylindrical process that conducts impulses *centrifugally* (away from the cell body).
- **Axoplasm:** The cytoplasm within the axon.
- **Axolemma:** The cell membrane of the axon.
- **Myelin Sheath:** A lipid-rich insulating layer produced by **Schwann Cells** (in PNS) or Oligodendrocytes (in CNS). It prevents signal leakage and increases speed.
- **Nodes of Ranvier:** Regular gaps in the myelin sheath where the axon membrane is exposed.
- **Synaptic Knobs:** Swellings at the end of axon terminals containing vesicles rich in neurotransmitters.

Motor Neuron



2. Functional Classification

- **Afferent (Sensory) Neurons:** Transmit impulses from receptors (skin, eyes) *to* the CNS. Structurally, they are often **Unipolar** (cell body sits to the side).
- **Efferent (Motor) Neurons:** Transmit impulses from the CNS *to* effectors (muscles/glands). Structurally, they are **Multipolar**.
- **Interneurons (Relay):** Located entirely within the CNS. They process information and connect sensory to motor neurons.

III. PHYSIOLOGY OF NERVE IMPULSE TRANSMISSION

A nerve impulse is not an electric current like in a wire; it is a propagating wave of electrochemical change.

1. Resting Potential (The Polarized State)

When a neuron is "at rest," it is electrically charged like a battery.

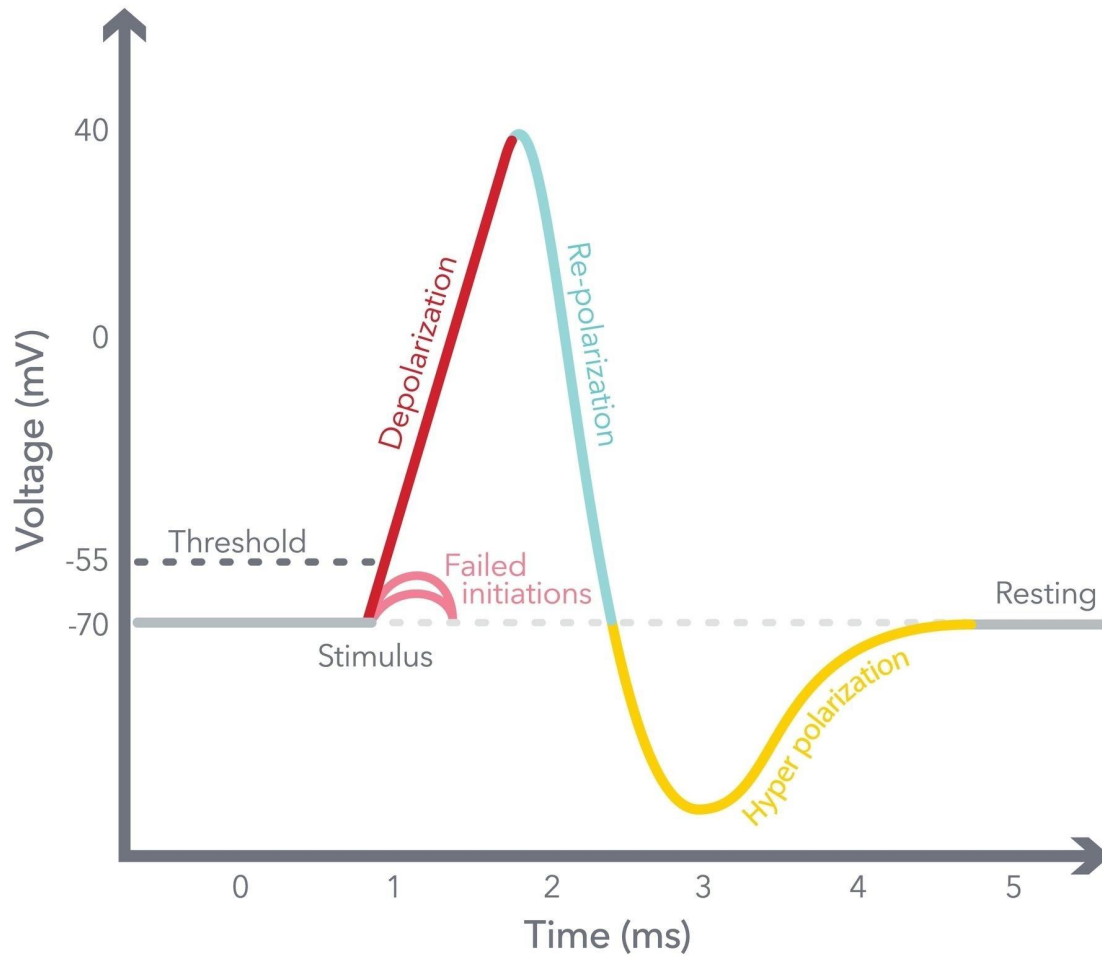
- **State:** The inside is negative () relative to the outside.
- **Mechanism:** The membrane is permeable to Potassium () but impermeable to Sodium (). The **Sodium-Potassium Pump** actively transports **3 out** and **2 in**. This differential distribution creates the resting potential.

2. Action Potential (The Impulse)

When a stimulus reaches the **Threshold Level** (approx):

1. **Depolarization:** Voltage-gated **Sodium channels open**. floods *into* the cell rapidly. The internal charge shoots up to ****.
2. **Repolarization:** Sodium channels close. **Potassium channels open**. rushes *out* of the cell to restore the negative charge.
3. **Hyperpolarization:** Potassium channels stay open slightly too long, making the cell temporarily *more* negative than resting potential ().
4. **Refractory Period:** The time during which the neuron cannot fire again. This ensures impulses travel in **one direction only**.

Action Potential



3. Saltatory Conduction

In myelinated neurons, ions cannot flow through the fatty sheath. Depolarization only occurs at the **Nodes of Ranvier**. The impulse "jumps" from node to node (Saltatory conduction), increasing transmission speed from ~2 m/s to ~120 m/s.

IV. SYNAPTIC TRANSMISSION

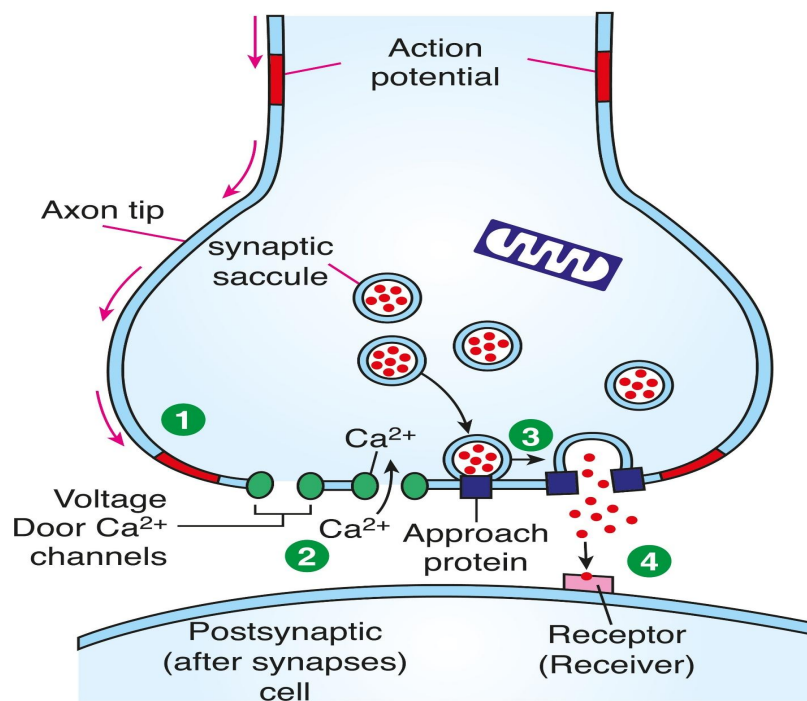
The synapse is the functional junction between two neurons.

1. **Arrival:** The electrical impulse reaches the pre-synaptic knob.
2. **Calcium Influx:** Voltage-gated channels open; Calcium ions enter.
3. **Exocytosis:** Calcium causes synaptic vesicles to fuse with the membrane, releasing **Neurotransmitters** into the synaptic cleft.
4. **Binding:** Neurotransmitters diffuse across the gap (20nm wide) and bind to specific receptors on the post-synaptic membrane.
5. **New Impulse:** This opens ion channels in the next neuron, triggering a new Action Potential.
6. **Cleanup:** Enzymes (e.g., **Acetylcholinesterase**) break down the neurotransmitter to stop continuous firing.

Common Neurotransmitters:

- **Acetylcholine:** Used in muscle contraction.
- **Dopamine:** Associated with pleasure and reward.
- **Serotonin:** Regulates mood and sleep.

Synapse (chemical transmission)



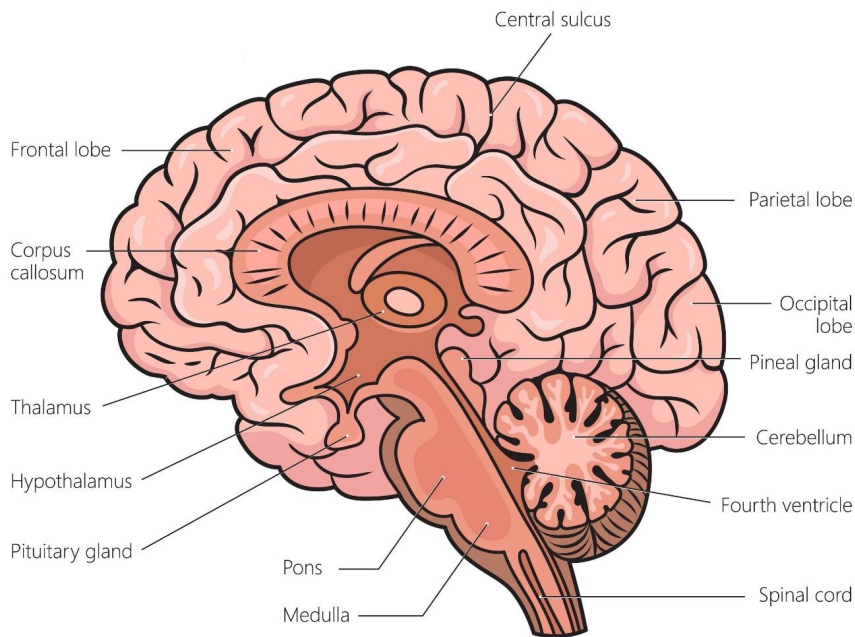
V. THE CENTRAL NERVOUS SYSTEM (CNS)

Protected by the skull/vertebrae and three membranes called **Meninges** (Outer Dura mater, Middle Arachnoid, Inner Pia mater).

1. The Brain

- **Forebrain:**
 - **Cerebrum:** Divided into left/right hemispheres. The outer layer is **Grey Matter** (Cerebral Cortex - cell bodies) and the inner is **White Matter** (axons). Highly convoluted (folded) to increase surface area for neurons.
 - **Thalamus:** The sensory relay station.
 - **Hypothalamus:** The homeostasis center (Temperature, Water balance, Endocrine control).
- **Midbrain:**
- **Optic Lobes:** Controls visual reflexes (e.g., pupil size).
- **Hindbrain:**
 - **Cerebellum:** Coordinates voluntary movements, posture, and balance. *Damage leads to jerky movements.*
 - **Medulla Oblongata:** Controls autonomic vital functions (Heartbeat, breathing, peristalsis, coughing).

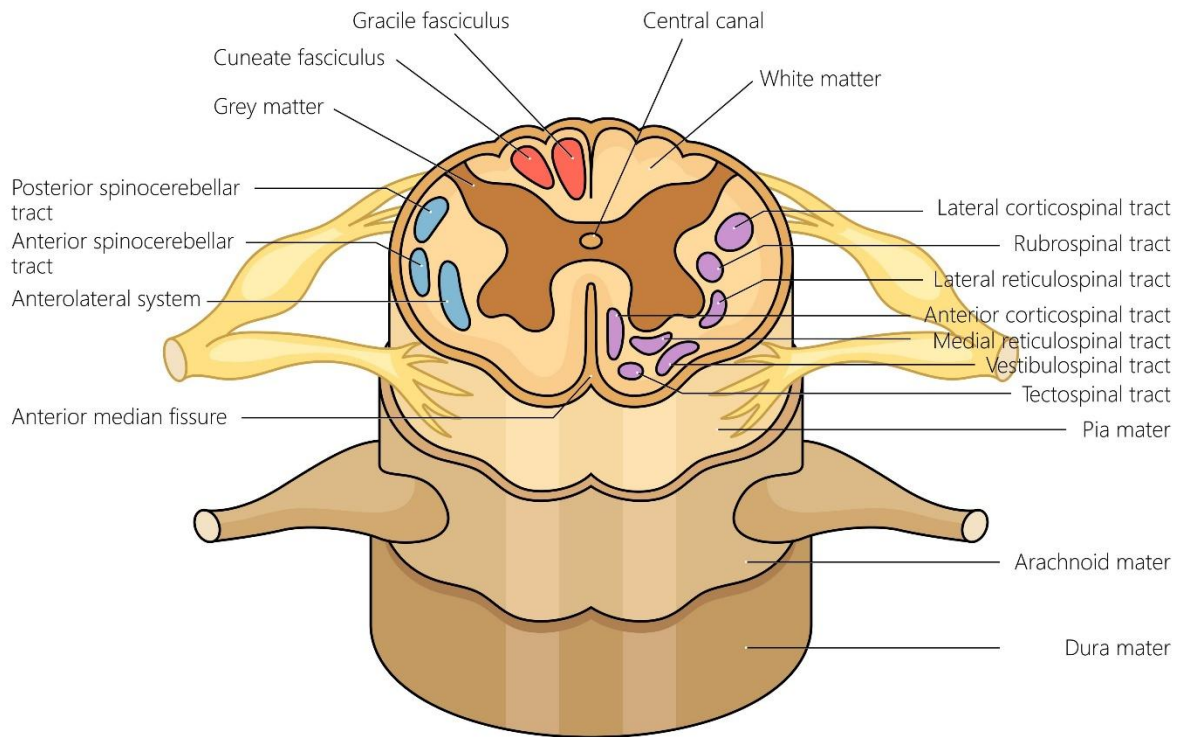
Human brain structure



2. The Spinal Cord

- A cylindrical bundle of nervous tissue.
- **Cross-Section Difference:** Unlike the brain, the spinal cord has **Grey Matter on the inside** (H-shape) and **White Matter on the outside**.
- **Roots:**
- **Dorsal Root:** Carries *Sensory* information into the cord (contains the Dorsal Root Ganglion).
- **Ventral Root:** Carries *Motor* information out to muscles.

Spinal Cord



VI. THE PERIPHERAL NERVOUS SYSTEM (PNS)

Connects the CNS to the rest of the body.

- **Cranial Nerves:** 12 pairs connected to the brain (e.g., Optic nerve, Vagus nerve).
- **Spinal Nerves:** 31 pairs connected to the spinal cord.

1. Somatic Nervous System

Voluntary control of skeletal muscles.

2. Autonomic Nervous System (ANS)

Involuntary control of visceral organs. It operates through two antagonistic systems:

Feature	Sympathetic (Fight or Flight)	Parasympathetic (Rest and Digest)
Neurotransmitter	Noradrenaline (Norepinephrine)	Acetylcholine

Feature	Sympathetic (Fight or Flight)	Parasympathetic (Rest and Digest)
Heart Rate	Accelerates	Slows down
Pupils	Dilates (to see danger)	Constricts (protects retina)
Digestion	Inhibits (Peristalsis stops)	Stimulates (Peristalsis active)
Bronchi (Lungs)	Relaxes/Dilates (more air)	Constricts
Liver	Converts Glycogen Glucose	Converts Glucose Glycogen
Bladder	Relaxes (holds urine)	Contracts (releases urine)

VII. REFLEX ACTIONS

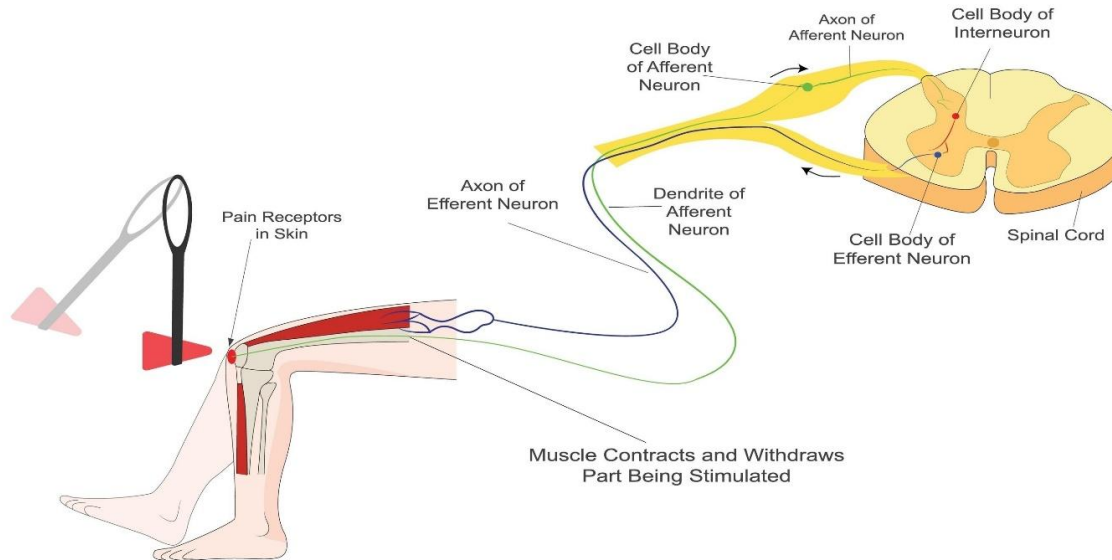
A reflex is a rapid, involuntary, and stereotyped response to a specific stimulus.

1. The Reflex Arc

The pathway taken by a nerve impulse during a reflex action.

1. **Receptor:** Detects stimulus (e.g., heat).
2. **Sensory Neuron:** Transmits to Spinal Cord (via dorsal root).
3. **Relay Neuron (Intermediate):** Passes signal within the Grey Matter.
4. **Motor Neuron:** Transmits from Spinal Cord (via ventral root).
5. **Effector:** Muscle contracts (hand pulls away). *Note: The brain is informed **after** the action has occurred (pain is felt later).*

Reflex Arc (Polysynaptic Reflex)



2. Types of Reflexes

- **Simple (Unconditioned) Reflex:** Inborn, instinctive. *Examples:* Knee-jerk, blinking, withdrawing hand from fire, pupil reflex.
- **Conditioned (Learned) Reflex:** Acquired through learning or association (Pavlovian). *Examples:* Walking, cycling, salivating at the sound of a dinner bell.

Evaluation:

1. Why is the transmission of a nerve impulse faster in a myelinated neuron than in a non-myelinated one?
2. Describe the state of the ion channels during the "Depolarization" phase of an action potential.
3. What specific role does Calcium (Ca^{2+}) play at the synapse?
4. A patient has suffered damage to the Cerebellum. What specific physical symptoms would you expect to observe?
5. Explain the "All-or-None" law as it applies to nerve impulses.

Assignment:

1. **Research:** Investigate the effects of one neurotoxic drug (e.g., Cocaine, Curare, or Organophosphates) on synaptic transmission. Write a one-page report.
2. **Diagram:** Draw a detailed cross-section of the Spinal Cord showing a reflex arc. Label the Dorsal Root Ganglion, Relay Neuron, and Central Canal.
3. **Analysis:** Tabulate the differences between a "Hormonal Response" and a "Nervous Response" in terms of speed, duration, and method of transmission.
4. **Case Study:** "A man sees a snake and runs." Describe the physiological changes occurring in his body driven by the Sympathetic Nervous System.
5. **Critical Thinking:** Why does a "Conditioned Reflex" require the Cerebrum, while a "Simple Reflex" does not?



WEEK 2: SENSE ORGANS

Specific Learning Objectives: By the end of this week, students should be able to:

1. **Dissect** the structure of the mammalian eye and explain the specific physiological roles of the retina, choroid, and ciliary body.
2. **Analyze** the mechanism of accommodation, describing the antagonistic action of ciliary muscles and suspensory ligaments.
3. **Diagnose** visual defects (Myopia/Hypermetropia) from ray diagrams and prescribe the correct corrective lens.
4. **Sequence** the transmission of sound from air waves to nerve impulses, highlighting the amplification function of the ossicles.
5. **Differentiate** between the mechanisms of static balance (otoliths) and dynamic balance (semicircular canals).

I. INTRODUCTION TO SENSORY PERCEPTION

Sense organs serve as the body's windows to the external world. They act as **Biological Transducers**, converting various forms of environmental energy (light, sound, pressure, chemical) into electrical nerve impulses that the brain can interpret.

- **Stimulus:** A detectable change in the internal or external environment.
- **Receptor:** Specialized cells that detect the stimulus.
- **Effector:** The organ (muscle or gland) that acts in response.

II. THE EYE (PHOTORECEPTION)

The mammalian eye is a complex, fluid-filled sphere protected by the bony orbit of the skull, eyelids, and lacrimal (tear) glands.

1. Anatomy of the Eye

A. The Outer Fibrous Tunic (Protection)

- **Sclera:** The "white of the eye." A tough, opaque, fibrous connective tissue that maintains the eyeball's shape and protects inner structures.

- **Cornea:** The transparent, bulging anterior extension of the sclera. It is avascular (no blood vessels) to remain clear, absorbing oxygen directly from the air. It provides approx. 70% of the eye's refractive power.
- **Conjunctiva:** A thin, transparent mucous membrane covering the sclera and lining the eyelids. Infection here causes *Conjunctivitis* (Apollo).

B. The Middle Vascular Tunic (Uvea)

- **Choroid:** A highly vascularized layer containing dark melanin pigments. *Function:* Nourishes the retina and absorbs stray light rays to prevent internal reflection (scattering) which would blur vision.
- **Ciliary Body:** A ring of tissue comprising:
 - *Ciliary Muscles:* Smooth muscles that control lens shape.
 - *Ciliary Processes:* Secrete aqueous humor.
- **Suspensory Ligaments:** Fine fibers connecting the ciliary body to the lens, holding it in place.
- **Iris:** The colored muscular diaphragm. It controls the diameter of the **Pupil** to regulate light entry.

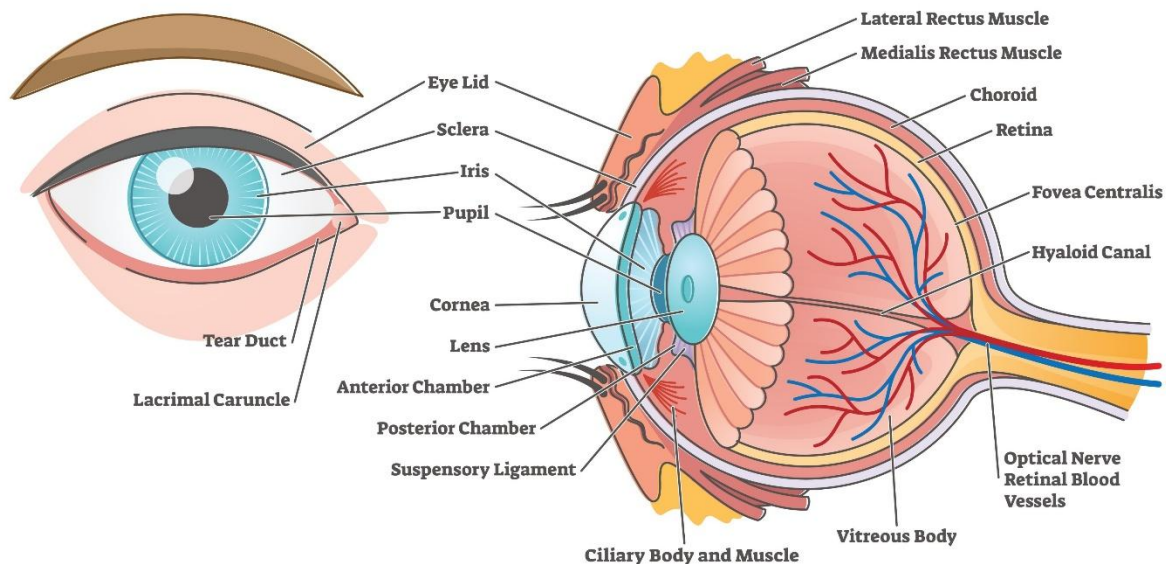
C. The Inner Nervous Tunic (Sensory)

- **Retina:** The photosensitive screen containing photoreceptors:
- **Rods:** ~120 million. highly sensitive to low light (scotopic vision). Contain the pigment **Rhodopsin** (Visual Purple). They provide black-and-white vision and motion detection. Located peripherally.
- **Cones:** ~6 million. Function in bright light (photopic vision). Contain **Iodopsin**. Responsible for color vision and high visual acuity. Concentrated centrally.
- **Fovea Centralis (Yellow Spot):** A small depression in the retina containing *only* cones. This is the point of sharpest vision.

- **Blind Spot (Optic Disc):** The point where the optic nerve exits the eye. No photoreceptors exist here, so no image is detected.

D. Refractive Media

- **Aqueous Humor:** A watery, alkaline fluid between the cornea and lens. It maintains Intraocular Pressure (IOP) and nourishes the lens/cornea.
- **Lens:** A transparent, biconvex, crystalline structure. It is elastic and changes shape to focus light.
- **Vitreous Humor:** A clear, jelly-like substance filling the posterior chamber. It supports the retina against the choroid.



EYE ANATOMY

2. Physiology of Vision

A. Image Formation Light passes through: **Cornea Aqueous Humor Pupil Lens Vitreous Humor Retina.**

- The refractive media bend (refract) the light rays to focus on the retina.
- The image formed is **Real, Inverted, and Diminished**. The brain processes this to appear upright.

B. Accommodation (Focusing Mechanism) This is the reflex ability of the eye to adjust the focal length of the lens to see objects at varying distances clearly.

Object Distance	Ciliary Muscles	Suspensory Ligaments	Lens Shape	Focal Length
Near (Reading)	Contract	Slacken (Relax)	Spherical (Fat)	Short (High power)
Far (Distance)	Relax	Tighten (Pull)	Flat (Thin)	Long (Low power)

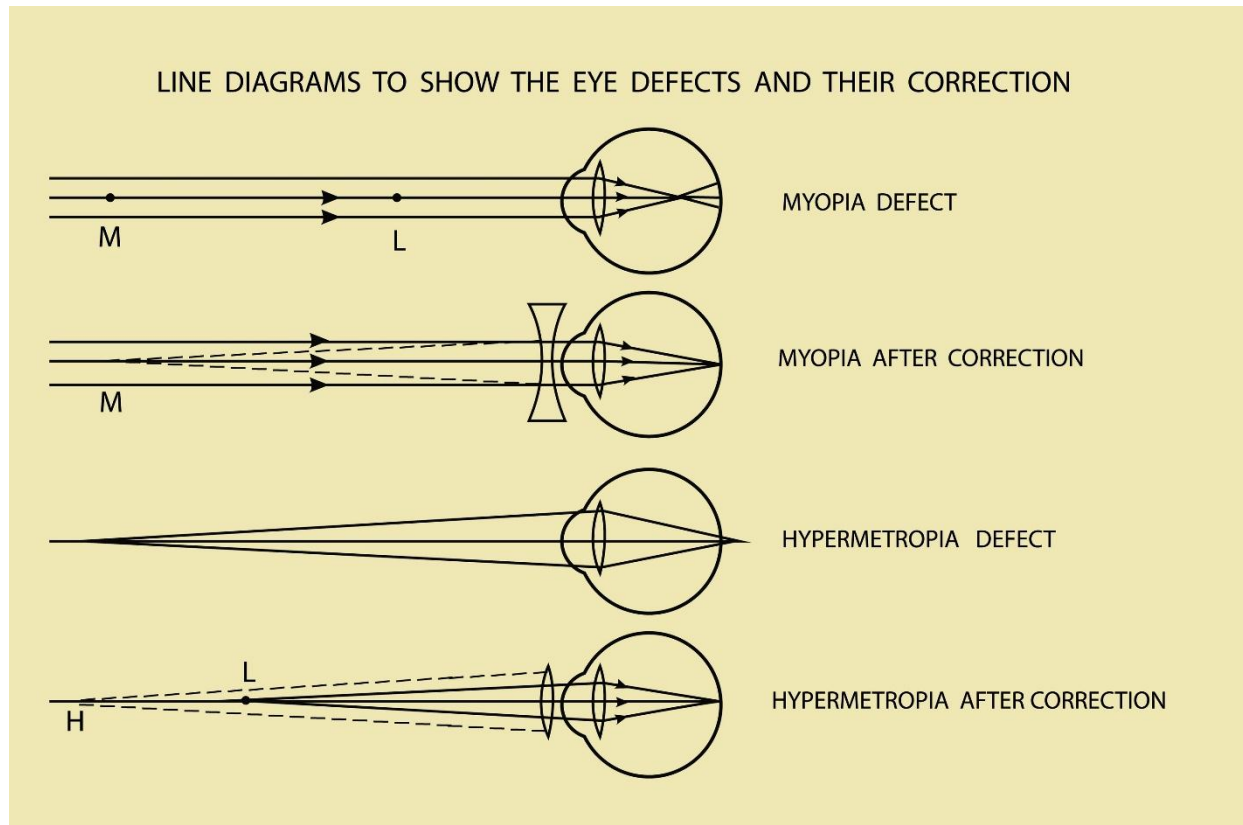
C. The Pupillary Reflex (Adaptation)

- **Bright Light:** Circular iris muscles contract; Radial muscles relax Pupil **Constricts** (prevents dazzling).
- **Dim Light:** Radial iris muscles contract; Circular muscles relax Pupil **Dilates** (maximizes light intake).

3. Defects of Vision and Corrections

- **Myopia (Short-sightedness):**
 - *Defect:* Can see near, but far is blurry.
 - *Cause:* Eyeball is too long (front-to-back) or lens is too strong/curved.
 - *Focus:* Image forms **in front** of the retina.
 - *Correction:* **Concave (Diverging) Lens.**
- **Hypermetropia (Long-sightedness):**
 - *Defect:* Can see far, but near is blurry.
 - *Cause:* Eyeball is too short or lens is too flat/weak.
 - *Focus:* Image forms **behind** the retina.
 - *Correction:* **Convex (Converging) Lens.**

- **Astigmatism:** Irregular curvature of the cornea or lens, causing distorted images.
Correction: Cylindrical lenses.
- **Presbyopia:** "Old age sight." The lens loses elasticity and cannot accommodate for near vision. *Correction:* Bifocals.



III. THE EAR (MECHANORECEPTION)

The ear is a dual-function organ: **Hearing (Auditory)** and **Equilibrium (Vestibular)**.

1. Anatomy of the Ear

A. Outer Ear

- **Pinna (Auricle):** Cartilaginous flap that collects and funnels sound waves.
- **Auditory Meatus (Ear Canal):** Tube lined with hairs and ceruminous glands (wax) to trap dust/insects.

- **Tympanic Membrane (Eardrum):** Thin, semi-transparent membrane separating outer and middle ear. It vibrates in response to sound waves.

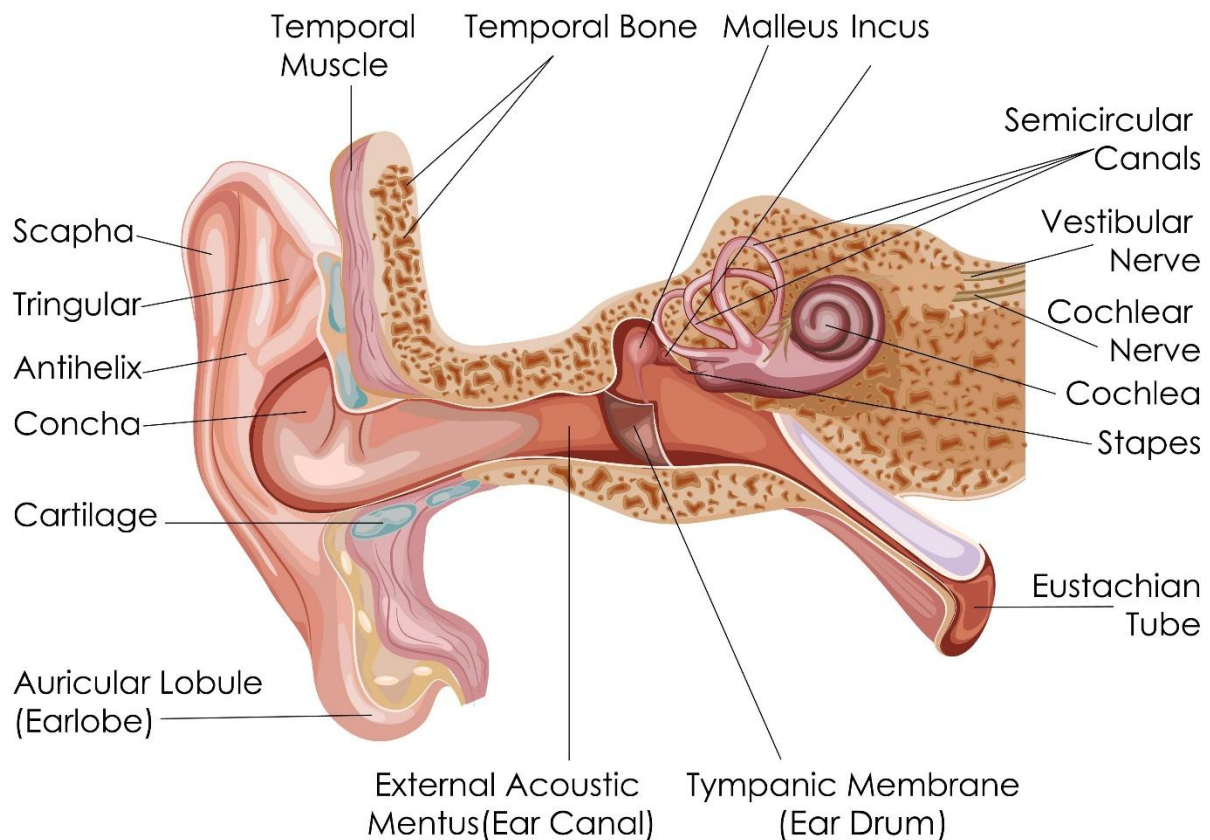
B. Middle Ear (The Amplifier)

- An air-filled cavity located in the temporal bone.
- **Ossicles:** Three tiny bones linked together—**Malleus (Hammer)**, **Incus (Anvil)**, and **Stapes (Stirrup)**. They transmit vibrations from the eardrum to the oval window, amplifying the force approx. 20 times.
- **Eustachian Tube:** Connects the middle ear to the nasopharynx (throat). It opens during swallowing/yawning to **equalize air pressure** on both sides of the eardrum.

C. Inner Ear (The Labyrinth)

- A fluid-filled complex embedded in bone.
- **Cochlea:** A coiled, snail-like tube responsible for **hearing**. It contains the **Organ of Corti** (the sensory unit).
- **Vestibular Apparatus:** Responsible for **balance**.
- **Semicircular Canals (3):** Oriented at right angles (X, Y, Z axes).
- **Vestibule (Utricle & Saccule):** Chambers containing otoliths.

Ear Diagram



2. Physiology of Hearing

1. **Collection:** Pinna funnels sound waves into the canal.
2. **Vibration:** Waves strike the Tympanic Membrane.
3. **Amplification:** Ossicles vibrate. The Stapes acts like a piston on the **Oval Window**.
4. **Fluid Wave:** Vibration of the Oval Window creates pressure waves in the cochlear fluid (Perilymph and Endolymph).

5. **Transduction:** These waves ripple the **Basilar Membrane** in the Organ of Corti. Sensory hair cells are bent against the Tectorial Membrane.
6. **Nerve Impulse:** Mechanical bending triggers action potentials sent via the **Auditory Nerve** to the brain.
7. **Release:** Pressure dissipates via the **Round Window**.

3. Physiology of Balance

- **Dynamic Equilibrium (Rotation):** Detected by **Semicircular Canals**. When the head rotates, the fluid (endolymph) inside lags behind due to inertia, bending the cupula (hair cells). This signals direction and speed of rotation.
- **Static Equilibrium (Gravity/Tilt):** Detected by **Utricle & Saccule**. These contain **Otoliths** (calcium carbonate crystals). When the head tilts, gravity pulls the crystals, bending the underlying hair cells.

IV. OTHER SENSE ORGANS

1. The Nose (Olfaction)

- **Chemoreceptors:** The olfactory epithelium in the roof of the nasal cavity detects volatile chemical molecules dissolved in mucus.
- **Significance:** Smell is 10,000 times more sensitive than taste and is directly linked to the limbic system (memory/emotion).

2. The Tongue (Gustation)

- **Taste Buds:** Located in papillae. They detect chemicals dissolved in saliva.
- **Primary Tastes:** Sweet (tip), Salty (front sides), Sour (back sides), Bitter (back), Umami (savory).
- *Flavor* is a combination of Taste + Smell + Texture.

3. The Skin (Somatosensation)

The largest organ, containing varied receptors:

- **Mechanoreceptors:** Meissner's corpuscles (light touch), Pacinian corpuscles (deep pressure/vibration).

- **Thermoreceptors:** Ruffini endings (heat), Krause end bulbs (cold).
 - **Nociceptors (Pain):** Free nerve endings. Crucial for survival as they warn of tissue damage.
-

Evaluation:

1. **Critical Thinking:** Why does it take a few minutes to see clearly when you walk from bright sunlight into a dark cinema hall? (Explain *Dark Adaptation*).
2. **Anatomy:** Name the three ossicles in order from the eardrum to the inner ear.
3. **Physiology:** Describe the shape of the lens and the state of the ciliary muscles when viewing a distant mountain.
4. **Pathology:** A patient cannot equalize pressure in their ears during a flight. Which structure is likely blocked?
5. **Optics:** Draw a ray diagram showing how a concave lens corrects short-sightedness.

Assignment:

1. **Research Project:** Investigate **Glaucoma** and **Cataracts**. Create a comparison table covering: Causes, Affected Part of Eye, Symptoms, and Treatment.
2. **Diagram:** Draw a detailed, labeled cross-section of the human ear. Color-code the Outer, Middle, and Inner ear sections.
3. **Experiment:** Perform the "Blind Spot Test" (draw a cross and a dot on paper, cover one eye, and move paper closer until the dot disappears). Explain the biological reason for this phenomenon.
4. **Essay:** Discuss the importance of **Binocular Vision** (Stereopsis) for predators compared to monocular vision for prey animals.
5. **Analysis:** Why is the sense of smell often lost when a person has a severe cold, and how does this affect the taste of food?

WEEK 3: ECOLOGY OF POPULATION

Specific Learning Objectives By the end of the lesson, students should be able to:

1. Define population and population density.
2. List four factors that affect the size of a population (Natality, Mortality, Immigration, Emigration).
3. Explain the difference between density-dependent and density-independent factors with examples.
4. Describe the phases of a Sigmoid (S-shaped) growth curve (Lag, Log, Stationary).
5. Calculate the population density of a species given the total number and land area.

I. MEANING OF POPULATION

In Biology, a **Population** is defined as a group of organisms of the **same species** living together in a specific area (habitat) at a particular time.

- **Key phrase:** "Same species" (e.g., all the tilapia fish in a school pond).
- **Community:** If we count the tilapia, the frogs, and the water weeds together, that is a *Community*, not a population.

II. POPULATION CHARACTERISTICS

To study a population, ecologists look at specific features known as "population parameters."

1. Population Size

This is the total number of individuals of the same species in the habitat.

- *Example:* 500 students in a school.

2. Population Density

This is the number of individuals per unit area (on land) or per unit volume (in water). It tells us how "crowded" an area is.

- **Formula:**

- *Significance:* High density leads to overcrowding and competition. Low density makes it hard to find mates.

3. Population Frequency

This refers to how often a particular species occurs in different samples of the habitat.

- *Example:* If you toss a quadrat 10 times and find *Tridax* weed in 6 of the tosses, the frequency is 60%.

4. Growth Rate

This refers to how fast a population is increasing or decreasing. It depends on four main factors:

- **Natality (Birth Rate):** Number of new offspring produced (Increases population).
- **Mortality (Death Rate):** Number of individuals dying (Decreases population).
- **Immigration:** Individuals moving *into* the area (Increases population).
- **Emigration:** Individuals moving *out of* the area (Decreases population).

5. Distribution (Dispersion)

How the organisms are arranged in the area:

- **Clumped:** Grouped together (e.g., a school of fish, a herd of cattle). This is the most common.
- **Uniform:** Evenly spaced (e.g., crops in a farm like maize or oil palms).
- **Random:** Scattered without a pattern (e.g., seeds blown by wind).

III. POPULATION GROWTH CURVES

When we graph population size over time, we get specific shapes.

1. Exponential Growth (The J-Shaped Curve)

This happens when conditions are perfect—plenty of food, no predators, and no diseases.

- The population grows slowly at first, then shoots up very fast.
- *Example:* Bacteria dividing in a lab or insects in the rainy season.

- *Problem:* It cannot last forever. Eventually, food runs out, and the population crashes.

2. Logistic Growth (The S-Shaped Curve)

This is the realistic growth seen in nature.

- **Lag Phase:** Growth is slow because the few parents are just settling in and finding mates.
- **Log (Exponential) Phase:** Growth becomes very fast because there is enough food and many young are being born.
- **Stationary Phase:** Growth stops or stabilizes. The number of births equals the number of deaths.
- **Carrying Capacity:** The maximum number of organisms the environment can support without breaking down.

IV. FACTORS AFFECTING POPULATION (ENVIRONMENTAL RESISTANCE)

Populations cannot grow forever because "resistance" stops them.

1. Density-Dependent Factors

These factors become severe only when the population is overcrowded.

- **Food Shortage:** More mouths to feed means less food for everyone.
- **Competition:** Organisms fight for space, water, and mates.
- **Disease:** Germs spread faster in crowded places.
- **Predation:** Predators easily find prey when there are many of them.

2. Density-Independent Factors

These factors kill organisms whether they are crowded or not.

- **Natural Disasters:** Flood, fire, drought, or earthquake.
- **Climate Changes:** Sudden extreme heat or cold.

V. METHODS OF ESTIMATING POPULATION SIZE

Since we cannot count every single ant or grass blade, we use sampling methods.

1. Quadrat Method (For Plants)

A quadrat is a square frame (usually $1\text{m} \times 1\text{m}$).

- **How to use:**

1. Throw the quadrat randomly on the field.
2. Count the number of a specific plant inside it.
3. Repeat this several times (e.g., 10 times).
4. Find the average number per quadrat.
5. Multiply by the total area of the field to get the estimated population.

2. Capture-Mark-Recapture Method (For Animals)

Used for animals that move (rats, fish, birds).

- **Steps:**

1. Catch a number of animals (First Catch).
2. Mark them (with non-toxic paint or tag) and release them.
3. Wait for them to mix with the others.
4. Catch a second group (Second Catch).
5. Count how many in the second group have marks.

- **Formula:**

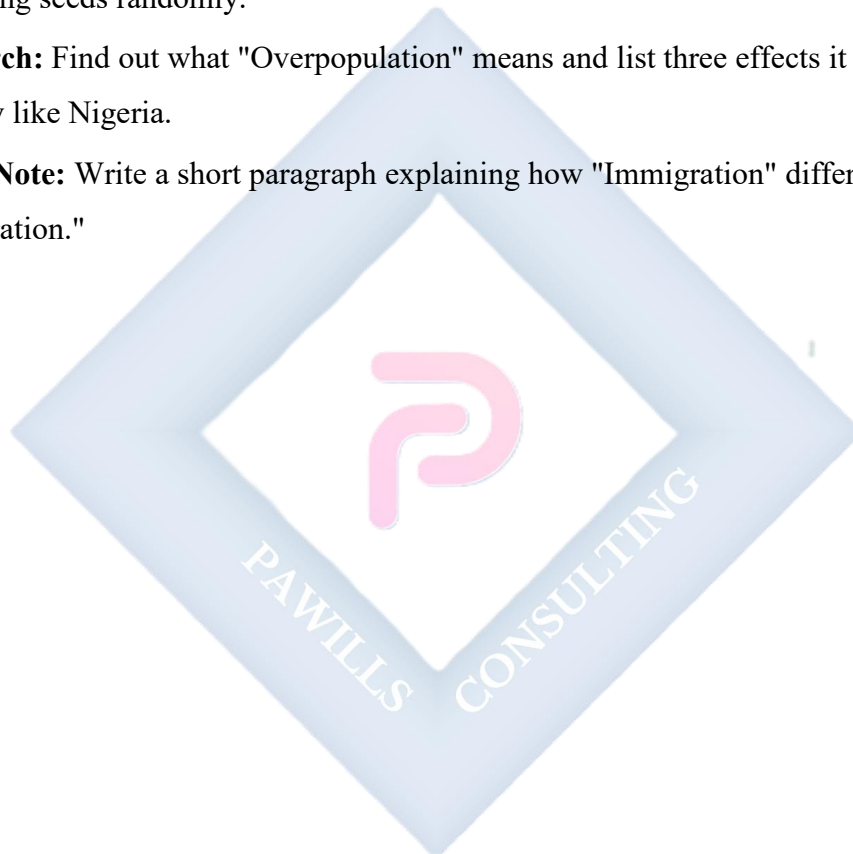
$$\text{Estimated Total Population} = \frac{\text{Number Caught in First Sample} \times \text{Number Caught in Second Sample}}{\text{Number of Marked Individuals Recaptured}}$$

Evaluation:

1. Define the term "Carrying Capacity."
2. In a school garden of , students counted 200 *Tridax* plants. Calculate the population density.
3. State two differences between the "Lag Phase" and the "Log Phase" of population growth.
4. Why is "Competition" called a density-dependent factor?
5. Mention one advantage of using the Capture-Mark-Recapture method.

Assignment:

1. **Calculation Practice:** A biologist captured 40 grasshoppers, marked them, and released them. Later, she captured 60 grasshoppers, and 12 of them had marks. Use the formula to estimate the total population of grasshoppers.
2. **Graphing:** Draw a standard S-shaped growth curve. Label the "Lag Phase," "Exponential Phase," and "Stationary Phase."
3. **Reasoning:** Explain why a farmer plants crops in rows (Uniform distribution) rather than scattering seeds randomly.
4. **Research:** Find out what "Overpopulation" means and list three effects it has on a country like Nigeria.
5. **Short Note:** Write a short paragraph explaining how "Immigration" differs from "Emigration."



WEEK 4: BALANCE IN NATURE (ECOSYSTEMS)

Specific Learning Objectives By the end of the lesson, students should be able to:

1. Define an ecosystem and list its biotic and abiotic components.
2. Differentiate between a food chain and a food web with examples.
3. Explain why energy flows in one direction while nutrients cycle.
4. Draw a Pyramid of Energy and explain why it is always broad at the base.
5. Describe the processes that add and remove Carbon Dioxide from the atmosphere.

I. THE CONCEPT OF AN ECOSYSTEM

An **Ecosystem** is a self-sustaining unit where living organisms (plants, animals, microbes) interact with one another and with their non-living environment.

- **Ecology:** The study of these interactions.
- **The Biosphere:** The part of the earth (land, water, and air) where life exists.

II. COMPONENTS OF AN ECOSYSTEM

Every ecosystem is made up of two main components:

1. Abiotic Components (Non-living)

These are the physical and chemical factors that affect organisms.

- **Climatic:** Temperature, rainfall, humidity, light intensity, wind.
- **Edaphic (Soil):** Soil texture, pH, water content, nutrient availability.
- **Topographic:** Shape of the land (slope, altitude).

2. Biotic Components (Living)

These are categorized based on how they get their food (Nutrition):

- **A. Producers (Autotrophs):** Organisms that manufacture their own food using sunlight (Photosynthesis) or chemicals (Chemosynthesis).
- *Examples:* Green plants, algae, phytoplankton.

- *Role:* They trap solar energy and convert it into chemical energy (food).
- **B. Consumers (Heterotrophs):** Organisms that cannot make their own food and must eat others.
- **Primary Consumers (Herbivores):** Eat plants directly (e.g., Grasshopper, Goat, Cow).
- **Secondary Consumers (Carnivores):** Eat herbivores (e.g., Lizard, Toad, Cat).
- **Tertiary Consumers (Top Carnivores):** Eat other carnivores (e.g., Lion, Hawk, Shark).
- **Omnivores:** Eat both plants and animals (e.g., Man, Pig).
- **C. Decomposers (Saprophytes):** Microorganisms that break down dead bodies and waste matter.
- *Examples:* Bacteria, Fungi (Mushroom, Mold).
- *Role:* They release nutrients (like nitrates) back into the soil for plants to use again. Without them, the ecosystem would be choked with dead bodies.

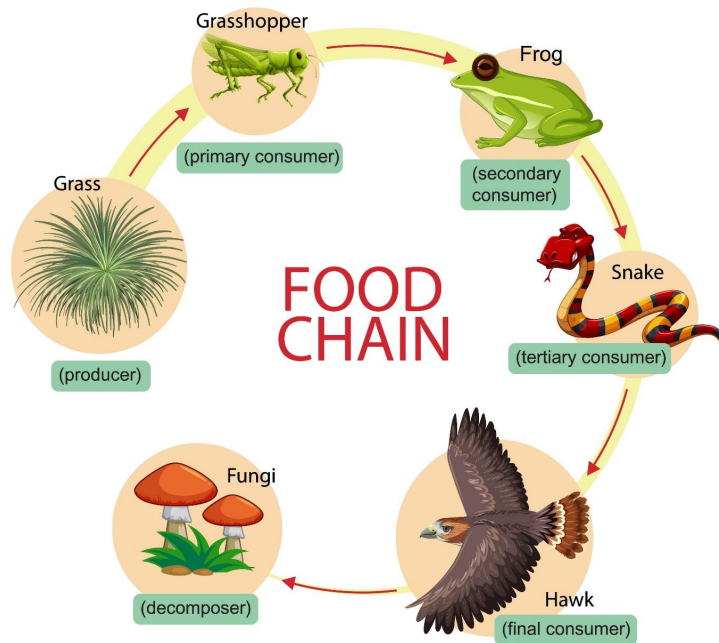
III. ENERGY FLOW IN THE ECOSYSTEM

Energy comes from the **Sun**. It flows in **one direction** only: from the Sun → Producers → Consumers. Unlike nutrients, energy does *not* cycle; it is used up and lost.

1. Food Chains

A linear pathway showing who eats whom in an ecosystem.

- *Structure:* Producer → Primary Consumer → Secondary Consumer → Tertiary Consumer.
- *Example (Terrestrial):* Grass → Zebra → Lion
- *Example (Aquatic):* Phytoplankton → Zooplankton → Small Fish → Large Fish → Shark



2. Food Webs

In real life, animals eat more than one thing. A lion eats zebras, antelopes, and buffaloes. A **Food Web** is a complex network of interconnected food chains.

- *Significance:* Food webs make the ecosystem more stable. If one food source fails, the animal can switch to another.

3. Trophic Levels

A trophic level is the feeding position an organism occupies in a food chain.

- **Level 1:** Producers (Plants).
- **Level 2:** Primary Consumers (Herbivores).
- **Level 3:** Secondary Consumers (Carnivores).
- **Level 4:** Tertiary Consumers.

IV. LOSS OF ENERGY (THE 10% LAW)

When a grasshopper eats grass, it does not store all the energy from the grass.

- Some energy is used for movement and flying.
- Some is lost as **Heat** during respiration.
- Some is lost in **Excretion** and feces.
- **Only about 10%** of the energy is stored in the grasshopper's body to be passed on to the lizard that eats it.

Consequence: Because energy decreases at each level, food chains are usually short (rarely more than 4 or 5 links). There isn't enough energy left to support a 6th level.

V. ECOLOGICAL PYRAMIDS

Diagrams that represent the relationship between organisms at different trophic levels.

1. Pyramid of Numbers

Shows the **number** of individuals at each level.

- Usually upright (Many grasses → fewer grasshoppers → one lizard).
- *Exception:* Can be inverted. One big Mango tree (Producer) can support hundreds of caterpillars.

2. Pyramid of Biomass

Shows the **total mass** (weight) of living matter at each level.

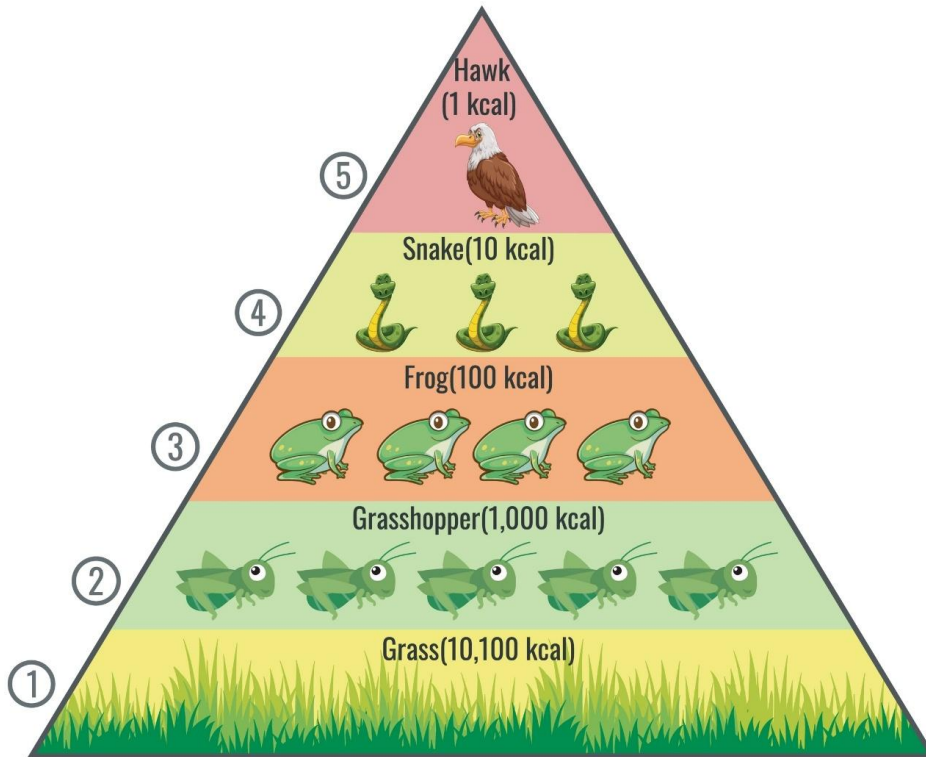
- Usually upright.
- *Exception:* Inverted in oceans, where tiny phytoplankton (low mass) reproduce fast enough to support massive whales.

3. Pyramid of Energy

Shows the **amount of energy** available at each level.

- **Always Upright.** Energy is always lost as you move up; it can never increase.

ENERGY PYRAMID



VI. NUTRIENT CYCLING (CYCLING OF MATTER)

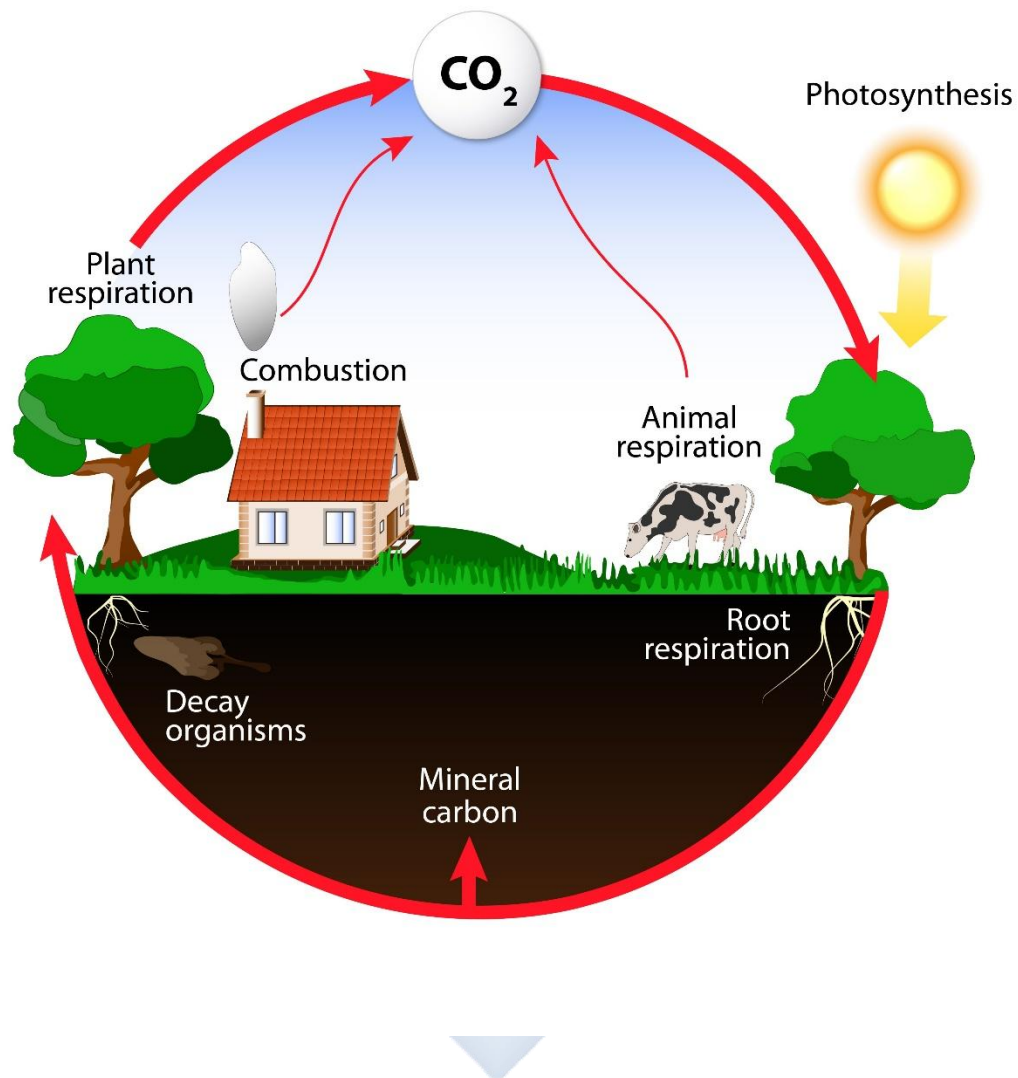
While energy is lost, chemical elements (nutrients) are recycled forever.

1. The Carbon Cycle

The balance of Carbon dioxide (CO_2) in the atmosphere.

- **Removal of CO_2 :** Plants take in CO_2 for **Photosynthesis**.
- **Addition of CO_2 :**
- **Respiration:** All living things breathe out CO_2 .
- **Decay:** Decomposers release CO_2 when breaking down dead matter.
- **Combustion:** Burning wood and fossil fuels (petrol/diesel) releases stored carbon.

CARBON CYCLE



2. The Water Cycle

The continuous movement of water between the earth and the atmosphere.

- **Evaporation:** Sun turns water from oceans/rivers into vapor.
- **Transpiration:** Plants lose water vapor from their leaves.
- **Condensation:** Vapor cools to form clouds.
- **Precipitation:** Water falls as rain, snow, or hail.

VII. BALANCE IN NATURE (HOMEOSTASIS)

An ecosystem is in a state of dynamic equilibrium.

- If the population of prey (e.g., rats) increases, the population of predators (e.g., snakes) will also increase because they have plenty of food.
 - Eventually, the snakes will eat so many rats that the rat population drops.
 - Then, the snakes will starve, and their numbers will drop.
 - This allows the rats to recover, and the cycle continues. This keeps nature balanced.
-

Evaluation:

1. What is the primary source of energy for all ecosystems?
2. Why are decomposers important in an ecosystem?
3. Given the food chain: **Maize → Grasshopper → Toad → Snake**. Which organism is the Secondary Consumer?
4. State the "10% Law" of energy transfer.
5. Name two processes that increase the amount of Carbon Dioxide in the atmosphere.

Assignment:

1. **Diagram:** Draw a Food Web involving the following organisms: *Grass, Rabbit, Grasshopper, Lizard, Hawk, Snake, Mouse*.
2. **Reasoning:** Explain why a "Pyramid of Numbers" can sometimes be inverted (upside down), but a "Pyramid of Energy" cannot.
3. **Application:** If a farmer kills all the Hawks in an area, explain what will likely happen to the population of Snakes and Mice in that area.
4. **Short Note:** Explain the terms "Abiotic Factor" and "Biotic Factor" with three examples for each.
5. **Cycle:** Draw a simple schematic diagram of the Water Cycle.

WEEK 6: REPRODUCTIVE SYSTEM AND REPRODUCTION IN HUMANS

Specific Learning Objectives By the end of the lesson, students should be able to:

1. Draw and label the male and female reproductive systems.
2. State the function of the testes, ovaries, fallopian tubes, and uterus.
3. Differentiate between a sperm cell and an egg cell in terms of size, mobility, and structure.
4. Define Fertilization, Implantation, and Gestation.
5. Explain the function of the placenta and amniotic fluid during pregnancy.

I. INTRODUCTION TO REPRODUCTION

Reproduction is the ability of living organisms to give rise to new individuals of the same species. It ensures the continuity of life. Humans exhibit **Sexual Reproduction**, which involves the fusion of male and female gametes (sex cells).

II. THE MALE REPRODUCTIVE SYSTEM

The function of the male system is to produce sperm (gametes) and transfer them into the female.

1. External Structures

- **Scrotum:** A sac of skin that hangs outside the body cavity.
- *Function:* It holds the testes. It keeps the testes at a temperature slightly **lower (about 2-3°C less)** than body temperature, which is necessary for healthy sperm production.
- **Penis:** The organ of copulation (mating). It contains erectile tissue that fills with blood to become stiff (erection), allowing it to enter the vagina.

2. Internal Structures

- **Testes (Singular: Testis):** Two oval organs inside the scrotum.
- *Function:* They produce **Sperm** and the male hormone **Testosterone**.
- **Epididymis:** A coiled tube located on top of each testis.

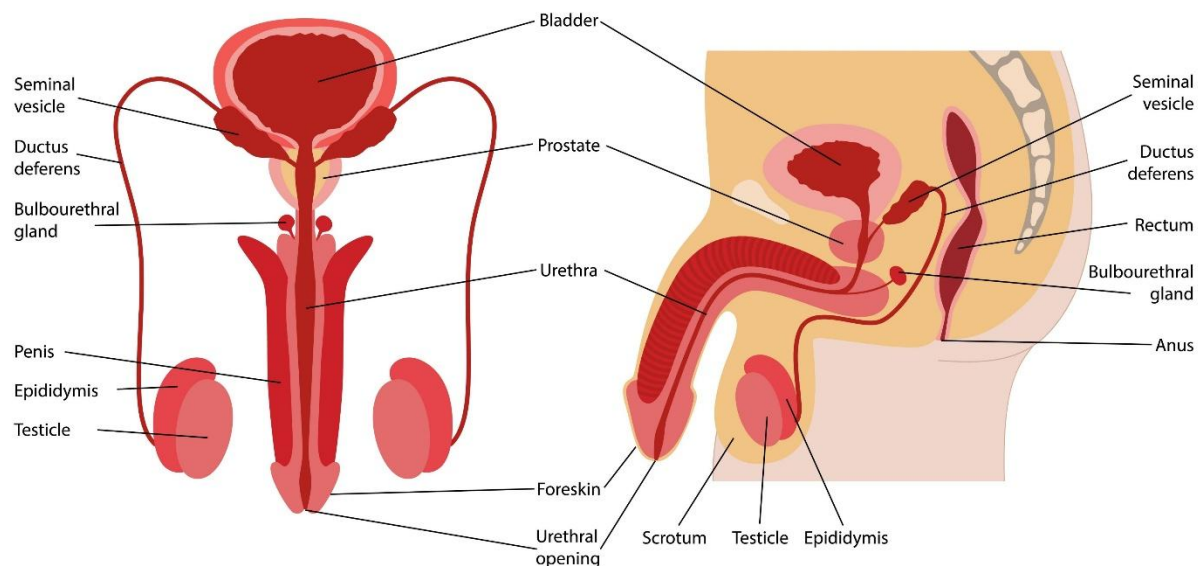
- *Function:* It stores sperm temporarily until they fully mature and learn to swim.
- **Vas Deferens (Sperm Duct):** A long tube that carries sperm from the epididymis towards the urethra.
- **Urethra:** A tube running through the penis.
- *Function:* It is a common passage for both **Urine** and **Semen** (but not at the same time).

3. Accessory Glands

These glands add fluids to the sperm to create **Semen**.

- **Seminal Vesicle:** Produces a sugary fluid (fructose) to give sperm energy.
- **Prostate Gland:** Produces an alkaline fluid to neutralize the acid in the female vagina, helping sperm survive.
- **Cowper's Gland:** Secretes a lubricant.

Male reproductive system



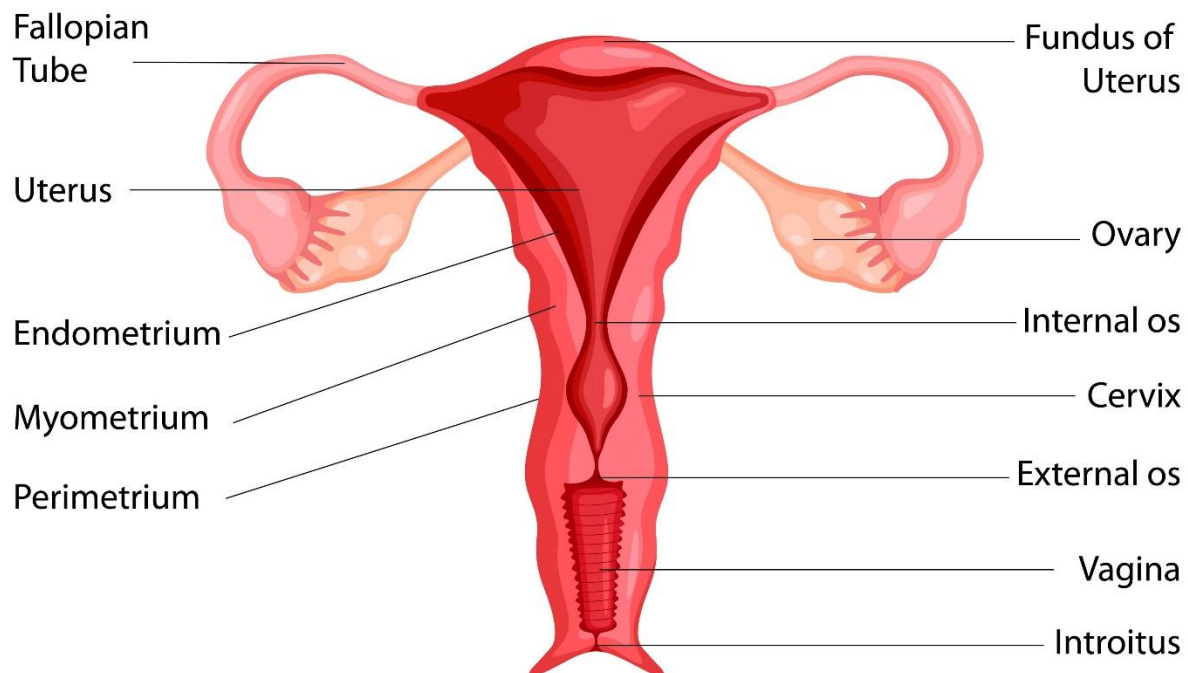
III. THE FEMALE REPRODUCTIVE SYSTEM

The function of the female system is to produce eggs (ova), receive sperm, and nurture the developing baby.

1. Internal Structures

- **Ovaries:** Two almond-shaped organs located in the lower abdomen.
- *Function:* They produce **Ova (Eggs)** and female hormones (**Estrogen** and **Progesterone**).
- **Fallopian Tubes (Oviducts):** Tubes connecting the ovaries to the uterus.
- *Function:* This is where **Fertilization** takes place. Cilia (tiny hairs) inside push the egg toward the uterus.
- **Uterus (Womb):** A muscular, pear-shaped organ.
- *Function:* This is where the fetus develops. The inner lining is called the **Endometrium**.
- **Cervix:** The narrow "neck" of the uterus that opens into the vagina.
- **Vagina:** A muscular tube leading to the outside.
- *Function:* It receives the penis during sex and serves as the **Birth Canal** during delivery.

Female Reproductive System



IV. STRUCTURE OF GAMETES

1. The Sperm Cell (Male Gamete)

It is microscopic and mobile.

- **Head:** Contains the Nucleus (genetic material) and the **Acrosome** (a cap containing enzymes to dissolve the egg's outer layer).
- **Middle Piece:** Contains many **Mitochondria** to provide energy for swimming.

- **Tail (Flagellum):** A long whip-like structure for movement.

2. The Ovum (Female Gamete)

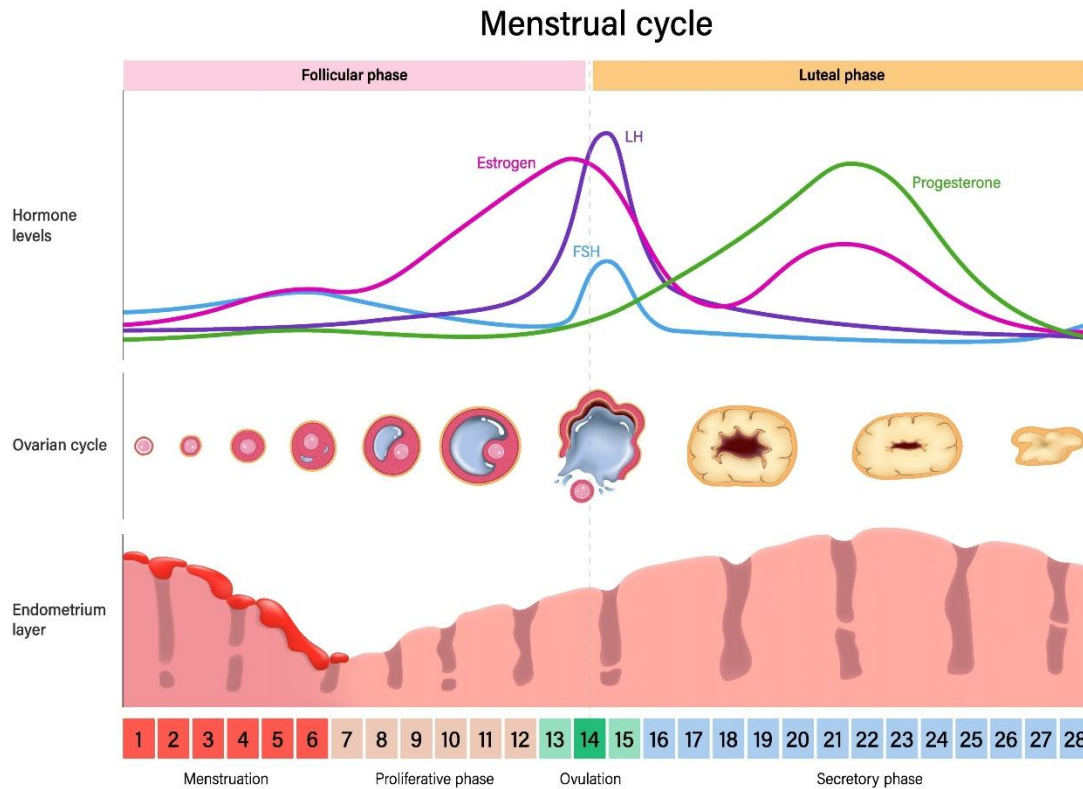
It is much larger than the sperm and spherical.

- **Cytoplasm:** Contains yolk (food store) for the early embryo.
- **Nucleus:** Contains genetic material.
- **Membrane:** Surrounded by a jelly coat that hardens after one sperm enters to stop others.

V. OVULATION AND THE MENSTRUAL CYCLE

The female body prepares for pregnancy every month in a cycle (usually 28 days).

1. **Menstruation (Days 1–5):** If no pregnancy occurs, the lining of the uterus breaks down and sheds as blood.
2. **Follicular Phase (Days 6–13):** The hormone **FSH** stimulates an egg to mature in the ovary. The uterus lining starts to rebuild.
3. **Ovulation (Day 14):** The hormone **LH** causes the ovary to release the mature egg. This is the most fertile time.
4. **Luteal Phase (Days 15–28):** The empty follicle becomes the **Corpus Luteum** and produces **Progesterone** to maintain the uterus lining. If the egg is not fertilized, the cycle restarts.



VI. FERTILIZATION AND DEVELOPMENT

1. Fertilization

- During sexual intercourse, millions of sperm are ejaculated into the vagina.
- They swim up through the cervix and uterus into the **Fallopian Tube**.
- Only **one** sperm fuses with the egg.
- The nucleus of the sperm (n) fuses with the nucleus of the egg (n) to form a **Zygote** ($2n$).

2. Implantation

- The zygote divides rapidly by mitosis to form a ball of cells.
- It travels down to the uterus and attaches itself to the wall of the womb. This is called **Implantation**.

3. Gestation (Pregnancy)

The period of development from fertilization to birth (approx. 9 months or 40 weeks).

- **Placenta:** A disc-shaped organ formed from both mother and embryo tissues.
- *Function:* It allows food, oxygen, and antibodies to pass from mother to baby. It allows waste (CO₂, urea) to pass from baby to mother. *Note: The mother's blood and baby's blood do not mix.*
- **Umbilical Cord:** The tube connecting the baby to the placenta.
- **Amniotic Sac:** A bag of fluid (Amnion) surrounding the baby.
- *Function:* It acts as a **shock absorber**, protecting the baby from bumps and temperature changes.

VII. BIRTH (PARTURITION)

- At the end of pregnancy, the hormone **Oxytocin** causes the uterus muscles to contract (Labour).
- The cervix dilates (opens).
- The amnion breaks ("water breaks").
- The baby is pushed out head first, followed by the placenta (afterbirth).

Evaluation

1. Why are the testes located outside the body in the scrotum?
2. In which specific part of the female reproductive system does fertilization take place?
3. What is the function of the acrosome in a sperm cell?
4. Name the tube that carries both urine and semen in males.
5. What substance prevents the fetus from physical shock and drying out?

Assignment

1. **Diagram:** Draw the male reproductive system (side view) and label the: Urethra, Prostate Gland, Testis, and Vas Deferens.
2. **Table:** Construct a table showing three differences between the Sperm and the Ovum.
3. **Reasoning:** Explain why a woman does not menstruate if she is pregnant.

4. **Process:** Describe the journey of a sperm cell from the testis until it meets the egg.
5. **Health:** Mention two sexually transmitted infections (STIs) and one way to prevent them.



WEEK 8: DEVELOPMENT OF NEW SEEDS (FLOWERING PLANTS)

Specific Learning Objectives By the end of the lesson, students should be able to:

1. **Define** Pollination and distinguish between Self and Cross-pollination.
2. **Relate** the features of a flower (e.g., sticky stigma, light pollen) to its agent of pollination (Insect or Wind).
3. **Describe** the path of the pollen tube from the stigma to the micropyle.
4. **List** the changes that occur in the Ovary, Ovule, and Zygote after fertilization.
5. **Differentiate** between Epigeal and Hypogeal germination with one example for each.

I. INTRODUCTION: THE FLOWER AS A REPRODUCTIVE UNIT

The flower is the sexual reproductive organ of an angiosperm (flowering plant).

- **Male Part (Androecium):** Consists of Stamens. Each stamen has an **Anther** (produces pollen grains) and a **Filament**.
- **Female Part (Gynoecium/Pistil):** Consists of Carpels. Each carpel has a **Stigma** (receives pollen), **Style** (tube), and **Ovary** (contains ovules).

II. POLLINATION

Before a seed can form, pollen must move from the male part to the female part.

Definition: Pollination is the transfer of mature pollen grains from the anther to the stigma of a flower.

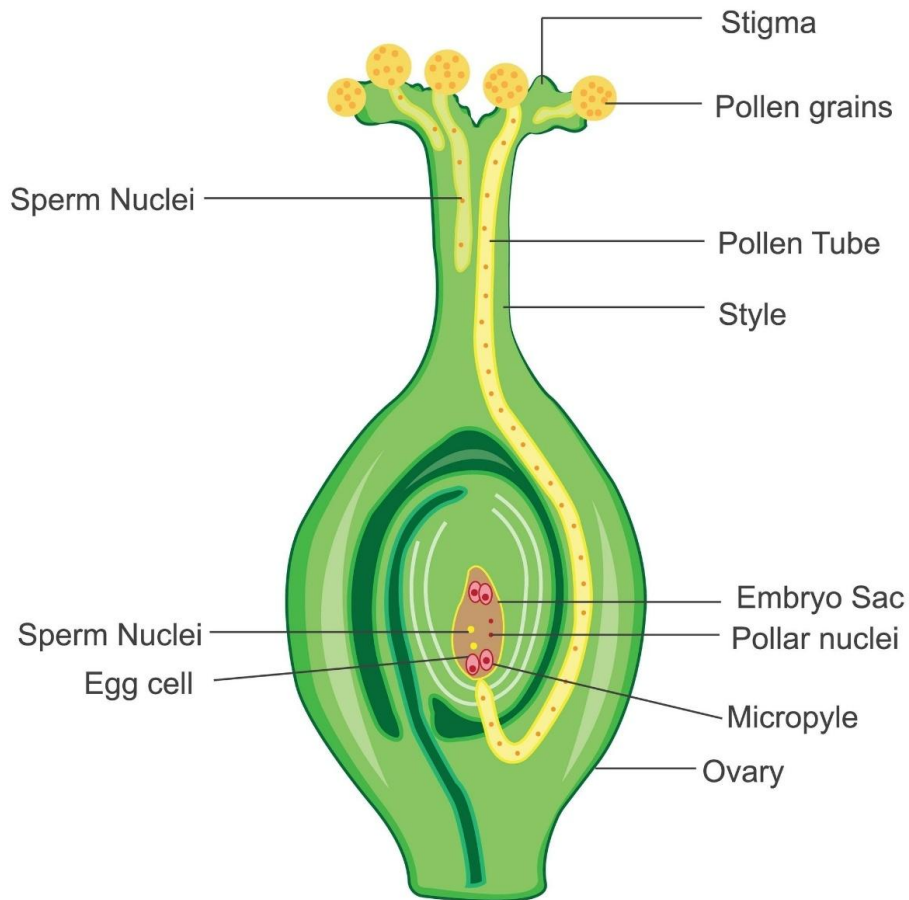
1. Types of Pollination

- **Self-Pollination:** Transfer of pollen from the anther to the stigma of the *same* flower or another flower on the *same* plant (e.g., Tomato).
- **Cross-Pollination:** Transfer of pollen from the anther of one flower to the stigma of a flower on a *different* plant of the same species (e.g., Pawpaw, Morning Glory).
Advantage: Produces healthier, more varied offspring.

2. Agents of Pollination

- **Insect Pollinated (Entomophilous):**

- Flowers are large, brightly colored, and scented to attract insects.
- They produce nectar (food for bees/butterflies).
- Stigma is sticky to trap pollen.
- Pollen is rough and sticky.
- *Examples:* Hibiscus, Pride of Barbados.
- **Wind Pollinated (Anemophilous):**
 - Flowers are small, dull, and scentless (no need to attract anyone).
 - Anthers hang outside the flower to release pollen into the wind.
 - Stigma is feathery and hangs outside to catch pollen.
 - Pollen is produced in massive quantities, light, and smooth.
 - *Examples:* Maize, Grasses, Rice.



III. THE PROCESS OF FERTILIZATION

Pollination is just the arrival. Fertilization is the actual fusion.

1. **Germination of Pollen:** When a pollen grain lands on a compatible sticky stigma, it absorbs a sugary liquid and swells.
2. **Pollen Tube Growth:** The pollen grain grows a long tube (pollen tube) that pushes down through the **Style** towards the ovary. It carries the male nuclei.
3. **Entry:** The pollen tube enters the **Ovule** through a tiny opening called the **Micropyle**.
4. **Fusion (Double Fertilization):**
 - One male nucleus fuses with the **Egg Cell** to form the **Zygote** (Future Plant).

- The second male nucleus fuses with the **Secondary Nucleus** to form the **Endosperm** (Food Store).
- *Note:* Because two fusions happen, it is called **Double Fertilization** (unique to flowering plants).

IV. FORMATION OF THE SEED AND FRUIT

After fertilization, the flower parts change dramatically. The petals and sepals usually wither and fall off.

Part of Flower	Changes Into...
Ovary	Fruit
Ovary Wall	Pericarp (Fruit coat/skin)
Ovule	Seed
Integuments (Ovule walls)	Testa (Seed coat)
Zygote	Embryo (Baby plant)
Triploid Nucleus	Endosperm (Food storage tissue)

V. STRUCTURE OF A SEED

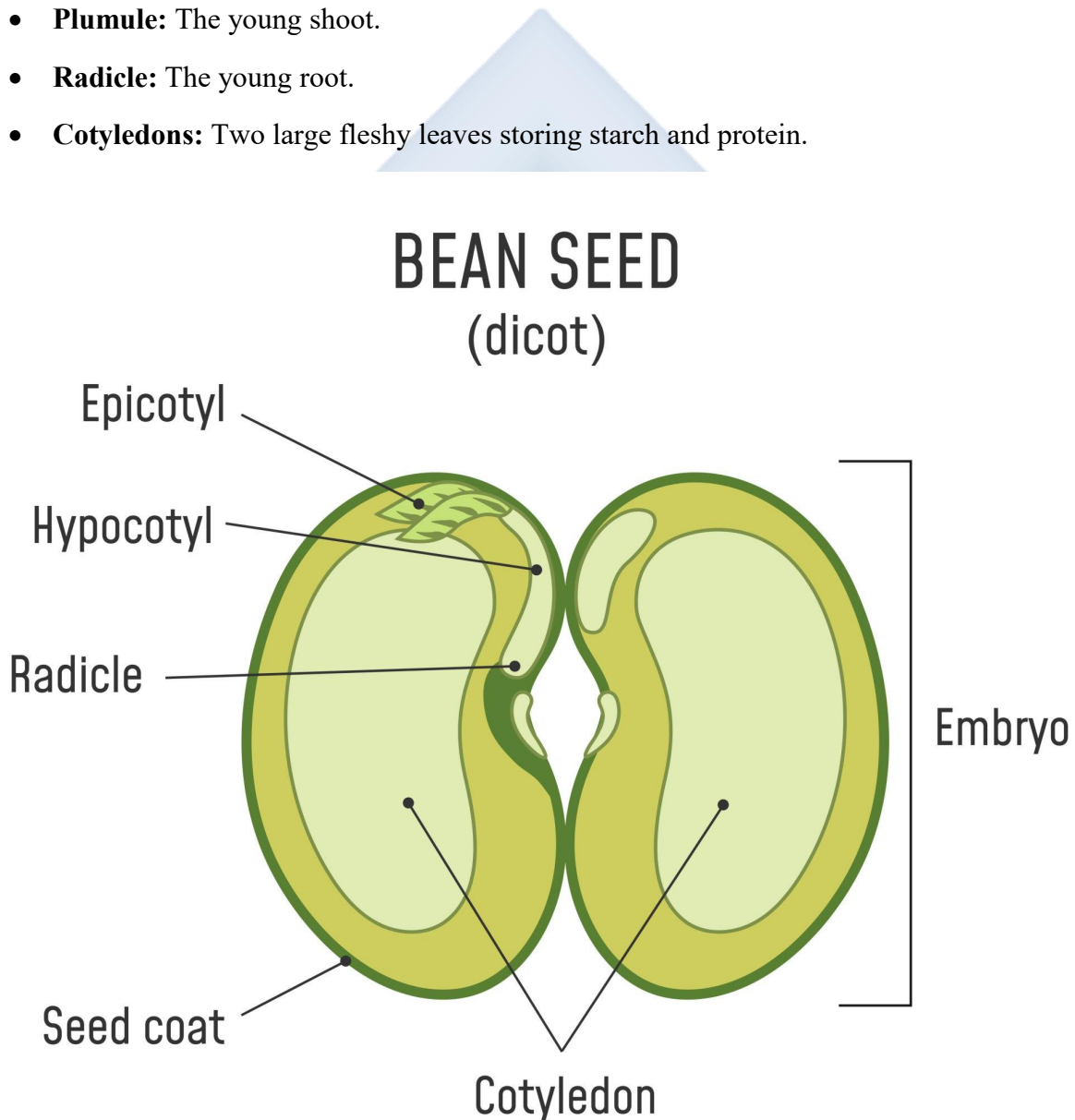
A seed is essentially a ripened fertilized ovule. It contains the embryo plant.

1. Types of Seeds

- **Monocotyledonous Seeds:** Have **one** seed leaf (cotyledon). The food is usually stored in a separate endosperm.
- *Example:* Maize, Rice, Wheat.
- **Dicotyledonous Seeds:** Have **two** seed leaves (cotyledons). The food is usually stored *inside* the massive cotyledons.
- *Example:* Bean, Groundnut, Mango.

2. Structure (The Bean Seed - Dicot)

- **Testa:** The tough, brown outer coat. *Function:* Protects the seed from fungi, bacteria, and insects.
- **Hilum:** A scar on the side. *Function:* Where the seed was attached to the fruit pod.
- **Micropyle:** A tiny hole near the hilum. *Function:* Allows water and air to enter during germination.
- **Embryo:** Consists of:
 - **Plumule:** The young shoot.
 - **Radicle:** The young root.
 - **Cotyledons:** Two large fleshy leaves storing starch and protein.



3. Structure (The Maize Grain - Monocot)

- It is actually a fruit (Caryopsis) because the fruit wall is fused with the seed coat.
- **Endosperm:** The large yellow area storing starch.
- **Scutellum:** The single cotyledon (absorbs food from endosperm for the embryo).
- **Coleoptile:** A sheath protecting the young plumule.
- **Coleorhiza:** A sheath protecting the young radicle.

VI. GERMINATION OF SEEDS

Germination is the process where the embryo wakes up from dormancy and grows into a seedling.

1. Conditions Necessary for Germination

Remember the acronym "**WOW**":

1. **Water:** To soften the testa and activate enzymes.
 2. **Oxygen:** For respiration to generate energy for growth.
 3. **Warmth (Suitable Temperature):** To allow enzymes to work (usually 25°C – 35°C).
- *Note:* Light is usually *not* required for the first stage of germination.

2. Types of Germination

- **Epigeal Germination:**
 - The **Cotyledons are carried above the soil** surface.
 - This happens because the *Hypocotyl* (stem below cotyledon) elongates rapidly.
 - *Examples:* Bean, Melon, Castor oil, Morning Glory.
- **Hypogeal Germination:**
 - The **Cotyledons remain underground** in the soil.
 - This happens because the *Epicotyl* (stem above cotyledon) elongates.
 - *Examples:* Maize, Guinea corn, Millet.

Evaluation

1. Name the part of the flower that develops into the **seed** and the part that develops into the **fruit**.
2. Why do wind-pollinated flowers produce millions of pollen grains?
3. What is the function of the **Micropyle** in a dry seed?
4. State the three external conditions necessary for seed germination.
5. In a Bean seedling, what happens to the cotyledons after the first green leaves appear?

Assignment

1. Diagram: Draw a large, labeled diagram of the Longitudinal Section of a Maize grain.
2. Experiment (Home Project): Plant 5 bean seeds and 5 maize seeds in two separate tins. Observe them daily for 7 days. Record which part comes out of the soil first for each. Sketch the seedlings on Day 5.
3. Table: Create a comparison table with 4 differences between Insect-pollinated flowers and Wind-pollinated flowers.
4. Critical Thinking: Why are tomato seeds formed *inside* the fruit, while the petals fall off?
5. Research: Find out what "Dormancy" means in seeds and list two ways farmers can break dormancy to make seeds grow.

WEEK 9: FRUITS

Specific Learning Objectives By the end of the lesson, students should be able to:

1. Differentiate between a True Fruit and a False Fruit with one example of each.
2. Classify specific fruits (e.g., Tomato, Mango, Maize) into their correct botanical groups (Berry, Drupe, Caryopsis).
3. Draw the longitudinal section of a drupe (e.g., Mango) and label the three layers of the pericarp.
4. Explain three structural adaptations of seeds dispersed by wind.
5. Justify why a Coconut is classified as a drupe and how it is adapted for water dispersal.

I. INTRODUCTION TO FRUITS

Biologically, a **Fruit** is a fertilized, ripened **ovary** containing seeds.

- After fertilization, the ovary swells and changes to protect the seeds and help scatter them.
- *Distinction:* In cooking, "fruit" usually means sweet (e.g., Orange), and "vegetable" means savory (e.g., Tomato). In Biology, **Tomatoes, Okra, and Peppers are all fruits** because they develop from flowers and contain seeds.

II. STRUCTURE OF A FRUIT

The wall of a fruit is called the **Pericarp**. In many fruits (especially fleshy ones), the pericarp is divided into three layers:

1. **Epicarp:** The outer skin (peel).
2. **Mesocarp:** The middle, fleshy layer (usually the edible part).
3. **Endocarp:** The inner layer surrounding the seed (sometimes hard/stony).

III. CLASSIFICATION OF FRUITS

Fruits are classified based on their origin (number of flowers involved) and the nature of their pericarp (dry or fleshy).

1. Classification by Origin

- **True Fruit:** Develops *only* from the ovary of a flower.

- *Example:* Mango, Orange, Cowpea.
- **False Fruit (Pseudocarp):** Develops from the ovary *plus* other floral parts (like the receptacle or calyx).
- *Example:* **Apple** (flesh is swollen receptacle), **Cashew** (flesh is swollen stalk), **Pineapple**.
- *Example:* **Strawberry** (red flesh is receptacle; tiny dots are the actual fruits).

2. Classification by Composition

- **Simple Fruit:** Formed from **one flower** with **one ovary** (either simple or fused carpels).
Example: Cherry, Bean.
- **Aggregate Fruit:** Formed from **one flower** with **many separate ovaries** (apocarpous). Each ovary becomes a tiny fruitlet clustered together. *Example:* Strawberry, Custard Apple.
- **Multiple (Composite) Fruit:** Formed from a **whole cluster of flowers** (inflorescence) fused together. *Example:* Pineapple, Breadfruit.

3. Classification by Pericarp Nature (Simple Fruits)

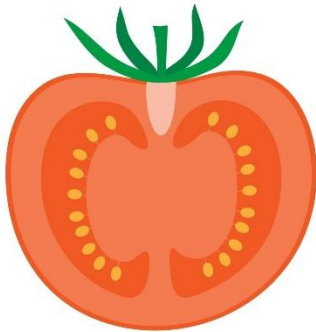
Simple fruits are divided into **Fleshy** (succulent) and **Dry**.

A. Fleshy Fruits (*Pericarp is soft and juicy*)

1. **Berry:** The entire pericarp is soft. Usually contains many seeds.
 - *Examples:* Tomato, Pepper, Guava, Watermelon, Orange (Hesperidium).
2. **Drupe:** Has a fleshy mesocarp but a hard, stony **endocarp** (pit) protecting the seed.
 - *Examples:* Mango, Coconut (Fibrous drupe), Oil Palm fruit.
3. **Pome:** The outer fleshy part is the receptacle (false fruit). The core is the ovary.
 - *Example:* Apple, Pear.

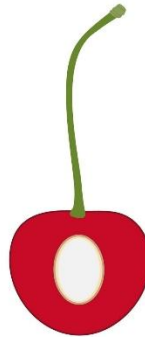
TYPES OF FLESHY SIMPLE FRUITS

BERRY



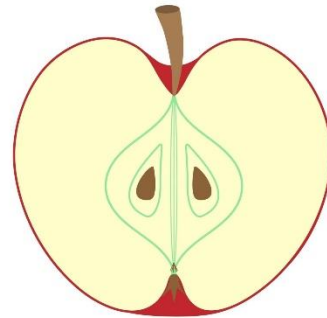
Tomato

DRUPE



Cherry

POME



Apple

B. Dry Dehiscent Fruits (Split open when ripe)

These fruits split to release seeds automatically.

1. **Legume (Pod):** Splits along **two** sides (sutures).
 - *Examples:* Bean, Flamboyant, Cowpea.
2. **Capsule:** Splits along **many** sides or pores.
 - *Examples:* Okra, Cotton, Poppy.
3. **Follicle:** Splits along **one** side only.
 - *Examples:* Kola, Rose periwinkle.
4. **Schizocarp:** Splits into separate one-seeded parts.
 - *Example:* Desmodium, Cassia.

C. Dry Indehiscent Fruits (Do not split)

These fruits do not open; the seed usually decays out or is eaten.

1. **Caryopsis (Grain):** The seed coat is fused to the fruit wall.

- *Examples:* Maize, Rice, Grasses.
- 2. **Cypsela:** Small, one-seeded fruit often with a hairy "parachute" (pappus).
 - *Example:* Tridax.
- 3. **Nut:** Has a very hard, woody pericarp.
 - *Example:* Cashew Nut.
- 4. **Samara:** The pericarp extends into a wing.
 - *Example:* Combretum.

IV. DISPERSAL OF FRUITS AND SEEDS

Dispersal is the scattering of seeds away from the parent plant.

- **Why Disperse?**
 1. To prevent overcrowding.
 2. To reduce competition for light, water, and nutrients.
 3. To colonize new habitats.
 4. To prevent the spread of disease among crowded plants.

1. Wind Dispersal

- **Adaptations:**
- **Lightweight:** Seeds are very small (Orchids).
- **Wings/Samara:** Extensions of the fruit wall to glide (Combretum).
- **Parachute/Pappus:** Hairs that catch the wind (Tridax, Dandelion, Cotton).

2. Water Dispersal

- **Adaptations:**
- **Buoyancy:** Must float.
- **Waterproof Epicarp:** Prevents rotting.
- **Air spaces:** Trapped air in fibrous mesocarp.
- *Example:* **Coconut** (can float across oceans), **Water Lily, Mangrove.**

3. Animal Dispersal

- **Adaptations:**
- **Edible Pulp:** Brightly colored, sweet-smelling, and tasty to attract animals. Seeds pass through digestion unharmed (Guava, Tomato, Pepper).
- **Hooks/Spines:** Attach to animal fur or human clothes. (Desmodium, Boerhavia).

4. Explosive Mechanism (Self-Dispersal)

- **Adaptations:**
- **Tension:** The fruit wall dries unevenly, creating tension. When triggered, it bursts open, flinging seeds away.
- *Examples:* Oil Bean, Balsam, Rubber, Okra.



V. PARTHENOCARPY

- **Definition:** The formation of fruit **without fertilization**.
- **Result:** These fruits are usually **seedless**.
- *Examples:* Banana, Pineapple.

- *Note:* Farmers can induce this artificially using hormones (Auxins) to create seedless grapes or watermelons.
-

Evaluation

1. Why is a Maize grain called a fruit and not just a seed?
2. Give one example of a fruit that is an "Aggregate Fruit."
3. Distinguish between a "Berry" and a "Drupe" based on their seed protection.
4. Name a local weed that uses hooks to disperse its seeds.
5. What is the biological advantage of seed dispersal?

Assignment

1. **Collection Project:** Collect 5 different weed seeds from around your house or school field. Tape them into your notebook. Identify how each one is dispersed (Look for hairs, hooks, or wings).
2. **Drawing:** Draw and label the internal structure of a Tomato (Transverse Section) and an Orange (Transverse Section).
3. **Observation:** Examine a Cashew fruit. Write a short paragraph explaining which part is the "Apple" and which part is the actual fruit.
4. **Reasoning:** Explain why plants that use "Explosive Mechanism" generally do not spread as far as those using "Wind Dispersal."
5. **Critical Thinking:** Why do unripe fruits tend to be green and sour, while ripe fruits are colored and sweet? (Hint: Think about when the plant *wants* the animal to eat it).

WEEK 10: REPRODUCTIVE BEHAVIORS

Specific Learning Objectives By the end of the lesson, students should be able to:

1. **Define** courtship behavior and parental care.
2. **Explain** three biological reasons why courtship is necessary before mating.
3. **Describe** the courtship display of the Agama Lizard and the Peacock.
4. **Establish the relationship** between the number of eggs produced and the level of parental care provided.
5. **Classify** animals (Fish, Toad, Bird, Mammal) based on their parental care strategies.

I. INTRODUCTION

Reproduction is not just about the anatomy (body parts); it involves complex **behaviors** that ensure the right mates find each other and that the offspring survive. These behaviors are largely **instinctive** (inborn).

II. COURTSHIP BEHAVIOR

Definition: Courtship refers to the series of distinct behavioral patterns exhibited by a male and female of the same species to stimulate each other prior to mating.

1. Importance of Courtship

Why don't animals just mate immediately? Courtship serves vital functions:

1. **Species Recognition:** It ensures that animals mate with their own kind. A female lizard will not respond to the dance of a male toad. This prevents wasted effort.
2. **Synchronization:** It prepares both partners physiologically. It excites the female and ensures she is ready to release eggs at the exact moment the male releases sperm.
3. **Mate Selection:** It allows the female to choose the strongest, healthiest male ("Survival of the Fittest").
4. **Territory Rights:** It establishes ownership of a safe area for mating.

2. Types of Courtship Displays

Different animals use different signals:

- **Visual Display:** Using bright colors or movements.
- *Example:* The **Male Agama Lizard** displays a bright blue head and orange tail. He nods his head rapidly (bobbing) to attract the female.
- *Example:* The **Peacock** spreads its magnificent tail feathers to impress the peahen.
- **Auditory (Sound) Display:** Using calls or songs.
- *Example:* **Frogs and Toads** croak loudly during the rainy season.
- *Example:* **Birds** sing specific mating songs.
- **Chemical Display (Pheromones):** Releasing scents.
- *Example:* Female **Moths** release pheromones that male moths can smell from kilometers away.
- *Example:* Dogs and Cats use urine to mark territory and signal readiness.
- **Territorial Display:** Fighting or aggression to secure a breeding ground.

III. PARENTAL CARE

Definition: Parental care involves any investment of time, energy, or resources by parents to ensure the survival of their offspring.

1. The Trade-off (Quantity vs. Quality)

There is a biological rule regarding reproduction:

- **Low Parental Care = Many Eggs:** Animals that do not care for their young must produce **thousands** of eggs because most will be eaten or die. (e.g., Fish, Amphibians).
- **High Parental Care = Few Eggs:** Animals that protect their young produce **very few** offspring because the survival rate is high. (e.g., Mammals, Birds).

2. Patterns of Parental Care in Animals

A. Class Pisces (Fish)

- *General:* Most fish lay eggs and leave them (No care).
- *Exception:* The **Tilapia** is a "Mouth Brooder." The female carries the fertilized eggs in her mouth until they hatch to protect them from predators. The **Seahorse** male carries eggs in a pouch.

B. Class Amphibia (Toads/Frogs)

- *General:* They lay jelly-covered eggs in water and abandon them. The tadpoles must survive on their own.
- *Exception:* The **Midwife Toad** (male) carries the egg strings wrapped around his legs.

C. Class Reptilia (Lizards/Snakes)

- *General:* Most bury their leathery eggs in the sand and leave.
- *Exception:* The **Crocodile** guards her nest fiercely and carries the hatchlings to the water in her mouth.

D. Class Aves (Birds)

- *High Care:* Birds build complex nests.
- **Incubation:** Parents sit on eggs to keep them warm.
- **Feeding:** Parents work tirelessly to bring worms/insects to the nestlings.
- **Protection:** They defend the nest against attackers.

E. Class Mammalia (Mammals)

- *Highest Care:* Mammals are Viviparous (give birth to live young).
- **Lactation:** Females produce milk to feed the young.
- **Protection:** Parents protect young for months or years.
- **Training:** Parents teach the young how to hunt and survive (e.g., Lions, Humans).

IV. IMPORTANCE OF REPRODUCTIVE BEHAVIORS

1. **Continuity of Species:** Without courtship (to find a mate) and parental care (to ensure survival), species would go extinct.

2. **Genetic Quality:** Courtship ensures that only the strongest males pass on their genes, keeping the population healthy.
 3. **Population Stability:** Parental care increases the chances of survival, stabilizing the population size.
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Evaluation

1. Why does a female fish lay thousands of eggs while a female cow produces only one calf a year?
2. Mention two ways the male Agama lizard attracts a female.
3. What is a Pheromone?
4. Define "Mouth Brooding" and name one organism that practices it.
5. Why is "Species Recognition" important during courtship?

Assignment

1. **Observation:** Observe the domestic fowl (chicken) in your neighborhood. Write a short report on how the hen protects and feeds her chicks.
2. **Comparison:** Create a table comparing the reproductive behavior of a **Toad** and a **Cat** under the headings: *Number of Eggs/Young*, *Method of Fertilization*, and *Level of Parental Care*.
3. **Drawing:** Draw a male Agama lizard in full courtship colors.
4. **Essay:** "Parental care is a sacrifice." Discuss this statement with reference to the energy cost for the parents.
5. **Research:** Find out about the **Cuckoo bird**, which is a "Brood Parasite" (it lays eggs in other birds' nests). How does this affect the host bird?